CLUSTERING ANALYSIS OF SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING

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In partial fulfillment of the requirements for the degree
Bachelor of Science in Computer Science
Specialized in Intelligent Systems

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The Researchers

DEDICATION

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N.S.D.

M.C.R

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ABSTRACT

This study analyzed two years of dengue case data in Cabuyao City, Laguna, with the aim of developing clear risk profiles for each barangay. Dengue cases from 2021 to 2022 were examined using three clustering algorithms: K-Means, Fuzzy C-Means, and Affinity Propagation. The analysis focused on key features: total number of cases, incidence rate, seasonal proportion, and peak month. To evaluate the models' performance, validation measures including the Silhouette Score, Davies-Bouldin Index, and Dunn Index were used. The results show that affinity propagation is more reliable than the other algorithms tested. The findings showed four distinct dengue risk profiles: (1) very high-risk barangays with high cases and severity, (2) high-risk barangays with early seasonal peaks, (3) moderate-risk barangays with strong rainy season patterns, and (4) low-risk barangays with fewer cases and irregular transmission. This study can help improve dengue prevention by guiding local health officials in focusing their actions and resources on the most at-risk barangays.

Keywords: Dengue incidence, high-risk barangays, Cabuyao City, machine learning, K-Means, Affinity Propagation, Fuzzy C-Means, clustering analysis, temporal trends, spatial trends, Davies-Bouldin Index, seasonal patterns, public health, Sustainable Development Goals (SDGs).

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DEFINITION OF TERMS

Affinity Propagation	A clustering method that finds natural groupings in data by letting the data points "communicate" with each other. It chooses cluster centers (called exemplars) based on how similar the data points are. In this study, it's used to automatically group barangays without needing to pre-set the number of clusters.
Choropleth Map	A type of map where areas are shaded based on the value of a certain variable, such as dengue case counts or risk level. It helps visually show which barangays are more affected.
Clustering	A machine learning technique that groups similar items together. In this study, it helps group barangays based on their dengue patterns so that health officials can better understand which areas face similar risks.
Davies-Bouldin Index (DBI)	A score used to evaluate how good the clusters are. A lower DBI means that the clusters are more distinct and well-separated, which is better.
Exemplar	A data point that represents the center of a cluster in Affinity Propagation. Exemplar data points differ from centroids in that they are actual data points and provide a more realistic representation of the cluster.
Feature Engineering	The activity of producing or choosing relevant variables (features) from raw data to help improve machine learning model performance. In this study, features including total cases, incidence rate, proportion of rainy season, and average month of peak incidence were developed to characterize patterns of dengue.

Fuzzy C-Means (FCM)	A clustering algorithm that allows one barangay to belong to more than one group at the same time, with different membership levels. It's useful when boundaries between clusters aren't very clear.
GeoJSON	A file format used for mapping. In this study, it contains the shapes and boundaries of barangays in Cabuyao and is used to visualize the clustered data on a map.
K-Means Clustering	A popular clustering method that groups data points (barangays) into K distinct clusters based on how similar their dengue trends are. It requires choosing the number of clusters in advance.
Machine Learning	A branch of artificial intelligence that involves computers learning from data to make decisions or predictions without being explicitly programmed. It is used in this study to automatically detect dengue risk patterns.
Pattern Recognition	The process of recognizing regular data trends, similarities, or forms. This is related to finding dengue activity frequency predictions among barangays in this study.
Standardization	A step in data preparation where numbers are adjusted to be on the same scale, making sure no single variable dominates the analysis unfairly.
Silhouette Score	A way to measure how well each barangay fits in its cluster compared to other clusters. Higher scores mean the barangay fits well in its assigned group.

Spatio-Temporal Analysis	A method that looks at both space (where something happens) and time (when it happens). In this study, it helps track dengue trends across different barangays over several years.
Unsupervised Machine Learning	A type of machine learning that identifies patterns, structures, or groupings in data without predefined labels or categories. In this study, clustering is performed as an unsupervised learning method to discover natural dengue risk groupings among barangays.

CHAPTER I

INTRODUCTION

Dengue fever remains a significant public health concern in the Philippines, affecting thousands of individuals annually, particularly during the rainy season when mosquito populations increase. According to the World Health Organization (2024), an estimated 390 million dengue infections occur worldwide each year. A significant portion of these cases is concentrated in Southeast Asia and is influenced by factors such as rapid urbanization, climate variability, and inadequate implementation of vector-control programs (Wilder-Smith et al., 2019).

In the Philippines, the Department of Health consistently monitors dengue. DOH (2022) reported more than 200,000 cases nationwide, emphasizing the persistent challenge despite ongoing prevention efforts such as mosquito control and public health advisories. One contributing factor to the continued spread is that case reports are frequently summarized only at the city level, resulting in limited analysis and delayed detection at the barangay level, where localized outbreaks often originate.

In Cabuyao City, Laguna, this issue is evident, as the city experiences periodic increases in dengue cases. Data from the Provincial Epidemiology and Surveillance Unit (PESU) of Laguna (Philippine Information Agency, 2025) indicate that dengue cases tend to rise every three years, making the disease a recurring concern for the province. This cyclical trend limits local health authorities' ability to anticipate high-risk periods and proactively implement targeted interventions.

During the first two months of 2025, Cabuyao City recorded 192 dengue cases. Although this number is lower than in nearby cities such as San Pedro and Santa Rosa, it still poses a significant threat to community health. In response to continued increases in reported cases, Laguna strengthened implementation of the 5S strategy: searching for and destroying mosquito breeding sites, taking self-protection measures, seeking early consultation, supporting fogging operations in hotspot areas, and maintaining adequate hydration. This approach aims to reduce transmission and improve case management (Philippine Information Agency, 2025).

This study aimed to help health agencies better identify high-risk areas by applying clustering techniques and analyzing spatial and temporal patterns. It used dengue data and case counts reported from 2021 to 2022. The findings revealed seasonal and geographical trends, showing how these patterns reflected the varying risk levels across the city. Insights from this study could help health officials improve their surveillance strategies, use resources more effectively, and better reduce dengue transmission in the community by enabling more informed, timely, and localized decision-making. It helps identify high-risk areas, allowing faster and more targeted dengue control.

Objectives of the Study

The main objective of this study is to develop and evaluate a machine learning framework for creating data-driven dengue risk profiles of the barangays in Cabuyao City using seasonal time-series data. This research specifically aims to:

1. Analyze the spatio-temporal patterns of dengue transmission in Cabuyao City by examining the historical (2014-2024) and seasonal (2021-2022) trends of dengue incidence at the city and barangay levels.
2. Develop a feature-based clustering model by engineering key epidemiological metrics (Burden, Risk, Seasonality, Timing) from the daily time-series data.
3. Evaluate and select an optimal clustering algorithm from a set of centroid-based methods (K-Means, Fuzzy C-Means, and Affinity Propagation) by conducting a rigorous comparative analysis using multiple internal validation indices.
4. Define and map the final barangay dengue risk profiles based on the selected optimal algorithm's results, and characterize these profiles using their unique temporal, demographic, and clinical severity characteristics to provide actionable insights for public health interventions.

Significance of the Study

To address dengue as an ongoing public health issue in Cabuyao City, this study used clustering methods to generate barangay-level risk profiles based on data collected from 2021 to 2022. The results obtained could direct tailored prevention efforts, support local health planning objectives, and contribute to the implementation of goals 3 (good health and well-being) and 11 (sustainable cities and communities) in the SDGs. Thus, the findings of this study have valuable implications for key stakeholders, including local health agencies, residents, and future researchers.

Local Health Officers/Agencies. Public health agencies identified barangays that continued to experience high dengue case counts. They then focused on fogging operations, removing mosquito breeding sites, and conducting information campaigns about mosquito control, especially targeting mosquito larvae (World Health Organization, 2024).

Residents of Cabuyao City. The results of the study would directly benefit the people of Cabuyao through targeted dengue prevention programs and enhanced public awareness of significant risk areas. The study promotes community engagement by encouraging citizens to take precautionary measures, such as effective waste disposal, cleaning stagnant water, and safe water storage (Centers for Disease Control and Prevention, 2024). By getting involved and addressing mosquitoes together, members could take action to combat mosquitoes and reduce the spread of dengue in their localities.

Scientists and Future Researchers. This research contributes to spatial epidemiology by demonstrating the application and evaluation of machine learning clustering techniques for analyzing long-term, geographically-based health data. The insight gained by this study into grouping locations based on multi-year incidence patterns may also point future researchers towards developing refined methodologies for mapping disease risk and provide new analytical paradigms for public-health surveillance (Louis et al., 2014), particularly by enabling more targeted, data-driven interventions and early warning systems for emerging disease hotspots. Future studies can build on this approach to explore additional environmental and socio-demographic factors that influence disease clustering.

Scope and Limitations of the Study

This study focused on the application and evaluation of three clustering algorithms, namely K-Means, Affinity Propagation, and Fuzzy C-Means, to identify dengue risk profiles in Cabuyao City. The analysis employed a feature-based time-series clustering approach on dengue incident data from 2021 to 2022. The clustering uses four engineered features that capture seasonal patterns and epidemiological characteristics: Total Cases (Burden), Incidence Rate (Risk), Rainy Season Proportion (Seasonality), and Average Peak Month (Timing).

This study limited its scope to 2021–2022, so it may not fully reflect the long-term, multi-year dengue epidemic cycles. Although the Cabuyao City Epidemiology and Surveillance Unit (CESU) of City Health Office I and II provided partial 2023 data, the study explicitly excluded them from the analysis to ensure data integrity and avoid potential biases arising from an incomplete temporal record. The validity of the results also depends on the accuracy and completeness of the official dengue case reports. The study did not include other possible methods, such as shape-based time-series clustering, or external factors such as environmental and socio-economic variables.

In the model analysis, researchers selected the final model after a multi-stage evaluation process that used three specific internal validation metrics: the Silhouette Score, Dunn Index, and Davies–Bouldin Index (DBI). Despite these limitations, the study remains appropriate for identifying barangay-level dengue risk profiles based on seasonal patterns using an interpretable feature-based clustering approach.

CHAPTER II

THEORETICAL FRAMEWORK OF THE STUDY

Review of Related Literature and Studies

This chapter presents the theoretical foundation and review of related literature and studies relevant to the research. It discusses existing knowledge, findings, and frameworks that inform the study's development. The review focuses on global and local perspectives on dengue incidence, clustering methods, and risk profiling, which provide a basis for understanding the problem and guiding the research process.

Epidemiology and Global Impact of Dengue

Dengue fever is one of the leading public health issues worldwide, with millions of cases each year. According to the WHO, the organization has cited that now half of the world's population is at risk for the disease, with an estimated 390 million infections taking place every year (World Health Organization, 2024). The dengue virus (DENV) causes dengue fever, and infected Aedes mosquitoes, such as Aedes aegypti and Aedes albopictus, mainly transmit it through bites (Centers for Disease Control and Prevention, 2023). These have mild to severe symptoms; the severe cases turn into dengue hemorrhagic fever (DHF) or dengue shock syndrome (DSS), which can be fatal if not treated well. Given the high transmissibility of the disease and the vector-borne nature of the pathogenic agent, urgent attention should be paid to dengue to lessen its impact on healthcare systems and economies (Gubler, 2012).

It is important to study dengue primarily because of its rising incidence and the strain it places on health care systems worldwide. Several countries regularly experience outbreaks, leading to hospital overload or congestion, increased health care costs, and economic losses (Shepard et al., 2013). Currently, dengue is endemic and a significant cause of morbidity and mortality in several regions, especially in Southeast Asia, the Americas, and Africa (Bhatt et al., 2013). "Climate change, urbanization, and poor vector control drive the corresponding transmission of dengue (Morin et al., 2013). The study of epidemiology will help identify targeted interventions effective in reducing transmission or preventing a future outbreak.

The Philippines is also among the countries that suffer from dengue. The Department of Health (DOH) reported over 220,000 dengue cases in 2022, along with hospitalizations and deaths (Department of Health, 2022). There are various reasons for this; one is the tropical climate and urbanization, coupled with sanitation issues, which create excellent breeding grounds for dengue mosquitoes (Capeding et al., 2014). Dengue control initiatives view vector control as a part of a larger program designed to raise awareness and create a local early warning system (Anders et al., 2015). However, to date, despite sustained public health efforts, dengue remains a significant threat, underscoring the need for more robust local research and more effective, locally targeted intervention strategies. Strengthening community engagement, enhancing surveillance systems, and integrating data-driven approaches can further improve prevention and control.

This study was conducted in Cabuyao City, Laguna, because it has experienced several dengue outbreaks in recent years. It is an urbanized area

with environmental concerns, including problems with waste management and poor waste disposal. There are also areas where water has become stagnant, including potential mosquito breeding grounds. Furthermore, Salapate (2019) stated that Barangay Banay-Banay is the largest barangay in Cabuyao City and faces issues in solid waste management, as it continues to experience environmental problems related to dengue fever.

According to the Provincial Epidemiology and Surveillance Unit (PESU) report published by the Philippine Information Agency (2025), Cabuyao City recorded 192 dengue cases in the last two months of 2025 alone. Not as many as in nearby Santa Rosa and San Pedro; however, Cabuyao City should be concerned about the continued spread of dengue in its area. The continued spread indicates they have not improved their environmental conditions to reduce the visible effects of a mosquito-borne disease (i.e., dengue). Therefore, this warrants a deeper investigation, using data collection and analysis to address geographical areas, especially barangays, before the spread occurs, and to allow local governments to take action.

Furthermore, according to the Laguna Provincial Health Office, dengue incidence in Cabuyao is likely to increase significantly during the rainy season (Department of Health Region IV-A, 2023). Given the seasonal pattern of increasing incidence, it is likely that the city will continue to experience dengue infections in the future. Therefore, it is urgent to have a sense of dengue incidence, contextually, as a city and in terms of individual barangays, in order to increase the preparedness of the local health care system. There is a need for localized, data-driven approaches that focus on the risk profile of individual barangays rather than generalized approaches.

Using available data, we analyze the spatio-temporal distribution and apply clustering methods to identify high-risk areas for monitoring dengue incidence in Cabuyao City. Spatio-temporal analysis is a powerful tool that can demonstrate how dengue transmission occurs over time and space, thereby helping identify areas of concern for the disease (Pasaribu et al., 2021). Approaches such as partitioning, exemplar-based, and fuzzy clustering, explored in this study through K-Means, Affinity Propagation, and Fuzzy C-Means, demonstrate distinct mechanisms of data grouping. Researchers can use them to determine groups of locations with similar infection patterns, which may later inform authorities in implementing targeted control measures in those specific areas (Ancheta, 2024; Addawe et al., 2021). Therefore, these methodologies can impact dengue prevention and control by improving resource management and informed decision-making.

Dengue at the local level is necessary to reduce the impact felt at the national level. This study will likely provide information-based recommendations for health authorities, policymakers, and community leaders in Cabuyao City. Strengthening surveillance and response strategies for dengue will not only benefit the city but also serve as a model for other areas in the Philippines prone to dengue. With sustained efforts and collaboration, combat against dengue can become more effective, culminating in reduced cases and improved public health outcomes (Wilder-Smith et al., 2019).

How Does Weather Affect Dengue Patterns in Metro Manila?

In Metro Manila, seasonal variations (rainfall, heat, humidity) do not immediately trigger dengue outbreaks, but they typically prompt a response before cases rise.

Sumi et al. (2020) found that dengue cases increase six to eight weeks after high temperatures, humidity, and rainfall. The explanation for this delay lies in the mosquitoes' developmental life cycle and transmission rate in the community.

Lumiguid et al. (2024) also validated this finding by analyzing weekly weather and dengue case data and identifying the most significant contributors. They found that maximum temperature, rainfall, and humidity were the most significant contributors. The use of weekly data also made clear and consistent patterns emerge in repeating intervals over time.

Researchers can leverage these patterns by grouping them using time-series clustering, which helps better explain temporal and spatial outbreaks. Therefore, this evidence suggests that seasonal dengue outbreaks in Metro Manila often follow time-based and weather-based patterns. By better understanding these patterns, we can begin identifying weeks or areas that will follow a similar outbreak behavior and align preparation and quicker response. This insight supports more targeted resource allocation and proactive public health interventions.

Spatio-Temporal Analysis of Dengue

Dengue fever remains a serious public health problem, particularly in tropical and subtropical regions where favorable climate and environmental conditions favor mosquito breeding. The World Health Organization (2024) reported that dengue cases have increased over the past decade, with approximately 390 million infections worldwide each year. As for spatial-temporal analysis, it is a crucial method for studying dengue diffusion, as it

allows researchers to track how outbreak patterns shift over time across different locations (Bhatt et al., 2013). It will also enable targeted interventions, such as mosquito control programs and early warning systems, in high-risk areas to reduce the spread of the disease. The WHO Early Warning and Response System has proven effective in preparedness and response to disease outbreaks (Cardenas et al., 2022).

Spatial Distribution and Dengue Clustering

Spatial analysis has mainly been used to examine dengue, especially to identify clusters or outbreaks. Understanding how dengue spreads in communities is important. These clusters often occur in cities where people are more likely to be affected by factors such as poverty, sanitation problems, and insufficient access to clean water (Louis et al., 2014). Using geographic information systems (GIS) and remote sensing has improved how we map dengue hotspots and outbreaks compared to the old methods (Nsoesie et al., 2016). In addition, spatial analysis can help public health officials target and control measures, such as vector control.

In the Philippines, spatial analysis has been used alongside other methods to help public health departments pinpoint high-risk areas in cities such as Manila, Cebu, and Quezon City. A study in Quezon City found that dengue outbreaks often hit the same neighborhoods repeatedly, showing that authorities need to conduct ongoing monitoring and implement targeted local actions in those spots (Medina et al., 2023). Moreover, the use of local Moran's I and Kulldorff's spatial scan statistics (SaTScan) revealed both the presence and movement of clusters over time, suggesting that dengue

hotspots can shift geographically but often persist in vulnerable urban neighborhoods. (Pangilinan et al., 2017)

Integrated Geographic Data Collection

Some people in the Philippines are using GIS to help monitor and control dengue outbreaks. The power of GIS lies in its ability to enable public health workers to create thematic maps that show spatial distributions and identify high-risk areas, thereby improving control strategies (Rubia & Bernardino, 2023).

In Quezon City, the researchers used GIS-based data to study the spatial and temporal distribution of reported dengue cases over five years, from 2010 to 2017. Using cluster and hotspot analyses, the researchers found that relative dengue case counts at various time points were helpful for local dengue prevention strategies (Medina et al., 2023).

GIS-based tools can therefore aid public health responses by enabling more targeted and efficient interventions. Authorities use scientific spatial analysis to pinpoint highly impacted areas and to justly channel resources, preventive actions, and initiatives that require significant intervention (Blanco, A., n.d.).

Dengue Clustering and High-Risk Areas

Studies show that population characteristics, breeding sites, and climate conditions drive dengue cases to form spatial clusters or aggregations rather than being randomly distributed (Getis & Ord, 1992; Gubler, 2012). GIS, along with spatial analysis methods, helps visualize clusters and reveal

relationships between dengue spread and community conditions. Clustering algorithms offer ways to improve the detection of high-risk regions; for instance, studies employing methods such as K-Means and DBSCAN have shown promise in predicting potential outbreak zones.

Studies of the spatio-temporal clustering of dengue cases have been vital in many regions, highlighting the value of combining the epidemiological data with spatial modeling. Researchers have conducted studies in both Southeast Asia and Latin America, where the disease burden is high. Scientists conducted a case study in Thailand using space-time models. The technique revealed areas with a high density of cas es, and successful interventions restricted virus transmission to susceptible individuals (Banu et al., 2011). In the same way, studies in Brazil employed clustering techniques to predict outbreaks, suggesting that, before any reported cases, the early identification of danger zones can help reduce the virus's spread (Lourenço & Recker, 2014).

The World Health Organization (2024) has stated that, instead of cure drugs, prevention programs, including surveillance and data-driven approaches, are the best ways to address dengue. Researchers disseminate studies of spatial clustering analysis, making them an essential aspect of an efficient response to outbreaks. For instance, A study in Rio de Janeiro State examined the spatio-temporal distribution of dengue, chikungunya, and Zika viruses from 2010 to 2019. The study described clusters of the same arboviral disease and overlapping clusters, which require established integrated surveillance systems to coordinate simultaneous reporting and response to different vector-borne diseases (Almeida et al., 2023).

The Philippines is a place where dengue clustering is now a significant problem, particularly in towns with higher populations, such as Metro Manila and Laguna, where poor drainage systems and high population density significantly support the reproduction of Aedes mosquitoes (Capeding et al., 2014). Poor waste disposal and stagnant water bodies in a few barangays of Cabuyao City, Laguna, primarily cluster dengue cases. This pattern reveals that authorities need to implement limited control measures in specific areas. The government tried to solve the problem by organizing community cleanups and using GIS-based tools to help citizens locate dengue sources. However, regular monitoring is needed to adapt to a changing environment (Department of Health, 2022).

Overall, clustering analysis in dengue surveillance is crucial for early detection and prevention. Location intelligence uses spatial data to identify areas with the highest risk of outbreaks, enabling health authorities to implement preventive measures such as fumigation, larvicides, and public information campaigns (Scott & Morrison, 2010). In the context of climate change and urbanization, these phenomena spread dengue infection, and researchers must solve the main problems of early and precise localization of high-risk areas (Bhatt et al., 2013).

Clustering Approaches to Disease Risk Profiles

Researchers use clustering, an unsupervised machine learning approach, to identify patterns and risk profiles in epidemiology. Clustering approaches group populations into clusters based on similar patterns of disease incidence. This approach is required if we are going to better plan for

public health and identify spaces that will require the limited health resources the most.

One of the classic clustering approaches is K-means clustering, introduced by MacQueen (1967). K-means partitions data into a fixed number of clusters that minimize variance, moving the data points towards the centroids iteratively. K-means clustering has become a widely popular method across various fields because of its ease of use, efficiency, and ability to handle moderate to large datasets.

K-means has been used to monitor various diseases, such as dengue fever, typically to identify high-incidence populations. For example, Siringi et al. (2020) used K-means clustering to identify areas with the highest dengue incidence, then mapped them and directed interventions toward high-risk populations.

There are many relevant attributes to K-means clustering. However, there are downsides, including the need to specify the number of clusters in advance and the clustering process's sensitivity to the placement of the centroids, which can yield many different clusterings of the same dataset (MacQueen, 1967). Additionally, K-means can often perform poorly when the clusters are not equally dense or spherical.

Alternative Clustering Techniques in Epidemiology

Besides K-Means, other clustering methods can address some of the problems when the data is noisy, complex, or overlapping.

Affinity Propagation. Frey and Dueck (2007) proposed a clustering algorithm that eliminates the need to preselect clusters. The algorithm

consists of a series of messages sent across data points to identify "exemplars" within the dataset, making it useful for unpredictable, heterogeneous datasets.

Fuzzy C-Means (FCM) clustering assigns data points to multiple clusters, with each data point having varying degrees of membership. Gupta (2020) stated that FCM can provide greater flexibility in classifying data points than K-Means or other clustering techniques, especially when dealing with overlapping boundaries or uncertainty, which are common characteristics of public health or epidemiological datasets. Furthermore, characterizing clusters or zones using FCM helps identify areas with varying risk levels, enabling more precise, targeted public health interventions.

Clustering Applications in Dengue Surveillance

According to research, clustering methods have been beneficial in better spatial and temporal analysis of dengue incidence. By identifying dengue clusters, health authorities can implement targeted interventions and use resources more effectively.

In the Philippines, researchers have used clustering methods in some local studies to improve dengue surveillance. For example, Ancheta (2024) demonstrated K-means clustering using dengue incidence and mortality data to identify regions with higher numbers of dengue cases and deaths. This information was helpful for health planning by enabling location-specific planning of dengue burden.

In another example, Addawe et al. (2021) used Fuzzy C-Means clustering on dengue case data from 127 barangays in Baguio City from 2009

to 2017. As a result, they identified distinct temporal trends and spatial clusters, enabling refined vector control and faster responses to outbreaks.

Although much less common in dengue studies in the Philippines, Affinity Propagation has promising research potential for more complex clustering conditions - such as dengue in vector control. The primary advantage of Affinity Propagation is to determine the number of clusters autonomously based on conditions of the specific data and is adaptable to many complex distributions of data, indicating the possibility of using Affinity Propagation on large-scale epidemiological datasets with noise (Frey & Dueck, 2007), making it suitable for uncovering meaningful disease patterns without prior cluster assumption.

Dengue Prevention and Control Strategies

To effectively prevent and manage dengue outbreaks, the community, the government, and technology must work together in a multifaceted initiative. Vector control, which involves methods for reducing mosquito populations, is a practical means of prevention through environmental management, larviciding, and insecticide-treated nets.

The World Health Organization (WHO) recognizes Integrated Vector Management (IVM) as the preferred long-term approach to biological, chemical, and environmental interventions (WHO, 2024). The government is committed to removing breeding areas where mosquitoes develop and reproduce to minimize dengue transmission. Mosquitoes will develop and reproduce much more quickly in clean, stagnant water tanks than in any tropical setting (Gubler, 2012). The government conducts its fogging and

cleanup work, but what constitutes clean is maintained by the community members through their own cleanup efforts.

Vaccination of humans against dengue fever is also a possible positive innovation. Dengvaxia, manufactured by Sanofi Pasteur, has been approved for use in many countries for individuals who already had a dengue infection (Hadinegoro et al., 2015). Also, second-generation vaccines, which have improved safety and efficacy profiles, are still undergoing research. An example of this is TAK-003 by Takeda Pharmaceuticals, which has shown positive results in clinical trials (Biswal et al., 2019). These advances will not only protect populations but also mitigate the severity of the virus's response.

Awareness and educational campaigns on dengue virus prevention are essential. Health authorities recognize that community involvement is the first step in preventing outbreaks. The 4S strategy used in the Philippines, search and destroy breeding sites of mosquitoes, Self-protection measures, seek early consultation, and support fogging operations during outbreaks, has also been implemented (Department of Health, 2022). Several schools and local governments have awareness campaigns to encourage community involvement in dengue prevention. Some early warning alert systems linking climate and disease variables will enable authorities to anticipate dengue outbreaks and act before the infection spreads within their cohort (Lourenço & Recker, 2014).

AI and GIS are increasingly used to improve the monitoring and targeting of intervention areas. These technologies enable more precise identification of high-risk zones and more efficient resource allocation. In

addition, community feedback from educational programs can help refine strategies to increase participation and improve the effectiveness of dengue prevention efforts.

Clustering Dengue Patterns Based on Time

Observing how dengue cases increase and decrease over time helps identify trends before outbreaks become more serious. Rather than simply counting how many cases each place has, time-series clustering focuses on timing: when cases begin to increase, how long they remain high, and how quickly they drop. The goal is to understand the pattern, not just the total numbers.

In Baguio City, analysts used dengue data from 129 barangays. They applied Fuzzy C-Means clustering. This method recognized that a barangay could belong to more than one group, demonstrating how outbreaks often spread simultaneously to nearby areas. Weekly data showed clearer trends across various periods, making it the best choice for studying outbreaks (Addawe et al., 2021). Other research studies employed Dynamic Time Warping (DTW) and correlation-based distance measures to improve trend matching across locations. These can exhibit the same pattern of outbreaks, although they may occur in different weeks (Aghabozorgi et al., 2015).

In another study, researchers pooled weekly data from 217 cities and identified dengue aggregations using hierarchical clustering. They then fed these clusters to an LSTM for training. The study found that predictions were more accurate when clustering was designed based on similar time-series patterns (Bogado-Machuca et al., 2021).

Recent Developments in Dengue Risk Profiling and Spatial Analysis

Previous studies have shown that dengue risk varies across spatial and socio-environmental contexts. Machine learning approaches, such as Random Forests in Singapore (Ong et al., 2018), and analyses in New Caledonia (Teurlai et al., 2015) and Brazil (Kikuti et al., 2015) have highlighted the influence of socio-economic and climate factors on dengue transmission. These findings demonstrate that localized risk profiling and spatial analysis can guide targeted interventions. This study uses a similar approach by analyzing dengue cases at the barangay level in Cabuyao City to identify high-risk areas and optimize control measures, thereby supporting more localized, data-driven public health decision making.

Literature Matrix

Table 1. Literature Matrix

Author(s) & Year	Title	Methodology	Key Findings
Addawe et al., 2021	A Fuzzy C-Means-Based Algorithm for the Surveillance of Dengue Cases Distribution in Local Communities	Applied Fuzzy C-Means clustering on spatiotemporal dengue data in 129 barangays in Baguio City	Identified temporal patterns of dengue cases, aiding in targeted intervention planning
Aghabozorgi et al. (2015)	*Time-series clustering – A decade review*	This paper reviewed various time-series clustering techniques, including Dynamic Time Warping (DTW) and correlation-based distance methods.	The authors highlighted that clustering methods like DTW help detect similar outbreak patterns even when they occur at different times, making them valuable.
Blanco, A. (n.d.)	Health-related Applications of Remote Sensing and GIS in the Philippines	Application of remote sensing and GIS technologies in health surveillance	Demonstrated the potential of these technologies in monitoring and modeling dengue outbreaks, aiming to improve disease surveillance and response strategies in the Philippines

Centers for Disease Control and Prevention (2023)	Data and Statistics on Dengue in the United States	Compilation and analysis of dengue case reports in the U.S.	Provided historical data on dengue cases, aiding in understanding transmission patterns and informing public health strategies
Lumiguid et al. (2024)	Determining meteorological factors influencing dengue cases in the National Capital Region using machine learning algorithms	This study used machine learning to analyze weekly dengue and weather data across Metro Manila.	Maximum temperature, rainfall, and humidity were found to be the strongest contributors to dengue outbreaks. The use of weekly data revealed consistent and
Medina et al., 2023	Spatial and Temporal distribution of reported dengue cases and hot spot identification in Quezon City, Philippines, 2010–2017	Analysis of barangay-level dengue case data from 2010 to 2017 using ArcGIS 10.3.1 for thematic mapping, global cluster analysis, and hot spot detection	Identified significant spatial and temporal variations in dengue incidence across Quezon City; eighteen barangays were consistently identified as hotspots, indicating areas for targeted interventions
PMC (2014)	Dengue disease surveillance: an updated systematic literature review	Systematic literature review on dengue surveillance methodologies	Emphasized the essential role of surveillance in detecting outbreaks and monitoring disease trends to trigger timely interventions
Sumi et al. (2016)	Effect of temperature, relative humidity, and rainfall on dengue fever in Manila, Philippines	The study analyzed weather data and dengue cases in Manila using statistical correlation with time-lag analysis.	Results showed that increases in temperature, humidity, and rainfall are followed by a rise in dengue cases after about 6–8 weeks. This delay is linked to the mosquito's life cycle and the time it takes for the virus to spread.
Teurlai et al. (2015)	Socio-economic and Climate Factors Associated with Dengue Fever Spatial Heterogeneity: A Worked Example in New Caledonia	Used maps and data to study dengue patterns based on weather and living conditions in New Caledonia.	Dengue risk was higher in crowded, low-income areas with specific weather conditions, underscoring the importance of both environmental and living conditions.
World Health Organization (2024)	Dengue Surveillance and Trends	A systematic review of global dengue surveillance data detection	Highlighted the increasing trend of dengue cases globally and emphasized the importance of robust surveillance systems

Synthesis and Relevance of the Study

This analysis demonstrates how spatial analysis and machine learning can identify high-risk barangays for dengue. Many studies have shown that GIS tools are crucial for monitoring, mapping, and predicting dengue outbreaks (Louis et al., 2014; WHO, 2024). At the local level, Medina et al. (2023) identified a consistent clustering of dengue cases in certain barangays in Quezon City. This underscores the significance of communities at high risk and the need for focused health interventions.

Historical data is important for improving dengue surveillance. The World Health Organization (2024) emphasizes that looking at past transmission patterns helps health officials predict future outbreaks more effectively. Similarly, Addawe et al. (2021) conducted a time-series study in Metro Manila, showing how weather patterns influence increases in cases, supporting the value of data-informed planning at the city and barangay levels.

This study provides practical knowledge for Cabuyao City, where the use of spatial clustering in dengue analysis helps improve local predictions and action planning. Annual hotspot identification enables local government units to more effectively focus their interventions, such as mosquito spraying, educational campaigns, and community cleanups. (Blanco, A. (n.d.).

Combining spatial analysis with historical case trends supports real-time monitoring systems that can issue early warnings, in line with precision public health principles emphasizing timely, targeted, and evidence-based interventions (Scott & Morrison, 2010). This study provides both theoretical and practical insights for dengue surveillance by identifying spatio-temporal patterns of cases within barangays.

Conceptual Framework

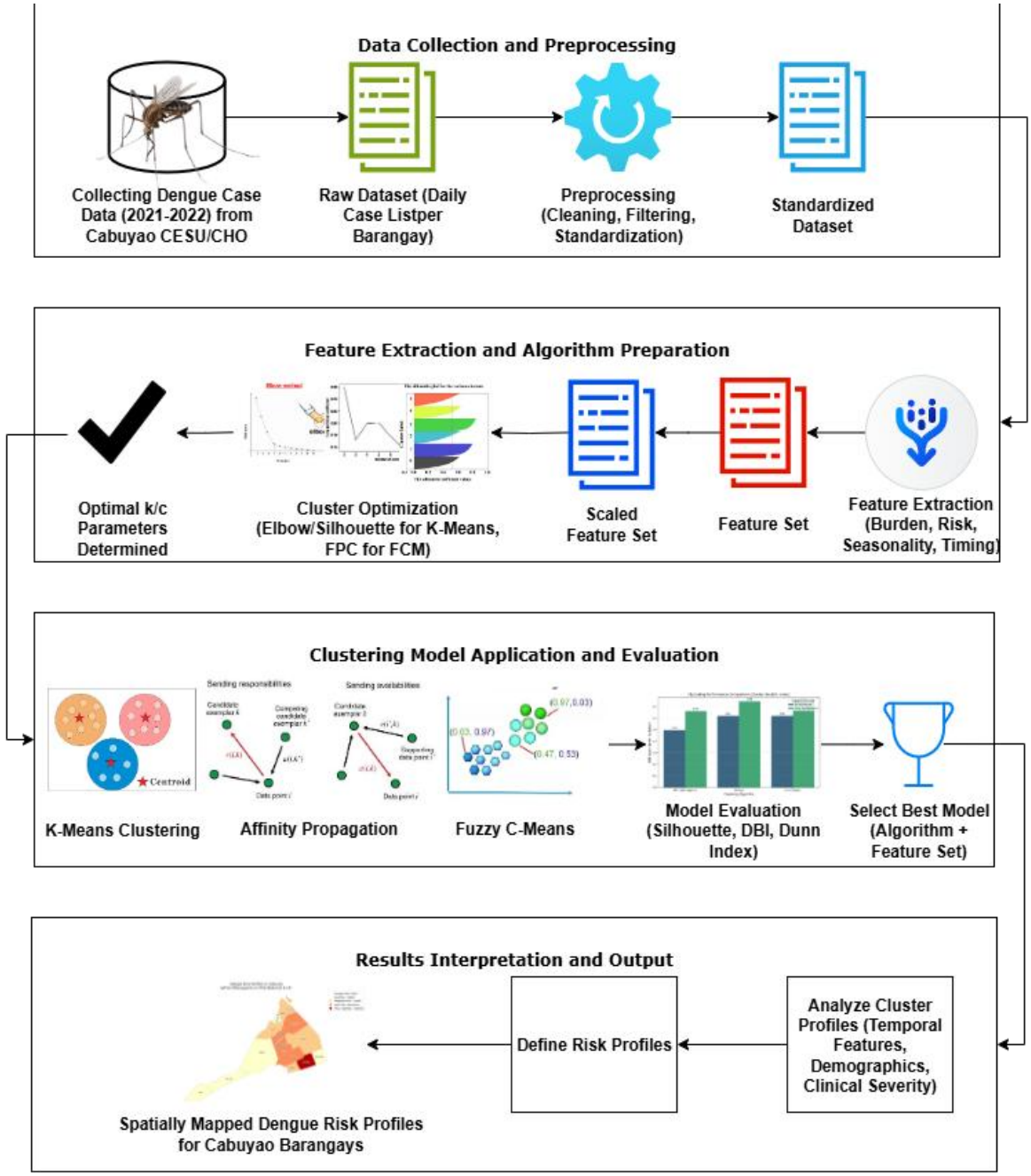


Figure 1. Conceptual Framework illustrating the feature-based clustering process of dengue cases in Cabuyao City from 2021 to 2022.

As shown in Figure 1, the analytical process begins with the collection of the 2021–2022 daily dengue case line list from the Cabuyao City Epidemiology and Surveillance Unit (CESU). The raw dataset then undergoes

a critical preprocessing stage that includes data cleaning, filtering, and standardization of barangay names, with close attention to missing or inconsistent entries to maintain data validity and reliability (Zhang, 2016).

Following preprocessing, the study applies a feature-based clustering framework. Rather than using a dimensionality reduction technique such as Principal Component Analysis (PCA), four epidemiologically meaningful features are derived from the standardized dataset to capture each barangay's dengue characteristics: Total Cases (Burden), Incidence Rate (Risk), Rainy Season Proportion (Seasonality), and Average Peak Month (Timing). These engineered variables are subsequently scaled to ensure equal influence during clustering.

To identify the optimal number of clusters for algorithms requiring predefined cluster counts (K-Means and Fuzzy C-Means), a cluster optimization process is conducted using multiple internal validation indices. The performance of three clustering algorithms, K-Means, Fuzzy C-Means, and Affinity Propagation, is then systematically assessed. Three well-established internal validation measures—the Silhouette Score (Rousseeuw, 1987), the Davies–Bouldin Index (Davies & Bouldin, 1979), and the Dunn Index (Dunn, 1974)—guide evaluation by collectively assessing cluster compactness and separation.

The final stage interprets cluster assignments using temporal, demographic, and clinical data. This process creates barangay-level risk profiles. Researchers visualize the results in a spatial risk map to guide targeted public health interventions in Cabuyao City (Medina et al., 2023).

CHAPTER III

METHODOLOGY

This chapter describes the methodical strategy used in this research. It outlines the geographical focus, the materials and resources used, the methods for data collection, the algorithms applied, and the model-creation process, including assessment and ethical considerations.

Study Area

This chapter describes the methodical strategy used in this research. It outlines the geographical focus, the materials and resources used, the methods for data collection, the algorithms applied, and the model-building process, including assessment and ethical considerations. The focus of this study was Cabuyao City, Laguna, Philippines, a first-class component city of Laguna Province. The town was strategically located in the Calabarzon region, bordered by the City of Santa Rosa to the northwest, Calamba to the south, and Laguna de Bay to the northeast. Cabuyao had a total land area of 43.30 square kilometers and a rapidly increasing population, primarily due to its industrial and residential growth (PhilAtlas, 2021).

Cabuyao served as an important economic center in the region. It contained several industrial parks, commercial lots, and residential subdivisions. However, urbanization and the tropical climate created a breeding ground for mosquitoes, increasing the risk of vector-borne diseases such as dengue. The mix of urban and rural areas also posed a challenge for local public health agencies as they worked to manage outbreaks of the

disease (Philippine Information Agency, 2025). The growing population, dense settlements, and limited sanitation in some areas further increase the risk of dengue, highlighting the need for targeted interventions.

The city had a total of eighteen (18) barangays, each with different population densities, environmental conditions, and levels of vulnerability to dengue epidemics (Philippine Statistics Authority, 2024). The more urbanized barangays showed varying levels of risk. It was important to understand the geographical distribution of dengue cases because this knowledge helped decision-makers develop effective intervention strategies and guide the proper allocation of health resources. Assessing these differences at the barangay level enables targeted, context-specific public health actions to reduce dengue transmission.

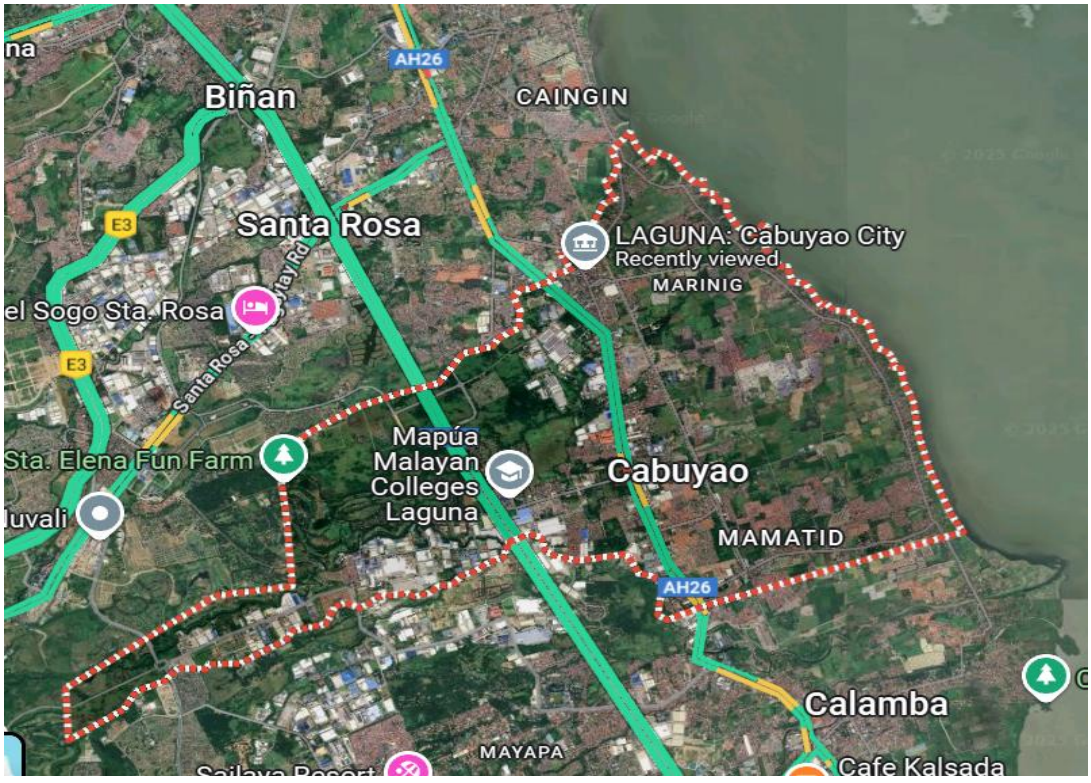


Figure 2. Boundary Map of Cabuyao, Laguna, Philippines, Google Maps (Accessed April 26, 2025)

Figure 2 presents a boundary map of Cabuyao City, Laguna, Philippines, clearly delineating the city's administrative limits in a prominent red outline against a satellite view backdrop from Google Maps. This map establishes the study area's physical and administrative scope, serving as an essential reference for understanding the distribution of dengue cases across Cabuyao's barangays and supporting spatially targeted public health interventions.

Research Design

This study employed a quantitative, cross-sectional design that applied a feature-based time-series clustering approach. Researchers recognize this methodology as a standard, highly interpretable approach for transforming complex time-series data into a structured format suitable for unsupervised machine learning. Unlike methods that compare raw time-series shapes, the feature-based approach allows for the incorporation of domain-specific knowledge. It produces clusters that are directly explainable in terms of their epidemiological characteristics (Aghabozorgi et al., 2015). The primary dataset consisted of a daily dengue case line list for Cabuyao City from 2021–2022, providing high-resolution temporal data for seasonal analysis. To establish a broader epidemiological context and verify long-term patterns, a dataset of annual totals from 2014–2024 was also utilized.

The core of the methodology involved a multi-stage machine learning process designed for robust and reproducible analysis. First, four epidemiologically relevant features, Burden, Risk, Seasonality, and Timing, were systematically engineered from the time-series data. This step is critical for converting the temporal dynamics of dengue transmission into a static,

multi-dimensional feature space that clustering algorithms can process. Subsequently, researchers conducted a comparative analysis of three distinct centroid-based clustering algorithms (K-Means, Fuzzy C-Means, and Affinity Propagation) to identify the most suitable model for this specific dataset. This "bake-off" approach is a standard practice in applied machine learning to ensure that the chosen algorithm is well-suited to the data's underlying structure (Rahman et al., 2022). Researchers guided the model selection process with a rigorous evaluation framework that included a baseline analysis to determine the optimal number of clusters and a hyperparameter optimization phase. They finalized the selection through a comparative assessment using three internal validation indices, since a multi-metric evaluation provides a more holistic and reliable assessment of cluster quality than any single score (Hussain et al., 2023). The best-performing model was then selected to define and map the final barangay risk profiles, providing a data-driven foundation for targeted dengue prevention and response strategies.

Data Collection and Preprocessing

This study acquired data using key parameters identified from a literature review on dengue surveillance and epidemiology. The primary data source was the Cabuyao City Epidemiology and Surveillance Unit (CESU). Researchers initially requested a five-year (2020–2024) daily line list to capture a broader temporal scope. The final dataset included complete, verified records for 2021 and 2022. Researchers excluded partial 2023 data to ensure data integrity and avoid potential biases from incomplete temporal

records. The resulting two-year dataset, comprising 1,113 records and 48 features, was deemed sufficient for analyzing seasonal patterns and served as the core of this study. To provide historical context, longer-term data from 2014 to 2024, available only as annual aggregates, were also acquired. Supplementary datasets were obtained for risk normalization and spatial analysis, including a Population Dataset (2020) with barangay-level counts from the Philippine Statistics Authority via PhilAtlas and a GeoJSON file of Cabuyao City's barangay boundaries from GADM.org. These supplementary datasets enabled the accurate calculation of incidence rates and the integration of spatial and demographic factors into the analysis.

Researchers conducted substantial preprocessing to ensure data quality, consistency, and ethical compliance. The first critical step involved de-identification: researchers permanently removed eight columns containing direct or indirect personally identifiable information (Streetpurok, PatientNumber, FirstName, FamilyName, FullName, DOB, EPIID, and UniqueKey), resulting in a 40-feature dataset for analysis. Researchers then consolidated the 2021 and 2022 daily case files and standardized barangay names across case, population, and geospatial datasets to ensure accurate merging. Missing values in dates were addressed by creating a final analysis date for each case, prioritizing the Date of Onset (DOnset) and using the Date of Admission (DAdmit) as a proxy where necessary, following standard epidemiological imputation practices (Zhang, 2016).

These preprocessing steps ensured that the dataset was accurate, consistent, and ethically compliant, providing a reliable foundation for subsequent analysis and modeling.

Data and Variables

From the de-identified, 40-column dataset, a specific subset of key variables was selected and processed for this analysis. Researchers selected these variables based on their established relevance in dengue epidemiology and their utility for constructing meaningful risk profiles.

The core variables for the analysis were the ClusterDate, which provides the essential temporal component for each case, and the Barangay, which serves as the fundamental spatial unit of analysis. Sociodemographic variables are critical for understanding which populations are most affected. For this, AgeYears was used, as age is a well-documented factor in dengue severity (Anders et al., 2015). This study stratified the continuous variable into a categorical Age_Group to enable more meaningful analysis beyond simple averages and included the patient's sex as a standard demographic descriptor.

Researchers included clinical variables to add a layer of public health relevance to the profiles, in line with World Health Organization (WHO) guidelines. The Dengue_Type provides critical information on clinical severity by distinguishing between classical Dengue Fever (DF) and the more severe forms, Dengue Hemorrhagic Fever (DHF) and Dengue Shock Syndrome (DSS). The Case_Classification provides context on the epidemiological certainty of each diagnosis (Confirmed, Probable, or Suspected).

These classifications are standard in infectious disease surveillance and are critical for understanding the nature of the data (WHO, 2024). An externally sourced variable, Population_2020, was included as it is essential for normalizing case counts and calculating the incidence rate, a standard and critical metric for comparing risk across geographic areas of different sizes.

These variables, formally defined in Table 2, provide essential information for constructing detailed and reliable barangay-level dengue risk profiles. They enable a comprehensive understanding of local transmission patterns and support targeted, actionable health strategies for more effective disease prevention and control.

Table 2: Description of Variables Used in the Analysis.

Variable Name	**Description**	**Values / Scale**
ClusterDate	The final analysis date for each case is derived from the Date of Onset or Date of Admission.	Date (YYYY-MM-DD)
Barangay	The standardized barangay of residence for the case.	Categorical (18 barangays)
AgeYears	The patient's age in completed years.	Continuous (0-78)
Age_Group	The patient's stratified age category.	Categorical ("0-5 yrs", "5 > - 19 yrs", "> 19 yrs")
Sex	The biological sex of the patient.	Categorical ("M" for Male, "F" for Female)
Outcome	The final clinical outcome of the case.	Categorical ("A" for Alive, "D" for Died)
Dengue_Type	The clinical severity classification of the dengue case.	Categorical: DF: Dengue Fever DHF: Dengue Hemorrhagic Fever DSS: Dengue Shock Syndrome
Case_Classification	The epidemiological certainty of the case diagnosis.	Categorical: C: Confirmed P: Probable S: Suspected
Population_2020	The official 2020 PSA census population for each barangay.	Continuous (per barangay)

Feature Engineering

This study adopted a feature-based approach to time-series clustering, transforming the complex 24-month time-series data into a concise, interpretable format suitable for clustering. This process is essential for representing each barangay's unique dengue situation as a set of quantitative metrics that a clustering algorithm can effectively compare. Unlike methods that directly compare the raw shapes of time series, such as Dynamic Time Warping (DTW), the feature-based approach allows for the incorporation of domain-specific epidemiological knowledge. By design, the feature-based approach ensures that the resulting clusters remain mathematically sound, epidemiologically meaningful, and directly interpretable for public health action (Aghabozorgi et al., 2015).

The objective of this stage was to engineer a set of features that effectively summarize each barangay's dengue profile across key public health dimensions. This study systematically developed four distinct features from the 2021–2022 daily case data to construct a multi-dimensional profile for each barangay, capturing its unique patterns of burden, risk, seasonality, and timing.

The first feature, Total Cases (Burden), was formulated as the absolute count of all reported dengue cases per barangay from 2021 to 2022. This feature serves as the most direct measure of the absolute scale and public health burden experienced by each barangay. While this is a primitive representation, it is a foundational metric in any epidemiological analysis for identifying the areas that contribute most to the total case load. The study expresses this relationship as:

$$\text{Burden} = \sum(\text{Total Dengue Cases per Barangay, 2021} - 2022)$$

To account for the significant variation in population sizes across the eighteen barangays of Cabuyao, a second feature, Incidence Rate (Risk), was calculated. This feature indicates the number of dengue cases per 10,000 population, based on the 2020 PSA census data. This measure represents the per-capita risk and enables fair comparisons among barangays of varying population sizes. This method allows for the normalization of case counts across populations of different sizes, ensuring comparability, a standard and critical practice in spatial epidemiology (The Open University, n.d.). The study expresses this relationship as:

$$Incidence\ Rate = \frac{Barangay\ Population}{Total\ Dengue\ Cases} \times 10{,}000$$

Recognizing that dengue in the Philippines is highly seasonal, the third feature, Rainy Season Proportion (Seasonality), was engineered to quantify this temporal pattern. This feature represents the percentage of a barangay's total dengue cases that occurred during the defined rainy season (June to November). It directly measures how closely a barangay's dengue transmission aligns with known climate drivers, a phenomenon well documented in the Philippines (Sumi et al., 2016). A higher value indicates a more pronounced seasonal pattern, while a lower value suggests more consistent, year-round transmission. This study calculates it as:

$$Seasonality = \frac{Cases\ (June - November)}{Total\ Dengue\ \ cases} \times 100$$

The final feature, Average Peak Month (Timing), was calculated to identify the average timing of peak dengue case counts, giving greater weight to months with higher case counts. This measure describes the general timing, or 'center of gravity,' of dengue activity in each barangay. It allows researchers to determine whether outbreaks characteristically occur earlier or later in the rainy season. The calculation follows a weighted mean month approach commonly used in epidemiological studies to examine seasonal patterns of disease (Paireau et al., 2016), and the study expresses this relationship as:

$$Timing = \frac{\sum(M \times C)}{\sum C}$$

Where M is the month number (1–12), and C is the case count for that month. By clustering on this feature, the methodology ensures that the resulting barangay risk profiles are defined not only by their total incidence but also by their crucial seasonal trends. This approach groups barangays with similar outbreak timing patterns.

Researchers standardized the four-dimensional feature set using scikit-learn's StandardScaler before developing the model. This critical step, also known as Z-score normalization, transforms the data so that its mean is zero and its standard deviation is 1.

This step ensures that all features contribute equitably to distance-based clustering calculations and prevents features with naturally larger numerical ranges (such as Total Cases) from disproportionately influencing the clustering outcome (Rahman et al., 2022). As a result, the clustering process becomes more accurate and balanced.

Model Development

The primary objective of the model development stage was to build, optimize, and compare three unsupervised machine learning algorithms—K-Means, Fuzzy C-Means (FCM), and Affinity Propagation—to identify the most suitable model for generating barangay-level dengue risk profiles. This study chose these algorithms for their complementary clustering capabilities: K-Means for simplicity and efficiency (Siringi et al., 2020), FCM for modeling overlapping cluster memberships common in public health data (Gupta, 2021), and Affinity Propagation for automatically determining the optimal number of clusters (Frey & Dueck, 2007)."

This study conducted all experiments in Python (3.x) within Google Colaboratory to ensure reproducibility. Data preprocessing and feature engineering used pandas and NumPy; K-Means, Affinity Propagation, and internal validation metrics were implemented with scikit-learn, while FCM used scikit-fuzzy. Researchers produced visualizations with Matplotlib, Seaborn, and Plotly. The clustering workflow (Figure 3) began with feature scaling and model execution, followed by internal validation to support objective comparison and robust model selection.

This study performed hyperparameter tuning and sensitivity analysis for all algorithms. It tested multiple K-Means and FCM cluster configurations to assess stability, adjusted Affinity Propagation parameters to examine their effects on clustering, and evaluated model performance using the Silhouette Score, Davies-Bouldin Index, cluster cohesion and separation measures, and visual inspection of cluster distributions. This multi-criteria framework ensured

the selected model was statistically robust, epidemiologically meaningful, and aligned with known dengue transmission patterns, supporting reliable barangay-level risk profiling.

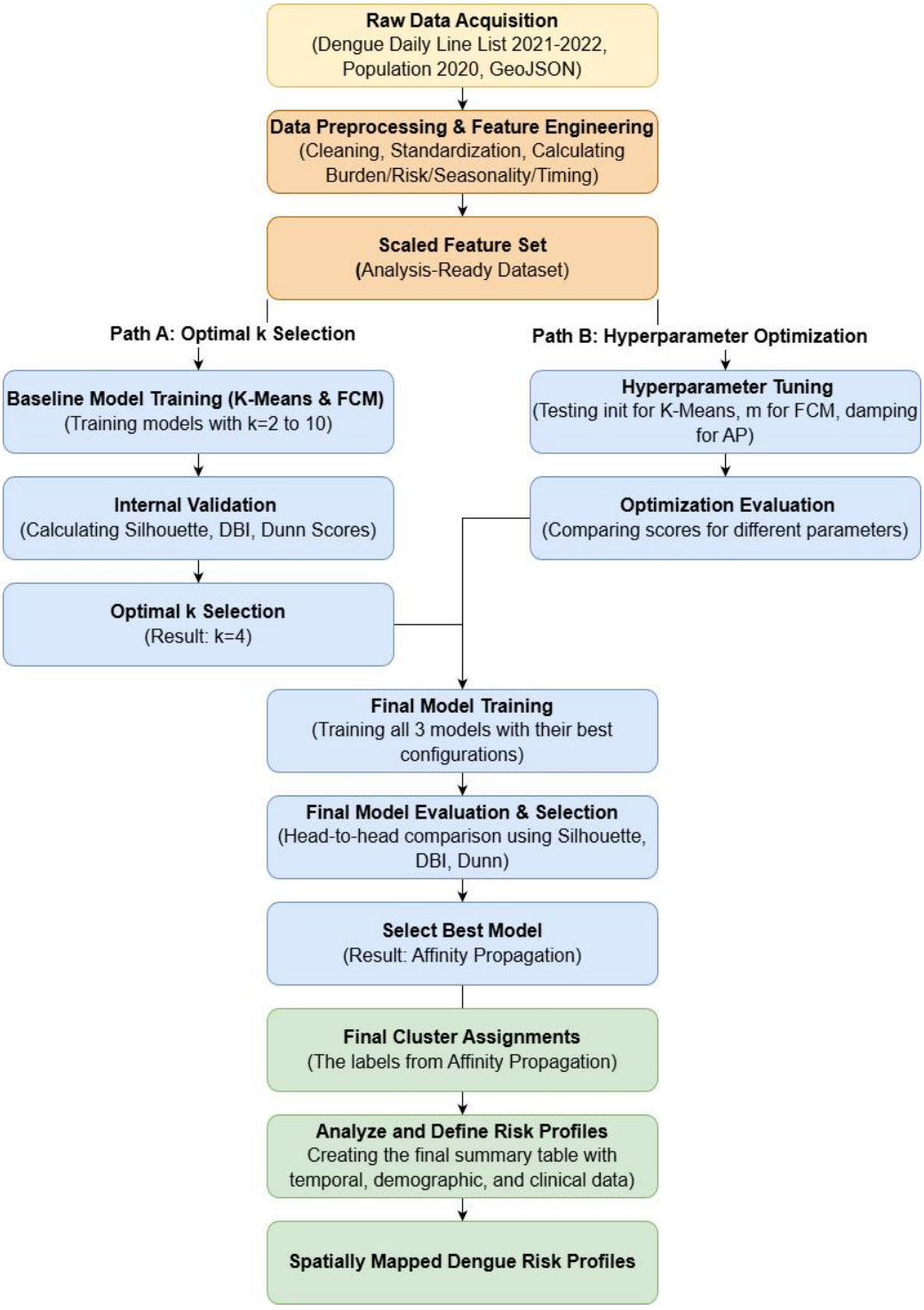


Figure 3. Analytical workflow illustrating the end-to-end process for dengue risk profiling using unsupervised clustering models.

Path A, the Baseline Analysis and Optimal k Selection, was conducted to determine the optimal number of clusters (k) for the K-Means and Fuzzy C-Means algorithms, which require this parameter to be specified before clustering begins. This study highlights the critical importance of choosing k appropriately, as an unsuitable value can oversimplify the data or produce non-generalizable clusters. The study ran both models with k values ranging from 2 to 10 and assessed cluster quality using a suite of internal validation metrics, a standard approach when a 'ground truth' is not available for external validation (Rousseeuw, 1987). This process is crucial to ensuring that the final cluster count is statistically significant and not arbitrary. The results of this analysis strongly indicated that k=4 was the most appropriate number of clusters, as it provided the best balance between cluster cohesion (compactness) and separation (distinctness) for this specific dataset, a crucial step for ensuring the final clusters are both statistically significant and epidemiologically interpretable (Kodinariya & Makwana, 2013).

Concurrently, Path B, the Hyperparameter Optimization phase, was conducted to ensure a fair comparison between the best possible configurations of each of the three models. This step is critical for demonstrating that the selected model's performance is not coincidental but rather a product of a well-suited configuration to the data's underlying structure. This study tested the init parameter in K-Means ('k-means++' vs. 'random') to evaluate how different centroid initialization strategies affected the final cluster quality. For Fuzzy C-Means, the fuzziness exponent m was tested at 1.5, 2.0, and 2.5 to determine the optimal degree of cluster overlap, a key parameter for soft clustering models (Bezdek, 1981). For Affinity

Propagation, the damping factor was set to 0.7, 0.8, and 0.9 to control message passing and convergence. The results of this phase indicated that the default or standard parameters were already highly robust and effective for this dataset, providing strong confidence in the models' stability.

In the final stage of development, Final Model Training, this study combined insights from both paths. It trained the three algorithms one last time using their determined optimal configurations, K-Means and Fuzzy C-Means with k=4, and Affinity Propagation with its optimized damping factor, producing three distinct sets of cluster assignments for the 18 barangays. The performance of these final, optimized models was then formally assessed using the internal validation metrics detailed in the following section.

This head-to-head comparison provided the basis for selecting the highest-performing model, which researchers then used to generate the barangay risk profiles, the radar chart visualization, and the thematic map.

Machine Learning Algorithms

This study utilized three distinct unsupervised machine learning algorithms to cluster barangays based on their engineered feature profiles. Researchers selected algorithms that represent complementary approaches to centroid-based clustering to provide a comprehensive evaluation of which methodology best suits the definition of dengue risk profiles. These include K-Means for its foundational partitioning method, Fuzzy C-Means for its ability to handle overlapping cluster boundaries, and Affinity Propagation for its capacity to automatically discover the number of clusters. The results from these algorithms were then compared to identify patterns.

K-Means Clustering is a non-parametric, partitioning algorithm that divides observations into a predefined number of clusters, assigning each observation to the cluster with the nearest mean (centroid) (MacQueen, 1967). Epidemiological studies widely use K-Means for its simplicity, computational efficiency, and effectiveness in identifying spherical clusters (Siringi et al., 2020). The algorithm minimizes the within-cluster sum of squares (WCSS), defined as:

$$J = \sum_{j=1}^{k} \sum_{x_i \in S_j} \|x_i - \mu_j\|^2$$

In this formulation, x_i i is a data point, μ_j is the centroid of the cluster S_j, and $\|x_i - \mu_j\|^2$ is the squared Euclidean distance between the point and its assigned centroid. As illustrated in Figure 4, the optimization proceeds iteratively through two steps. First, in the assignment step, each data point is assigned to the cluster whose centroid is closest. Second, in the update step, each centroid is recalculated as the mean of all data points assigned to it. These steps repeat until the cluster assignments stabilize, indicating convergence. Researchers must select the optimal number of clusters k as a critical prerequisite for K-Means, typically guided by internal validation metrics such as the Silhouette Coefficient, Davies–Bouldin Index, and Dunn Score (Rousseeuw, 1987; Kodinariya & Makwana, 2013). Choosing the correct value of k ensures meaningful, well-separated clusters, thereby enhancing the interpretability of the results. Once the algorithm converges, the final cluster centroids define the distinct group characteristics within the dataset.

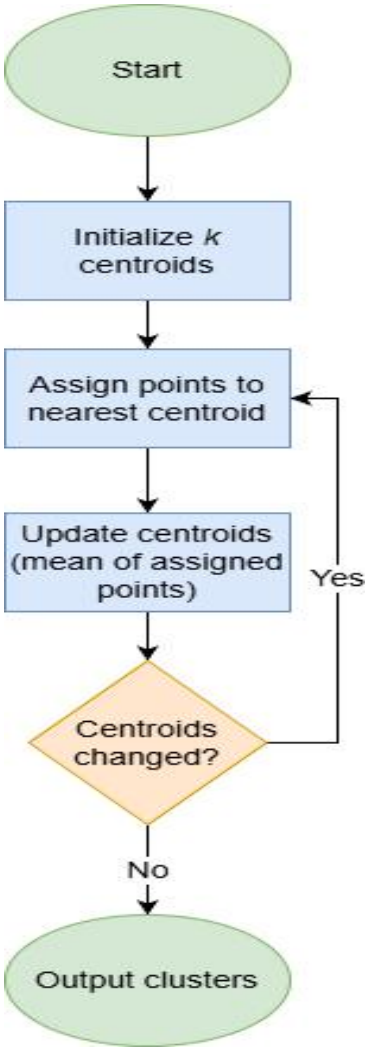


Figure 4. Flowchart of the K-Means clustering algorithm based on MacQueen (1967).

Affinity Propagation (AP) is an exemplar-based clustering algorithm that identifies cluster centers directly from the data through a message-passing procedure (Frey & Dueck, 2007). Unlike K-Means, AP does not require the number of clusters to be specified beforehand. Instead, as shown in Figure 5, it treats all data points as potential exemplars and iteratively exchanges two message types between them: responsibility and availability.

This approach allows Affinity Propagation to adaptively discover clusters of varying sizes and densities, making it particularly useful for complex datasets where natural groupings are not obvious.

The algorithm's iterative message-passing ensures that each data point is evaluated in the context of the entire dataset, improving the accuracy of cluster assignments. As a result, AP can reveal subtle patterns in dengue incidence across barangays that traditional clustering methods, such as K-Means, might miss.

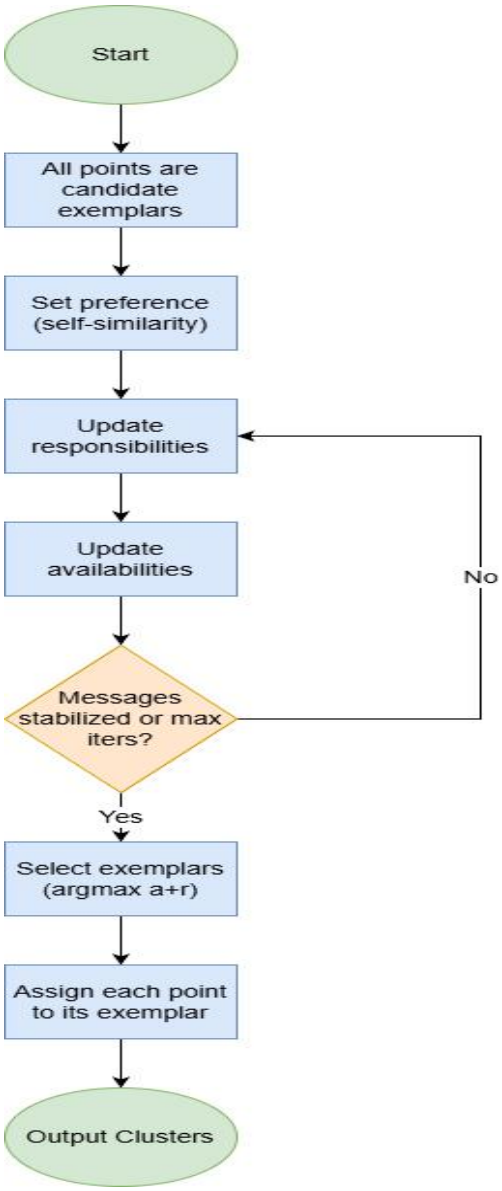


Figure 5. Flowchart of Affinity Propagation clustering algorithm based on Frey & Dueck (2007).

Figure 5 presents a flowchart illustrating the iterative message-passing process of the Affinity Propagation clustering algorithm, as initially proposed

by Frey and Dueck (2007). Unlike methods that require specifying the number of clusters in advance, Affinity Propagation treats every data point as a potential cluster exemplar (representative). It determines both the number and the members of clusters through the iterative exchange of two message types between points. The algorithm starts by considering all points as candidate exemplars. This study sets the preference values (the diagonal of the similarity matrix, representing self-similarity), which control the likelihood that the algorithm chooses each point as an exemplar; these values are often set to the median similarity or tuned to control the desired number of clusters. This elegant, data-driven approach naturally discovers the number and structure of clusters without requiring a pre-specified k, producing clusters centered on actual data points (exemplars) that are often more interpretable and representative of the dataset. The flowchart effectively captures the distinctive message-passing nature of Affinity Propagation, making it a valuable visual reference for understanding this exemplar-based clustering method, which this study evaluates as an advanced alternative in the analytical workflow.

The responsibility reflects how well-suited the point is as the exemplar for another point, and the algorithm updates it as follows:

$$r(i,k) = s(i,k) - \max_{k' \neq k} \left\{ a(i,k') + s(i,k') \right\}$$

Availability reflects how appropriate it is for one point to choose another as its exemplar, and the algorithm updates this value as follows:

$$a(i,k) = \min \left\{ 0, r(k,k) + \sum_{i' \notin \{i,k\}} \max \left\{ 0, r(i',k) \right\} \right\}$$

In these equations, $s(i,k)$ is the similarity between points i and k, typically defined as the negative squared Euclidean distance. The optimization begins by initializing all similarities and message values. Then, in each iteration, responsibilities and availabilities are updated based on the current state of the messages. This process continues until the values converge, at which point the algorithm selects exemplars and forms clusters. The damping factor is a key hyperparameter that controls the influence of previous message values and helps prevent oscillations during convergence. Affinity Propagation is particularly robust for heterogeneous datasets where the number of natural clusters is not apparent (Shang et al., 2012).

Fuzzy C-Means (FCM) is a soft clustering algorithm that allows each data point to belong to multiple clusters, with membership values ranging from 0 to 1. This flexibility is especially useful in epidemiological studies, where risk boundaries are rarely clear-cut and often overlap (Gupta, 2021; Addawe et al., 2021). Rather than forcing each observation into a single group, FCM reflects gradual differences in risk by assigning varying degrees of membership across clusters. The algorithm minimizes a fuzzy objective function, as shown in Figure 6, producing clusters that better capture uncertainty and variation in public health data. As a result, researchers can interpret barangays with intermediate dengue risk levels more realistically. Additionally, FCM enables public health officials to prioritize interventions based on membership strength, targeting areas that show higher potential risk.

The method also facilitates comparative analysis with other clustering techniques, helping validate the robustness of the identified risk profiles. This

representation supports more flexible decision-making, as researchers closely monitor barangays with moderate membership in high-risk clusters even if they do not classify them as extreme hotspots.

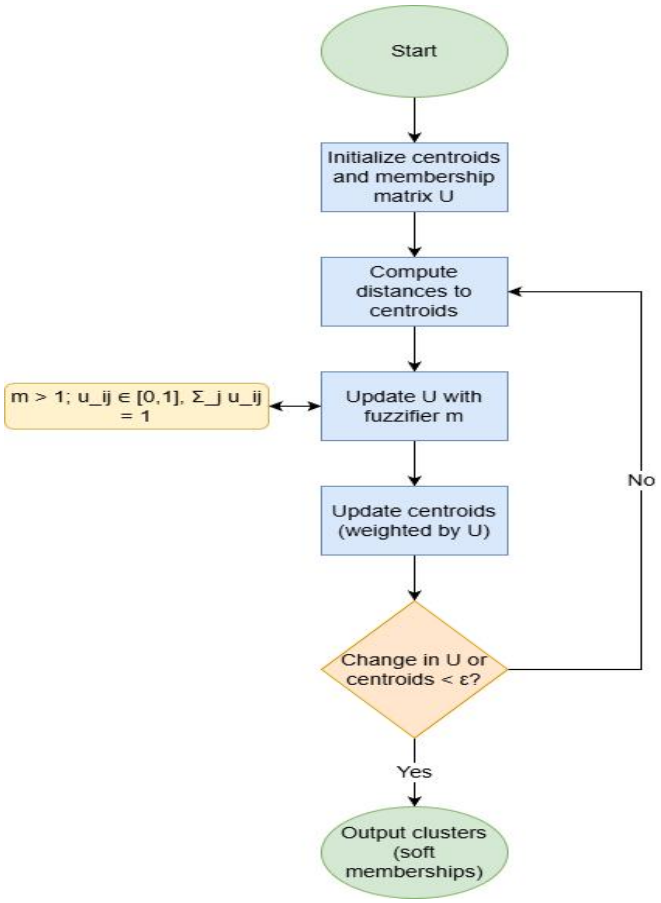


Figure 6. Flowchart of the Fuzzy C-Means clustering algorithm based on Bezdek (1981).

Figure 6 presents a flowchart of the iterative Fuzzy C-Means (FCM) clustering algorithm, as initially proposed by Bezdek (1981). The algorithm begins with the initialization of c centroids (where c is the specified number of clusters) and a membership matrix U, where each element u_ij indicates the degree of membership of data point i in cluster j, with values constrained to the interval [0,1] and summing to 1 across all clusters for each point. Data points are then repeatedly assigned fuzzy memberships to the centroids based on their distances (typically Euclidean), adjusted by the fuzzifier

parameter m (m > 1), which controls the softness of the partitioning. Higher m values produce fuzzier, more overlapping memberships. The algorithm subsequently updates each centroid as the weighted mean of all data points, with weights given by the corresponding membership values raised to the power m. The algorithm continues the membership and centroid update cycle in a loop until convergence occurs, specifically, when the change in the membership matrix U or the centroids falls below a predefined small threshold ε. Upon convergence, the final soft clusters are output as the membership matrix U, providing probabilistic assignments rather than crisp partitions. This fuzzy, soft-partitioning structure highlights the flexibility of Fuzzy C-Means in handling ambiguous or overlapping data distributions, making it a powerful unsupervised learning technique widely used for applications such as the barangay-level dengue risk features in this study. The flowchart serves as a visual reference for understanding one of the advanced clustering methods evaluated in the analytical workflow.

The objective function is defined as:

$$J_m = \sum_{i=1}^{N} \sum_{j=1}^{c} u_{ij}^{m} \cdot \|x_i - v_j\|^2$$

Here, N is the number of data point,c is the number of clusters, c is the centroid of cluster j, m is the fuzziness exponent (typically $m > 1$), and u_{ij} is the membership degree of x_i in cluster j. The optimization begins by initializing membership values randomly or heuristically. In each iteration, the algorithm updates the centroids by averaging the data points, with membership degrees determining the weights. Then, the membership values

are recalculated based on the updated centroids and the distances between each data point and each centroid. This process continues until the change in membership values falls below a predefined threshold. Membership values are constrained such that the sum across all clusters equals 1 for each data point.

Model Evaluation

Since no pre-existing “ground truth” or expert-defined risk classification exists for the barangays of Cabuyao City, the quality of the final cluster outputs from the three optimized models was assessed using established internal validation metrics. This approach is standard in unsupervised learning, where evaluation relies on the data's inherent properties and the geometric structure of the resulting clusters (Hussain et al., 2023). The comparative assessment served as the basis for selecting the best-performing model for the final barangay risk profiling.

To ensure a holistic and reliable assessment of cluster quality, three distinct metrics were applied. Depending on a single metric can be misleading, as each index is sensitive to different aspects of clustering, such as shape, density, and separation.

By combining metrics that evaluate quality from complementary perspectives, point-level cohesion and separation, average cluster compactness, and worst-case cluster distinctiveness, a more reliable basis for model selection was established. (Rahman et al., 2022). The chosen metrics were the Silhouette Score, the Davies-Bouldin Index (DBI), and the Dunn Index.

The **Silhouette Score** measures how well-defined and distinct the clusters are by evaluating, for each data point, how similar it is to its own cluster (cohesion) compared to how similar it is to the nearest neighboring cluster (separation). The step-by-step process for calculating a single-point score is illustrated in Figure 7. The average Silhouette Score over all data points provides a single measure of overall clustering quality, with values ranging from −1 to 1. Higher scores indicate that data points are well matched to their own clusters and poorly matched to neighboring clusters, values near 0 suggest overlapping clusters, and negative values may indicate potential misclassification.(Rousseeuw,1987).

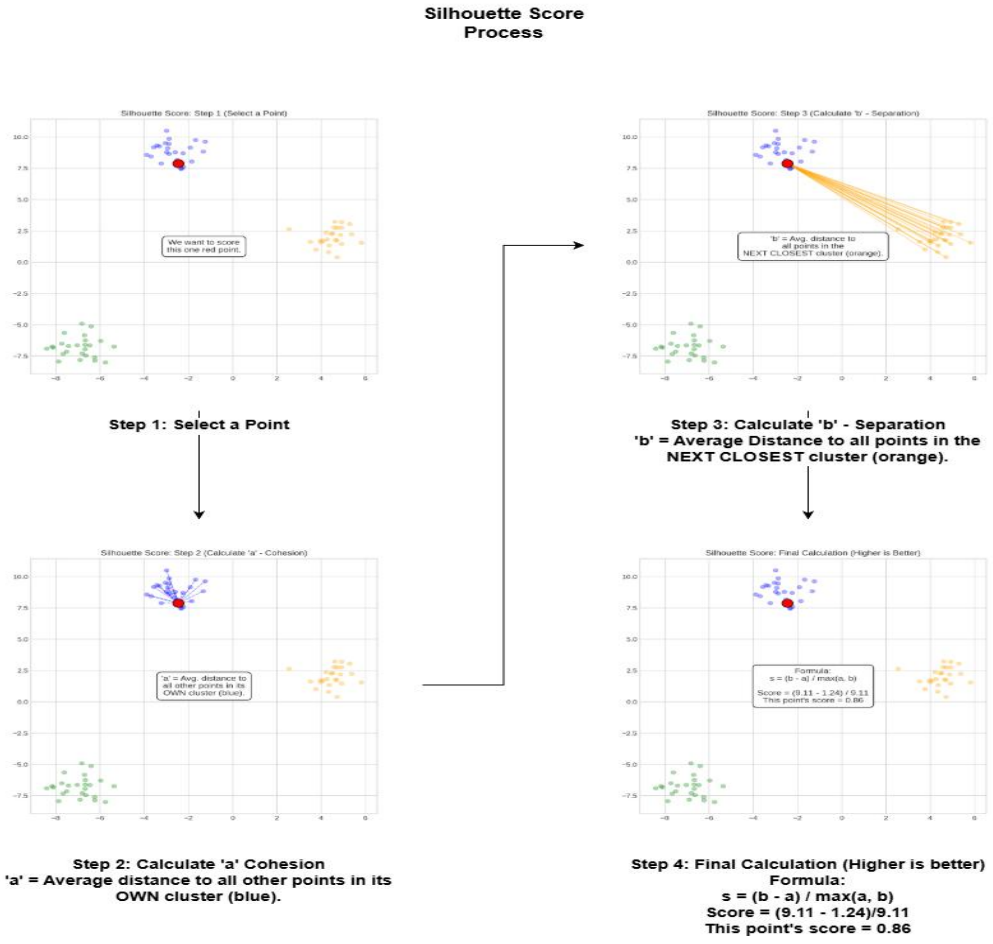


Figure 7. Diagram of the Silhouette Score process based on Rousseeuw (1987).

Figure 7 presents a multi-panel diagram illustrating the step-by-step calculation of the Silhouette Score for an individual data point, as introduced by Rousseeuw (1987). The Silhouette Score is an internal validation metric used to evaluate clustering quality by measuring how well each point fits within its assigned cluster (cohesion) while remaining distant from points in the nearest neighboring cluster (separation). The diagram uses scatter plots to demonstrate the four sequential steps of the computation, highlighting the selected point in red, its own cluster in blue, and the nearest neighboring cluster in orange. Step 1 involves selecting a single data point for scoring. In Step 2, the cohesion term aaa is calculated as the average distance from the selected point to all other points in its own cluster. Step 3 computes the separation term bbb as the average distance from the selected point to all points in the nearest neighboring cluster.

In Step 4, the silhouette score for the point is calculated using the formula s=(b−a)/max(a,b)s = (b - a) / \max(a, b)s=(b−a)/max(a,b), which yields a value between −1 and +1. A score close to +1 indicates that the point is well-clustered, demonstrating strong cohesion and clear separation from other clusters. A score near 0 suggests the point lies near the boundary between clusters, while a negative score implies potential misassignment. In the example shown, the selected point achieves a silhouette score of 0.86, reflecting excellent assignment to its cluster with strong internal cohesion and distinct separation from the nearest neighboring cluster.

This visual, sequential breakdown effectively communicates the intuitive meaning behind the silhouette coefficient, making it easier to interpret

how individual points contribute to overall cluster quality. The diagram serves as an explicit, practical reference for understanding silhouette analysis and its application in assessing the performance of clustering methods evaluated in the analytical workflow of this study.

The score for a single data point i, denoted s_i, is calculated as:

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}}$$

In this formula, a_i is the mean intra-cluster distance (the average distance between point i and all other points in the same cluster), and b_i is the mean nearest-cluster distance (the average distance between point ii and all points in the next closest cluster).

The average of s_i over all data points provides a single score for the entire clustering solution. The score ranges from -1 to 1, where a high value indicates that a data point matches its own cluster well and mismatches neighboring clusters. A score near 0 suggests overlapping clusters, and a negative score indicates potential misclassification (Rousseeuw, 1987).

The **Davies–Bouldin Index (DBI)** provides an average-case measure of cluster quality by calculating the average similarity between each cluster and its most similar neighbor. The index is computed by considering both cluster cohesion (average within-cluster dispersion) and separation (distance between cluster centroids), as illustrated in Figure 8. Lower DBI values indicate better clustering performance, as they reflect compact, well-separated clusters. Unlike the Silhouette Score, which evaluates clustering

quality at the individual data point level, the DBI summarizes the overall clustering structure.

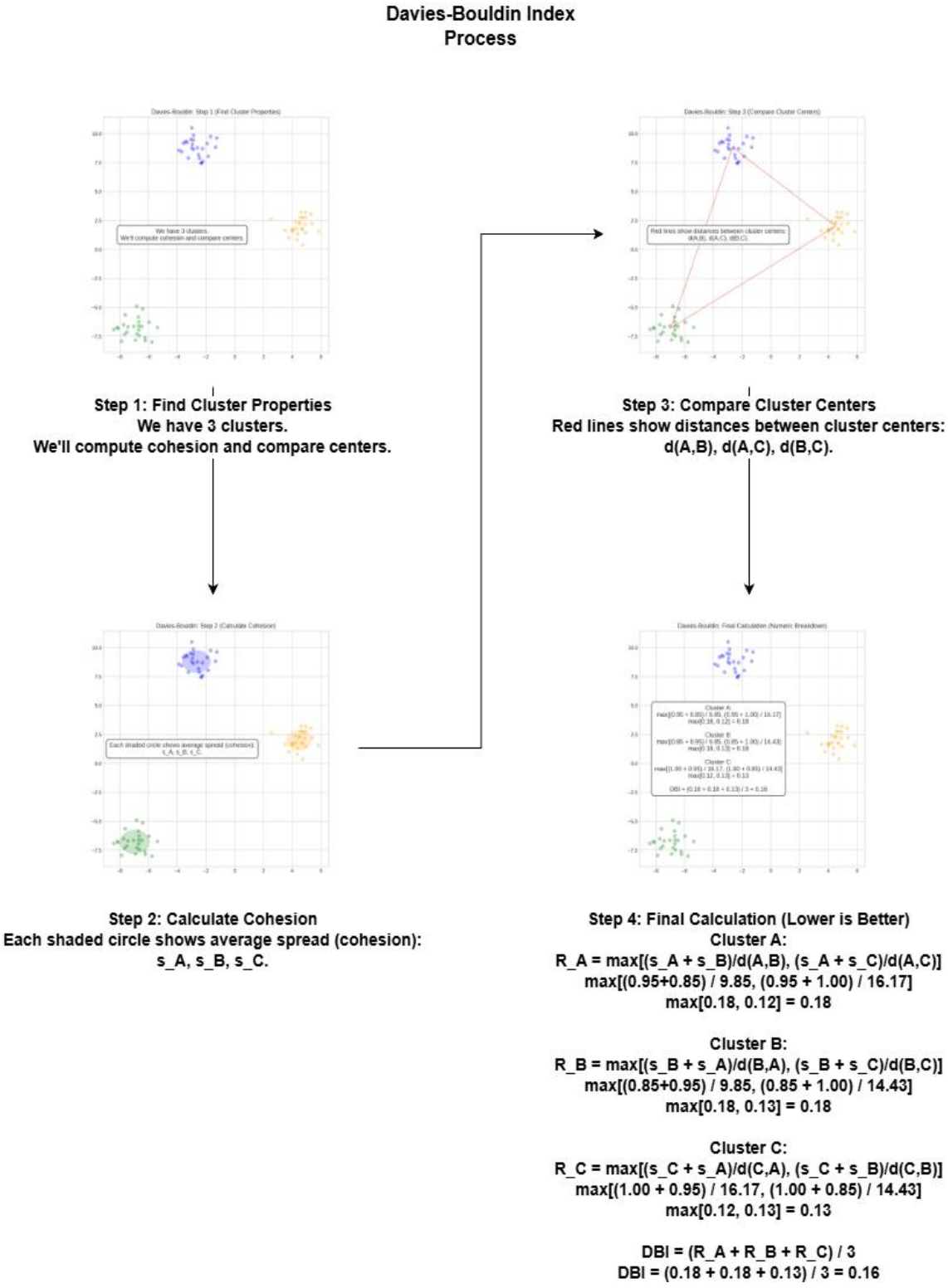


Figure 8. Diagram of the Davies-Bouldin Index process based on Davies and Bouldin (1979).

Figure 8 presents a multi-panel diagram illustrating the step-by-step calculation process for the Davies-Bouldin Index (DBI), as initially proposed by Davies and Bouldin (1979). The Davies-Bouldin Index is an internal clustering validation metric that evaluates the overall quality of a clustering solution by measuring the average ratio of the intra- and inter-cluster similarities; lower index values indicate better clustering performance, characterized by compact, well-separated clusters. The diagram uses a series of scatter plots to demonstrate the computation for an example dataset with three clusters (labeled A, B, and C), with points color-coded accordingly and cluster centers clearly marked.

The process begins in Step 1 with the identification of cluster properties, where the algorithm prepares to compute intra-cluster cohesion and inter-cluster separation. Step 2 calculates the cohesion (intra-cluster spread) for each cluster, represented by shaded circles that indicate the average dispersion of points around their respective centroids (denoted as s_A, s_B, and s_C). Step 3 measures separation by computing the distances between cluster centroids, as illustrated by the red lines showing d(A, B), d(A, C), and d(B, C).

In Step 4, the final step, the similarity ratio R_i for each cluster i is calculated as the maximum value of (s_i + s_j)/d(i, j) over all other clusters j, capturing the worst-case similarity to any neighboring cluster. The algorithm then obtains the overall Davies-Bouldin Index as the average of these R_i values across all clusters. In the illustrated example, the pairwise ratios yield R_A = 0.18, R_B = 0.18, and R_C = 0.13, resulting in a final DBI of 0.16.

This sequential visual explanation effectively conveys how the Davies-Bouldin Index balances intra-cluster compactness against inter-cluster separation to provide a single, interpretable score for comparing clustering solutions. The diagram serves as an explicit, practical reference for understanding this key validation metric and its application in assessing the performance of the clustering methods evaluated in the analytical workflow of this study.

We define similarity as the ratio of the sum of within-cluster scatter to the distance between cluster centroids. For a clustering of clusters, the algorithm calculates the Davies–Bouldin Index (DBI) as:

$$DBI = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} \left(\frac{s_i + s_j}{d(c_i, c_j)} \right)$$

Here, s_i is the average distance of all points in cluster i to their centroid, and $d(c_i, c_j)$ is the Euclidean distance between the centroids of clusters i and j. Lower DBI values indicate better clustering, as they reflect dense clusters that are well-separated from their nearest neighbors (Davies & Bouldin, 1979).

The **Dunn Index** provides a stricter, worst-case measure of cluster quality, complementing metrics such as the Silhouette Score and the Davies–Bouldin Index. It evaluates clustering performance by identifying the minimum inter-cluster distance relative to the maximum intra-cluster diameter, thereby emphasizing cluster separation while penalizing internally dispersed clusters. Higher Dunn Index values indicate better clustering, with compact, well-separated clusters, whereas lower values suggest overlapping or poorly defined groupings.

As illustrated in Figure 9, the index is computed by first determining the most significant distance between any two points within the same cluster (intra-cluster diameter) and then finding the smallest distance between points from different clusters (inter-cluster distance).

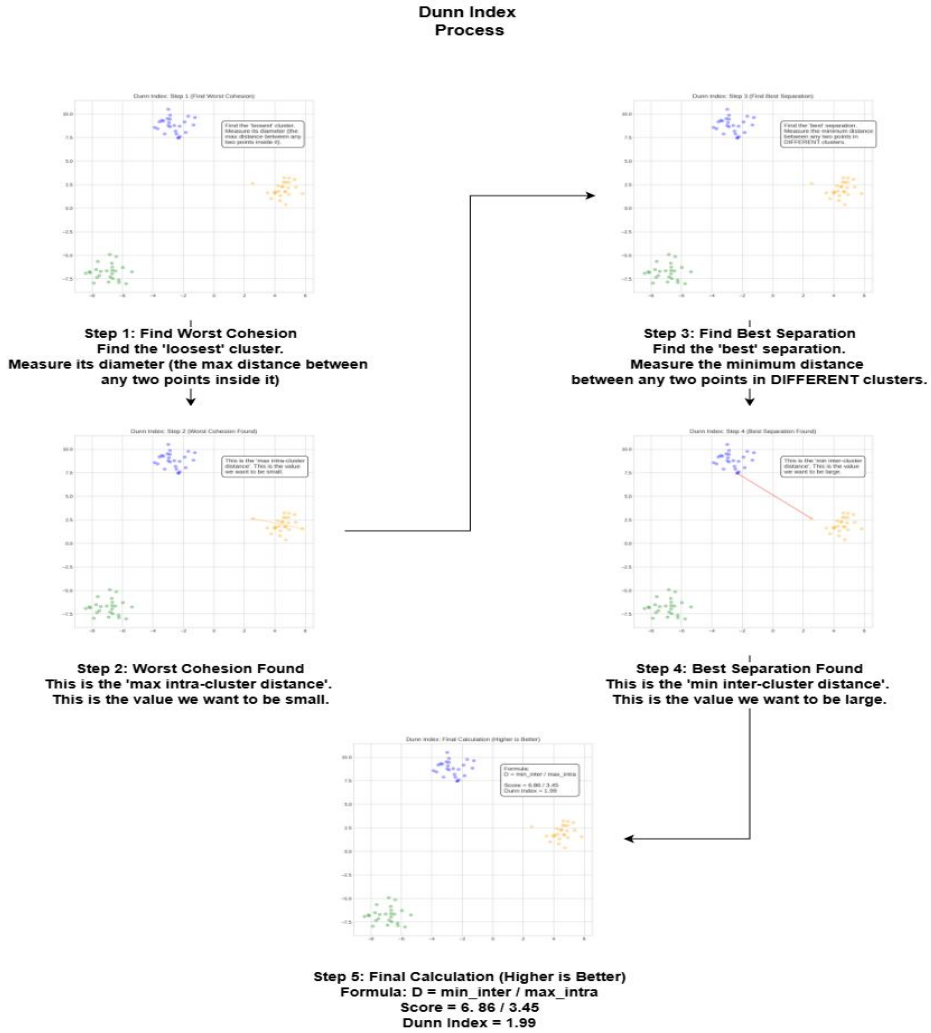


Figure 9. Diagram of the Dunn Index process based on Dunn (1974).

Figure 9 presents a multi-panel diagram illustrating the step-by-step calculation process for the Dunn Index, as initially proposed by Joseph C. Dunn (1974). The Dunn Index is an internal clustering validation metric that evaluates the quality of a clustering solution by comparing the separation between clusters to the compactness within clusters; higher values indicate

better-defined clusters with large inter-cluster distances and small intra-cluster spreads. The diagram uses a series of scatter plots with three color-coded clusters (blue, orange, and green) to demonstrate the computation across five sequential steps. The process begins in Step 1 by identifying the worst-case cohesion — locating the cluster with the largest diameter, defined as the maximum distance between any two points within the same cluster (the "loosest" or most spread-out cluster). Step 2 confirms this worst intra-cluster distance (max_intra), which the index seeks to minimize. Step 3 then identifies the best separation by finding the minimum distance between any two points belonging to different clusters (the smallest inter-cluster distance, min_inter), which the index aims to maximize. Step 4 confirms this best separation value. Finally, in Step 5, the Dunn Index is calculated as the ratio D = min_inter / max_intra. In the example shown, the minimum inter-cluster distance is 6.86 and the maximum intra-cluster diameter is 3.45, resulting in a Dunn Index of 1.99 (higher values being better). This sequential visual explanation effectively highlights how the Dunn Index rewards clustering solutions that produce compact, tightly grouped clusters that are well separated from one another, providing a single interpretable score for assessing cluster validity. The diagram serves as an explicit, practical reference for understanding this important validation metric and its application in evaluating the performance of the clustering methods used in the analytical workflow of this study.

It is defined as:

$$D = \frac{\min_{1 \le i < j \le k} d(C_i, C_j)}{\max_{1 \le l \le k} \delta(C_l)}$$

In this expression, the numerator $min_{1\leq i<j\leq k}d(C_i, C_j)$ is the minimum distance between any two clusters C_i and C_j , and the denominator $max_{1\leq l\leq k}\delta(C_l)$ is the maximum intra-cluster distance (diameter) within any cluster C_l. A higher Dunn Index value is more favorable, as it indicates that even the closest clusters are well-separated and the most dispersed cluster is still tightly grouped. Although sensitive to outliers, this metric provides a strong guarantee of cluster quality when the score is high (Dunn, 1974).

This study selected the model that demonstrated the most consistently favorable performance across the three metrics as the final, optimal model. This selection ensured that the solution achieved mathematical optimality on one dimension and robust superiority across multiple criteria of cluster quality. By doing so, the resulting barangay risk profiles are both statistically reliable and practically meaningful for guiding dengue prevention and control efforts.

Ethical Consideration

This study was conducted with the highest regard for ethical principles, focusing on patient privacy, responsible use of data, and public health benefit. Patient confidentiality was the priority. The dataset provided by the Cabuyao City Health Offices contained sensitive personal information. To protect privacy, the research team immediately and permanently removed all details that could identify individuals using Google Spreadsheet before conducting any analysis. Specifically, the team deleted eight columns containing names, addresses, birth dates, and unique codes. The dataset imported into Google

Colaboratory was already fully de-identified. After this step, only the non-identifiable information needed to meet the research objectives was retained, following the principle of keeping data use to the minimum required.

The primary ethical justification for the study is its potential public health benefit. By identifying high-risk areas and their unique epidemiological patterns, the research provides an evidence-based tool for the City Health Office to allocate resources more effectively, design targeted interventions, and strengthen dengue prevention and control.

This study also considered equity and non-stigmatization. While the study's outputs label certain areas as "Very High Risk," this is not intended to stigmatize communities. Instead, it guides the fair and efficient distribution of public health resources to where they are most needed.

Finally, the researchers designed the study to be open and repeatable. They sourced all supplementary data from public, open-access repositories. The entire analysis process, from the use of the anonymized dataset to the final clustering and visualization, was carried out using free, publicly available tools (Python in Google Colaboratory). This study ensures methodological transparency and enables the reproduction of results, upholding high standards of scientific integrity.

Additionally, the study prioritized clear, practical outputs, presenting findings and visualizations in a way that decision-makers can readily interpret and use to support effective dengue prevention and control, ensuring that the research translates into actionable public health strategies.

CHAPTER IV

RESULTS AND DISCUSSION

This chapter presents the study's results, beginning with an overview of the dengue situation in Cabuyao City based on the raw data. It then reports the findings from the exploratory spatio-temporal analysis, emphasizing key factors relevant to dengue risk levels. The chapter further describes the features used in the clustering analysis. It summarizes the four distinct risk profiles identified and the process for selecting the most appropriate model. These findings are supported by quantitative metrics and visual maps, providing a robust foundation for targeted public health actions.

Analysis of Spatio-Temporal Patterns of Dengue Transmission

This section presents the foundational analysis of spatio-temporal dengue patterns in Cabuyao City. It begins with a long-term historical overview to provide a broader context for dengue as a public health issue in the city. The analysis then presents a detailed descriptive epidemiology of the 2021–2022 study period. Finally, the section examines the specific temporal and spatial patterns observed during the study period. Together, these analyses highlight the complex and heterogeneous nature of dengue transmission, providing a clear rationale for the application of the clustering methodology.

These findings also provide a clear baseline for understanding cluster results and for planning more focused and effective public health actions, helping guide interventions where they are needed most.

Historical Trends of Dengue in Cabuyao City

To provide context for the focused analysis of 2021–2022, the researchers first examined the total number of dengue cases reported annually in Cabuyao City from 2014 to 2024.

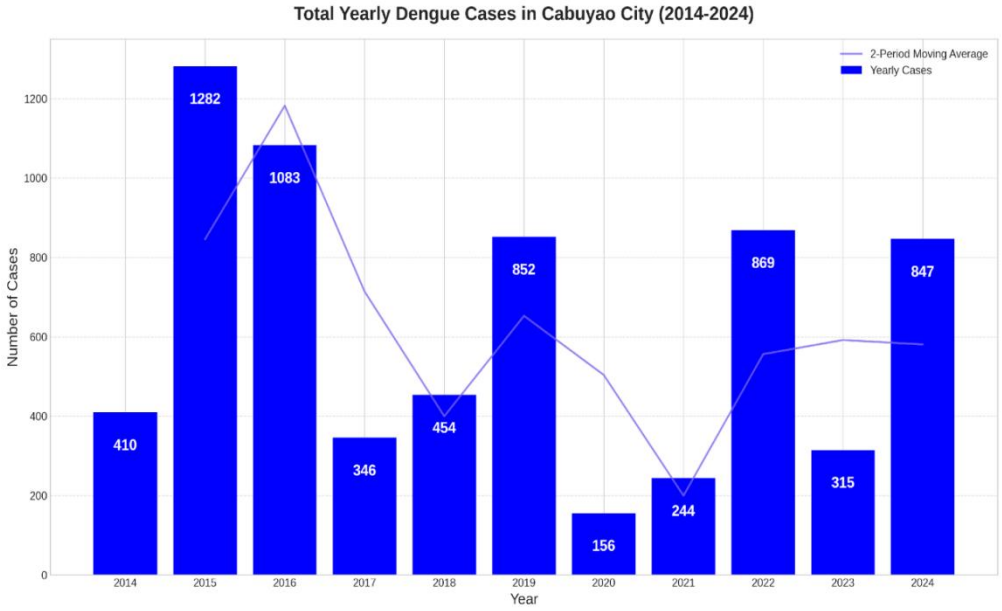


Figure 10: Total Yearly Dengue Cases in Cabuyao City (2014–2024).

Figure 10 shows a consistent cyclical pattern of dengue outbreaks in Cabuyao City, with pronounced peaks in 2015 (1,282 cases) and 2019 (852 cases), highlighting the persistent public health threat of the disease. An anomaly occurred in 2020, with only 156 cases, likely due to reduced exposure, changes in healthcare-seeking behavior, and altered mobility during COVID-19 restrictions. The 2021–2022 period captures a resurgence, with a transition from a low-transmission year (244 cases) to a high-transmission year (869 cases). This long-term trend underscores the endemic nature of dengue in the city. It provides a clear rationale for this study, focusing on identifying the factors that define high-risk barangays to support more effective public health interventions.

Descriptive Epidemiology of Dengue Cases (2021–2022)

To establish a baseline for analysis, this section presents a detailed statistical summary of the 1,113 dengue cases recorded during the 2021 to 2022 study period in Cabuyao City.

Table 3. Comprehensive Descriptive Profile of Dengue Cases (Adapted from Onuh and Cabanacan-Salibay, 2021).

Characteristic	Metric	Value
Age Distribution	0-5 yrs	184 (16.5%)
	5 > - 19 yrs	607 (54.5%)
	> 19 yrs	322 (28.9%)
Sex Distribution	Male	611 (54.9%)
	Female	502 (45.1%)
Outcome	Survived	1109
	Died	4
	Case Fatality Rate	0.36%
Clinical Severity	Dengue Fever (DF)	990 (90.1%)
	Dengue Hemorrhagic Fever (DHF)	103 (9.4%)
	Dengue Shock Syndrome (DSS)	6 (0.5%)
	Total Severe Cases (DHF+DSS)	109 (9.9%)
Case Classification	Confirmed	4 (0.4%)
	Probable	706 (64.2%)
	Suspected	402 (36.6%)

As presented in Table 3, the descriptive statistics offer a comprehensive overview of the patient population. The age distribution indicates that dengue disproportionately affects the youth; the 5 to 19-year-old age group accounts for the majority of cases (607 cases, 54.5%), suggesting that school-aged children and adolescents are the most vulnerable segment.

The data also shows a slight predominance of male cases (54.9%) over female cases. From a clinical perspective, the overall case fatality rate was low at 0.36% (4 fatalities). Notably, this study classified 9.9% of the 1,099 cases with complete clinical data as severe dengue (Dengue Hemorrhagic Fever or Dengue Shock Syndrome). This finding indicates that approximately 1 in 10 patients experienced a more serious clinical outcome, underscoring the substantial burden on healthcare resources.

Table 4. Age Distribution by Clinical Severity (Adapted from Onuh and Cabanacan-Salibay, 2021).

Severity **Age Group**	**Non-Severe (DF)**	**Severe(DHF/DSS)**
0-5 yrs	170 (93.4%)	12 (6.6%)
5>-19 yrs	538 (89.7%)	62 (10.3%)
>10 yrs	282 (89.0%)	35 (11.0%)

Table 4 presents a deeper analysis stratified by age group, revealing a critical pattern with significant public health implications. Although children and adolescents aged 5–19 years had the highest total number of cases, the likelihood of developing severe dengue increased with age. Adults (>19 years) had the highest proportion of severe cases (11.0%), followed by those aged 5–19 years (10.3%) and children under 5 years (6.6%).

This important finding suggests that while the youth population drives the overall volume of cases, adult patients may be at a higher relative risk for developing severe dengue, a critical consideration for clinical triage and patient monitoring in healthcare facilities.

Temporal Trends and Barangay-Level Spatial Distribution

An analysis of the monthly case distribution for the entire city, presented in Figure 11, confirms a strong seasonal pattern.

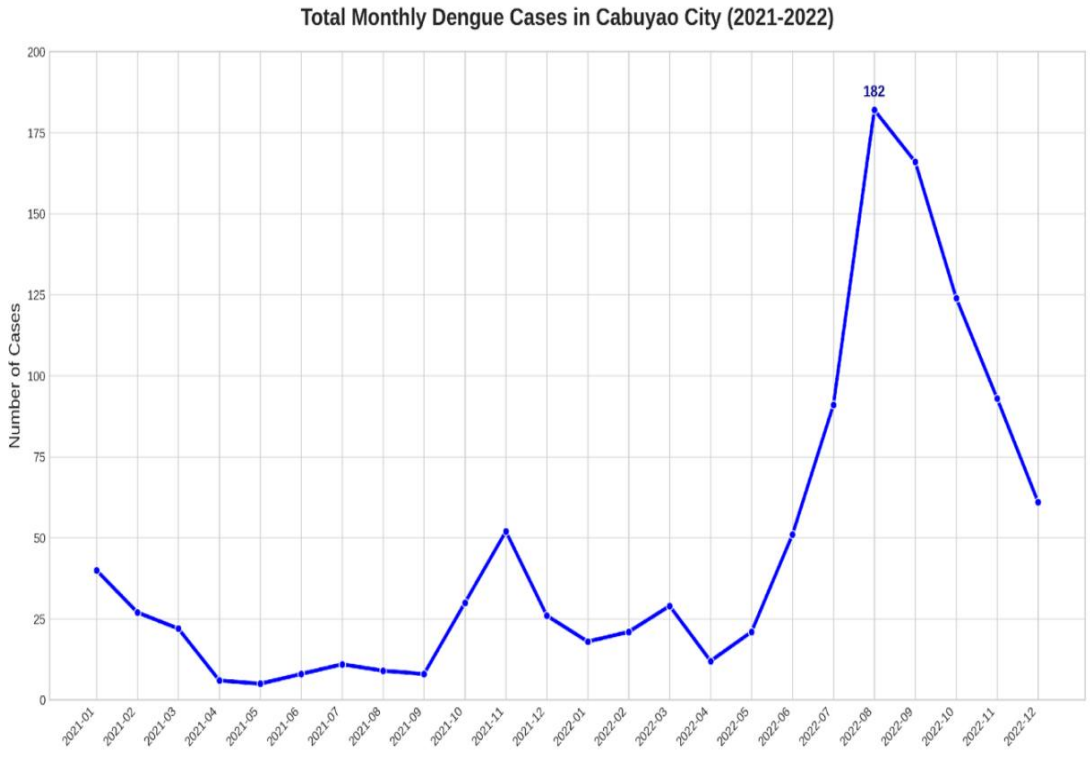


Figure 11. Total Monthly Dengue Cases in Cabuyao City (2021–2022).

Figure 11 shows that cases peaked in August 2022 at 182, consistent with the expected increase during the rainy season. This city-level view, however, aggregates all barangay data and can mask significant local variations. To deconstruct this trend, the monthly dengue cases for each barangay were plotted simultaneously in Figure 12. This analysis helps the team identify barangays that need closer monitoring. Examining these barangay-level trends also allows for the detection of early warning signals in specific areas, informing targeted interventions and resource allocation to prevent larger outbreaks.

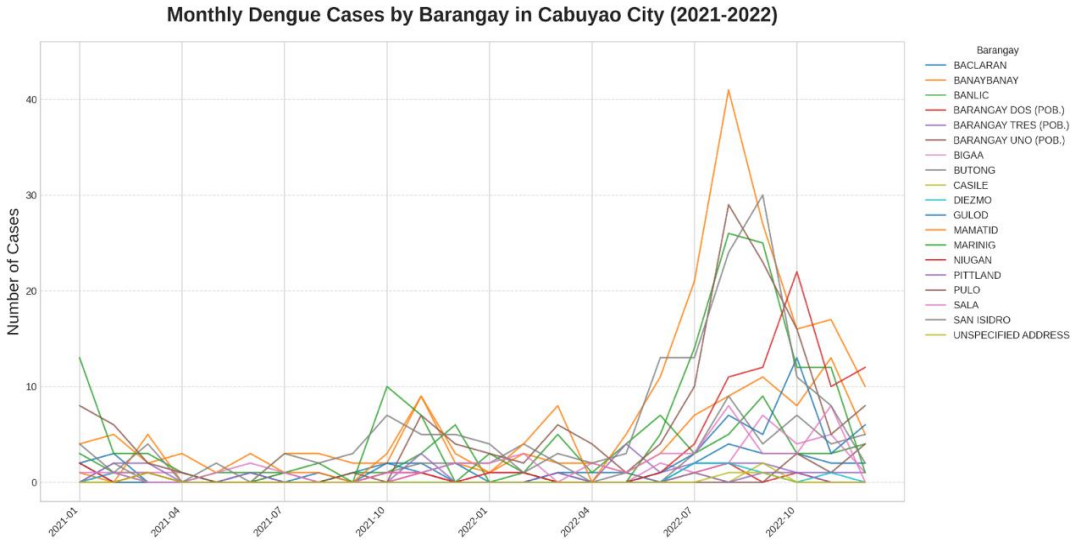


Figure 12. Monthly Dengue Cases in Cabuyao (2021–2022).

Figure 12 illustrates the significant heterogeneity in seasonal dengue patterns across Cabuyao City. The plot shows that a few barangays with pronounced peaks, such as Mamatid and San Isidro, are the main drivers of the overall city-wide trend, while many other barangays display different patterns. Some exhibit moderate peaks but at different times, whereas others show consistently low and sporadic transmission with no apparent seasonality. This visual evidence of heterogeneous temporal patterns provides a strong rationale for the study's core methodology. It shows that a one-size-fits-all approach is insufficient, underscoring the need for a formal clustering analysis to group barangays with similar epidemiological profiles systematically.

Thematic Maps of Dengue Burden and Incidence Rate

To better visualize the spatial distribution of dengue cases, this section presents thematic maps that highlight both the total number of reported cases (burden) and the per capita relative risk (incidence rate). Table 5 summarizes the data used for these maps.

Table 5. Total Dengue Cases and Incidence Rates by Barangay.

Barangay	Total Cases (2021-2022)	Incidence Rate (per 10k)
BACLARAN	30	20.54
BANAYBANAY	92	26.85
BANLIC	64	31.00
BARANGAY DOS	31.00	31.79
BARANGAY TRES	14	32.76
BARANGAY UNO	9	24.39
BIGAA	29	21.22
BUTONG	48	32.51
CASILE	3	8.29
DIEZMO	7	10.57
GULOD	47	27.30
MAMATID	196	32.09
MARINIG	139	30.66
NIUGAN	79	20.48
PITTLAND	12	19.83
PULO	140	39.87
SALA	46	42.19
SAN ISIDRO	150	64.31

The burden of dengue, defined as the total number of cases per barangay, is shown in Figure 13.

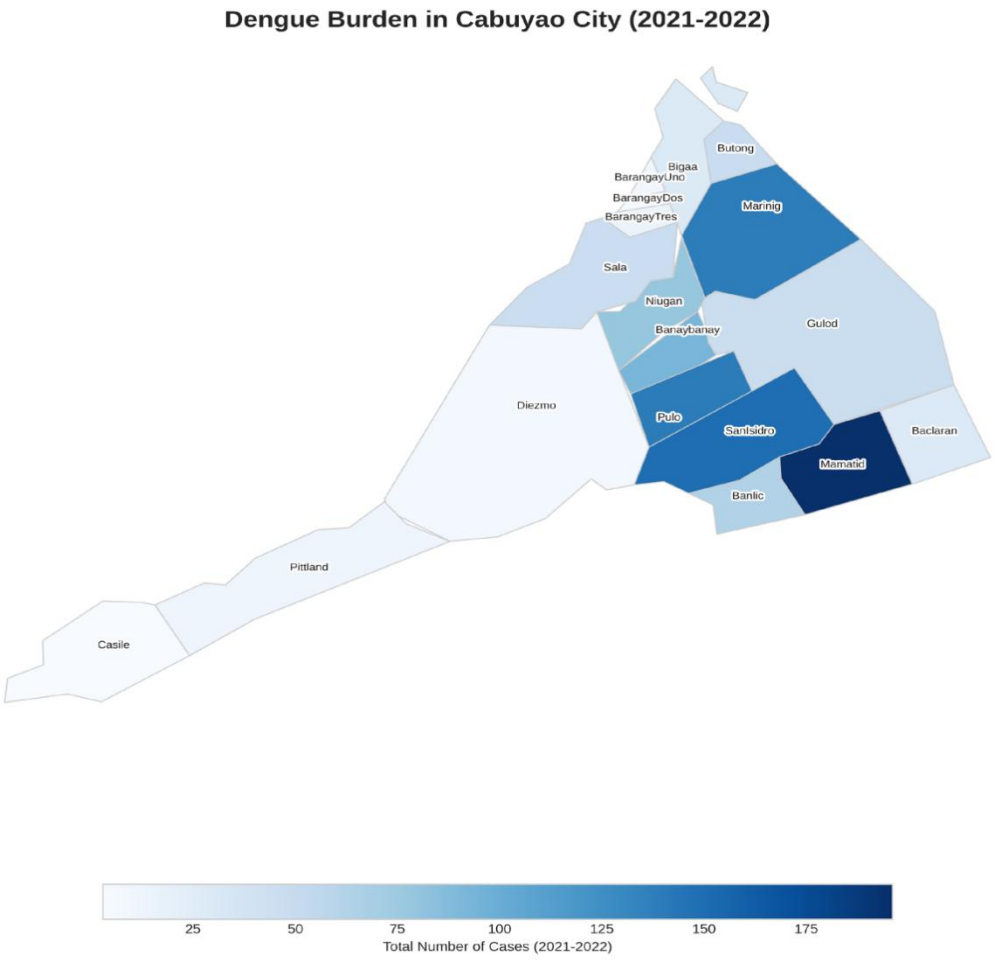


Figure 13. Dengue Case Burden per Barangay (2021–2022).

The map shows that the highest case counts occurred in Mamatid (196 cases), San Isidro (150 cases), Pulo (140 cases), and Marinig (139 cases), indicating concentrated transmission in these more urbanized and populous areas. To offer a more risk-adjusted perspective, the incidence rate per 10,000 population is presented in Figure 14, allowing for a more precise comparison of dengue risk across all barangays, including those with lower absolute case counts but higher relative risk due to smaller

populations.

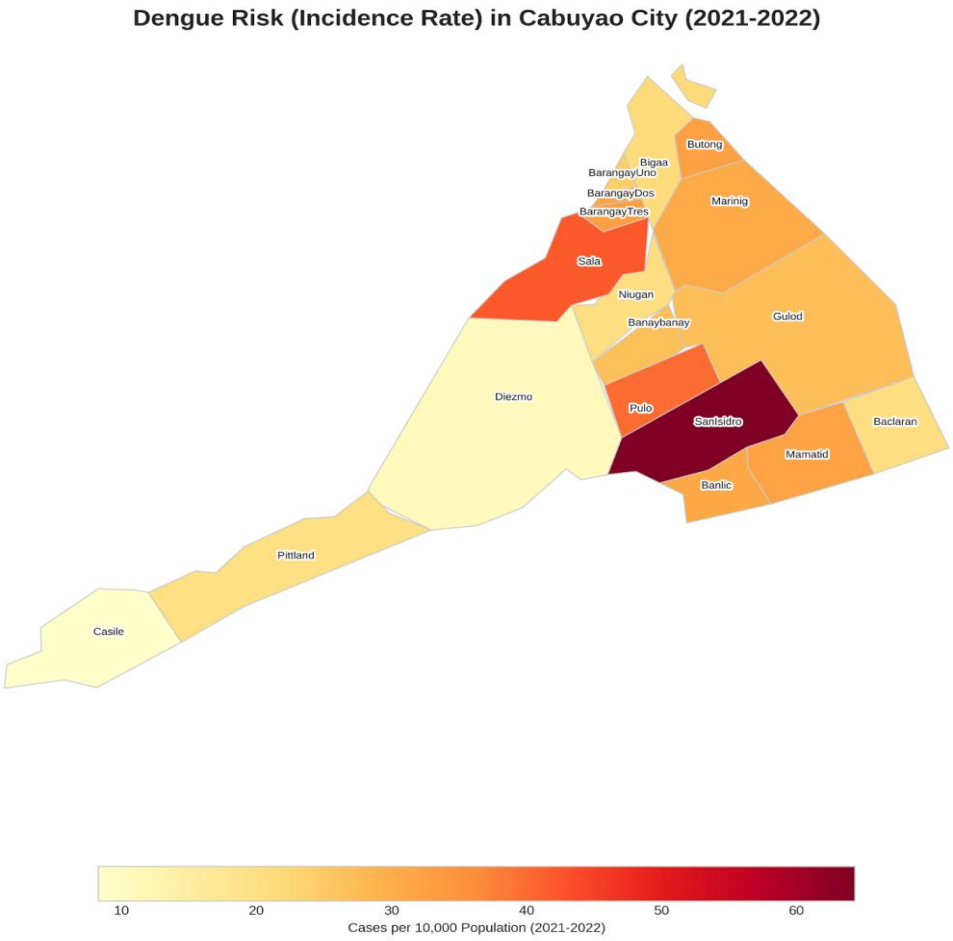


Figure 14. Dengue Incidence Rate per Barangay (2021–2022).

In this analysis, San Isidro (64.31) and Sala (42.19) emerge as the barangays with the highest incidence rates, indicating a greater relative risk to residents, even after adjusting for population size. This distinction highlights that the areas with the highest absolute burden are not necessarily those with the highest individual risk, a key observation that informed the subsequent clustering analysis, which incorporates both risk metrics.

Feature Engineering for Model Development

This section presents the results of the feature engineering stage. Before applying clustering algorithms to group barangays with similar dengue

patterns, this study transformed the raw 24-month time-series data into a clear, meaningful, and quantitative format.

To achieve this, it employed a feature-based approach. This process is essential for representing each barangay's unique and complex dengue profile as a set of static numerical features, enabling clustering algorithms to compare and assess similarity between barangays effectively.

Engineered Features

Four key features were systematically derived from the 2021–2022 dataset to characterize each barangay's dengue situation across public health-relevant dimensions. This feature-based approach integrates domain-specific epidemiological knowledge, ensuring that the resulting clusters are both mathematically robust and interpretable in a public health context (Aghabozorgi et al., 2015). The four engineered features were:

Total Dengue Cases (Burden): This represents the total number of reported cases per barangay from 2021 to 2022. It provides a direct measure of the public health burden and the potential strain on resources within each barangay. For instance, Mamatid recorded the highest burden of 196 cases, highlighting its disproportionate contribution to the city's overall caseload and serving as a foundational metric for identifying high-burden areas.

Incidence Rate (Risk): Calculated as the number of dengue cases per 10,000 population based on the 2020 PSA census, this feature enables a per-capita comparison of risk, independent of population size. Incidence Rate (Risk): This study calculates the number of dengue cases per 10,000 population using the 2020 PSA census.

This feature enables a per-capital comparison of risk, independent of population size, and follows standard practice in spatial epidemiology (The Open University, n.d.). The analysis shows that San Isidro recorded the highest incidence rate (64.31 per 10,000), despite having the lowest absolute burden, underscoring the importance of this normalization.

Rainy Season Proportion (Seasonality): This feature is defined as the percentage of cases that occur during the rainy season (June–November), reflecting the climate-driven nature of dengue transmission in the Philippines (Sumi et al., 2016). It quantifies the degree of seasonality in each barangay. The results showed substantial variation: Barangays such as Dos and Casile exhibited extreme seasonality (1.00), with all cases occurring during the rainy months, whereas Barangay Uno had a lower proportion (0.44), indicating more sporadic, year-round transmission.

Average Peak Month (Timing): Calculated as a weighted mean of monthly case counts, this feature identifies the "center of gravity" of dengue activity. It distinguishes barangays with early versus late outbreak peaks, a method commonly employed in epidemiological studies (Paireau et al., 2016). For example, a value of 7.44 (mid-July) indicates an early peak, whereas 10.11 (early October) reflects a later peak in dengue activity. This feature allows for comparison of temporal patterns across different barangays.

Standardization and Final Dataset

All four engineered features were standardized using scikit-learn's StandardScaler, which scales data to a mean of 0 and standard deviation of 1. This ensures fair comparisons across barangays and prevents features with

larger ranges, like Total Cases, from dominating the clustering. Table 6 summarizes all features used for accurate and meaningful clustering.

Table 6. Final Feature Table for Clustering (Sorted by Barangay).

Barangay	Feat_Total_Cases	Feat_Incidence_Rate	Feat_Rainy_Proportion	Feat_Avg_Peak_Month
BACLARAN	30	20.54	0.67	7.53
BANAYBANAY	92	26.85	0.76	8.36
BANLIC	64	31.00	0.58	7.59
BARANGAY DOS (POB.)	5	31.79	1.00	8.80
BARANGAY TRES (POB.)	14	32.76	0.64	7.64
BARANGAY UNO (POB.)	9	24.39	0.44	10.11
BIGAA	29	21.22	0.79	8.10
BUTONG	48	32.51	0.62	8.00
CASILE	3	8.29	1.00	8.33
DIEZMO	7	10.57	0.86	7.43
GULOD	47	27.30	0.77	9.17
MAMATID	196	32.09	0.77	7.88
MARINIG	139	30.66	0.82	7.64
NIUGAN	79	20.48	0.78	9.28
PITTLAND	12	19.83	0.58	6.50
PULO	140	39.87	0.68	7.53
SALA	46	42.19	0.72	7.41
SAN ISIDRO	150	64.31	0.79	7.79

Table 6 presents the final, structured datasets that serve as the input for the machine learning models. Each of the 18 barangays now has a four-

dimensional vector that quantitatively describes its unique dengue profile. This table provides the foundation for clustering algorithms to identify and group barangays that share these complex, multidimensional characteristics, enabling a more nuanced, evidence-based approach to public health planning in Cabuyao City.

Evaluation and Selection of Clustering Algorithm

This section presents the results of the multi-stage evaluation process conducted to identify the most effective clustering algorithm. To determine how well different algorithms could capture meaningful patterns within the engineered feature set, the researchers employed a systematic, multi-stage approach. The process began with an internal validation of K-Means and Fuzzy C-Means to determine the optimal number of clusters, followed by a hyper-parameter optimization phase. It concluded with a final head-to-head comparison of the three selected algorithms.

Performance metrics such as silhouette scores and cluster cohesion were used to assess the quality of each algorithm’s output. The results were also visualized to facilitate the interpretation and comparison of cluster patterns across barangays. This comprehensive evaluation ensured that the selected algorithm not only fit the data well but also provided actionable insights for dengue risk assessment.

Internal Validation of K-Means and Fuzzy C-Means (k = 2 to 10)

To determine the optimal number of clusters (k) for algorithms that require it, this study tested K-Means and Fuzzy C-Means over a range of k values from 2 to 10. It evaluated the quality of the resulting clusters using

three internal validation metrics: the Silhouette Score (higher values indicate better clustering), the Davies-Bouldin Index (DBI) (lower values indicate better clustering), and the Dunn Index (higher is better).

The combination of these metrics provided a balanced assessment of cluster compactness, separation, and overall structure. By analyzing trends across different k values, the researchers identified the point at which adding more clusters no longer significantly improved clustering quality.

Table 7. Internal Validation Scores for K-Means and Fuzzy C-Means (k = 2 to 10).

K-Means Baseline Evaluation Scores

K	Silhouette	Davies-Bouldin Index	Dunn Index
2	0.202	1.338	0.136
3	0.282	1.145	0.247
4	0.282	1.019	0.378
5	0.297	0.796	0.378
6	0.238	0.779	0.378
7	0.288	0.574	0.457
8	0.225	0.603	0.438
9	0.213	0.637	0.445
10	0.144	0.593	0.445

Fuzzy C-Means Baseline Evaluation Score

K	Silhouette	Davies-Bouldin Index	Dunn Index
2	0.228	1.455	0.229
3	0.282	1.145	0.247
4	0.303	0.981	0.321
5	0.150	1.007	0.378
6	0.151	0.978	0.362
7	0.088	0.891	0.362
8	0.163	0.878	0.362
9	0.057	0.906	0.254
10	0.045	0.777	0.254

Looking at the results, we can see that K-Means performed best when using around 5 to 7 clusters. The highest Silhouette Score was at k = 5, suggesting that these clusters were relatively well-defined. Meanwhile, the lowest DBI and highest Dunn Index were at k = 7, indicating that clusters were both compact and well-separated. This finding suggests that the optimal number of clusters falls within that range, balancing cohesion and separation."

For Fuzzy C-Means, the patterns were slightly different. The best Silhouette Score and DBI occurred at k = 4, while the Dunn Index peaked at k = 5. Beyond k = 6 or 7, clustering quality decreased, with lower Silhouette

Scores and Dunn Index values, suggesting that too many clusters may have fragmented the data and made interpretation more difficult.

Overall, both algorithms show that moderate cluster numbers, roughly 4 to 7, produce the most meaningful and stable clusters. These insights guided the selection of the final model. They informed the number of clusters used to generate barangay-level dengue risk profiles, ensuring that the results were both statistically sound and interpretable for public health decision-making.

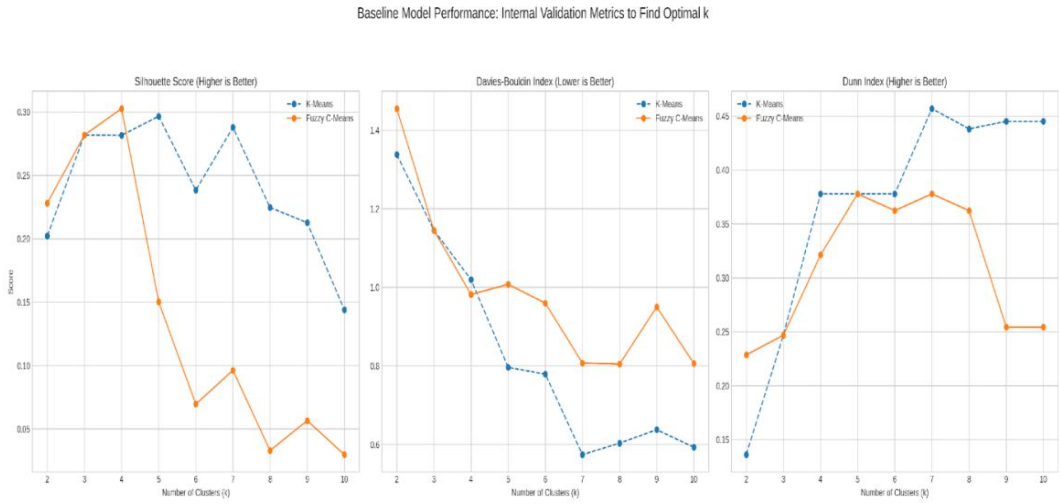


Figure 15. Internal Validation Scores for K-Means and Fuzzy C-Means.

As shown in Figure 15, the plot shows how each model performed with different numbers of clusters using the Silhouette Score, Davies–Bouldin Index, and Dunn Index. Both K-Means and Fuzzy C-Means worked best at k = 4, which seems to be the most meaningful number of clusters. Fuzzy C-Means had the highest Silhouette Score and lowest DBI, while K-Means did better on the Dunn Index, showing good separation between clusters. These results suggest that both algorithms are effective, but highlight different strengths in terms of cluster compactness and separation.

Performance declined for Fuzzy C-Means when k was higher than five, meaning more clusters were less valuable. Based on this, the researchers chose k = 4 as the best number of clusters for creating barangay-level dengue risk profiles.

Hyperparameter Optimization

This section presents the results of the hyperparameter optimization phase. To ensure a fair and robust comparison, this study evaluated each of the three clustering models using internal validation metrics to identify the most suitable configuration.

It also optimized each model's hyperparameters to improve performance: K-Means with different initialization methods, Fuzzy C-Means with different fuzziness coefficients, and Affinity Propagation with different damping factors.

Table 8. Internal Validation Scores from Hyperparameter Optimization (k = 4).

Model	Silhouette	Davies-Bouldin Index	Dunn Index
K-Means (init=k-means++)	0.282	1.019	0.378
K-Means (init=random)	0.301	0.822	0.400
Fuzzy C-Means (m=1.5)	0.303	0.981	0.321
Fuzzy C-Means (m=2.0)	0.303	0.981	0.321
Fuzzy C-Means (m=2.5)	0.272	1.018	0.253
Affinity Prop (damp=0.7, k=4)	0.301	0.822	0.400
Affinity Prop (damp=0.8, k=4)	0.301	0.822	0.400

Table 8 shows that for Fuzzy C-Means, fuzziness coefficients (m) of 1.5 and 2.0 produced the highest Silhouette Score (0.303), while performance degraded at higher fuzziness levels (m=2.5). For K-Means, the random initialization method yielded the best values for both the Davies-Bouldin Index (0.822) and the Dunn Index (0.400), suggesting it produced a more compact, well-separated clustering solution than the k-means++ default for this dataset. Affinity Propagation's performance was stable across the tested damping factors.

Final Model Selection

This section presents a direct comparison of the three clustering models using their previously determined best-performing configuration. The evaluation incorporates two complementary approaches: numerical scores from internal validation metrics and visual inspection of the cluster shapes. This combined analysis ensures that the selected model is both statistically robust and interpretable in a public health context.

Table 9 first shows the numerical performance of the optimized models (K-Means, Fuzzy C-Means, and Affinity Propagation).

Table 9. Final Model Performance Comparison.

Model	Silhouette	Davies-Bouldin Index	Dunn Index
K-Means (Optimized)	0.282	1.019	0.378
Fuzzy C-Means (Optimized)	0.303	0.981	0.321
Affinity Propagation (Optimized, k=4)	0.301	0.822	0.400

To make the comparison more straightforward to understand, the scores from Table 9 are also shown as bar charts in Figures 16, 17, and 18, with each model using a consistent color for clarity.

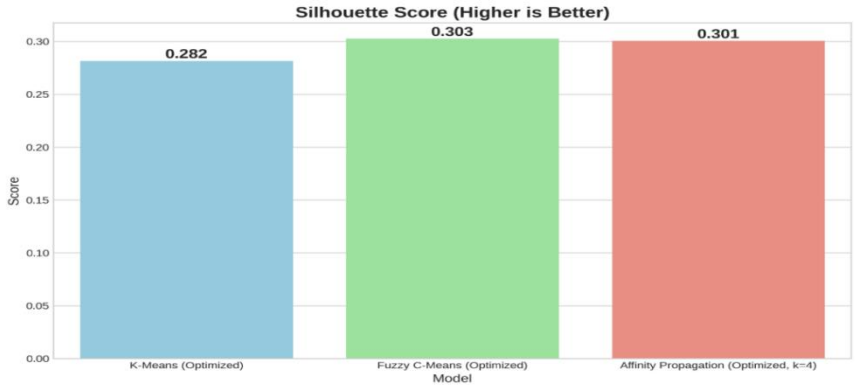


Figure 16. Silhouette Score Comparison. (Higher scores mean the clusters are more clearly defined.)

Figure 16 presents the Silhouette Score results, which measure how well each barangay fits within its assigned cluster compared to other clusters. Fuzzy C-Means obtained the highest score (0.303), indicating slightly better-defined clusters, followed closely by Affinity Propagation (0.301). K-Means recorded the lowest score (0.282), suggesting weaker cluster cohesion and separation.

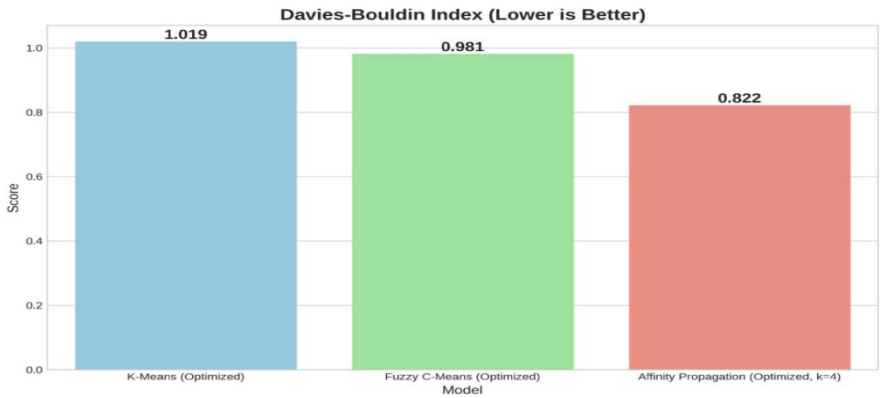


Figure 17. Davies-Bouldin Index Comparison. (Lower scores mean the clusters are tighter and more distinct from each other.)

Building on the Silhouette Score results, Figure 17 evaluates cluster compactness and separation using the Davies–Bouldin Index, where lower values indicate better clustering quality. Affinity Propagation achieved the lowest DBI value (0.822), indicating more compact and distinct clusters than Fuzzy C-Means (0.981) and K-Means (1.019). This result suggests improved internal cohesion among barangays when using Affinity Propagation.

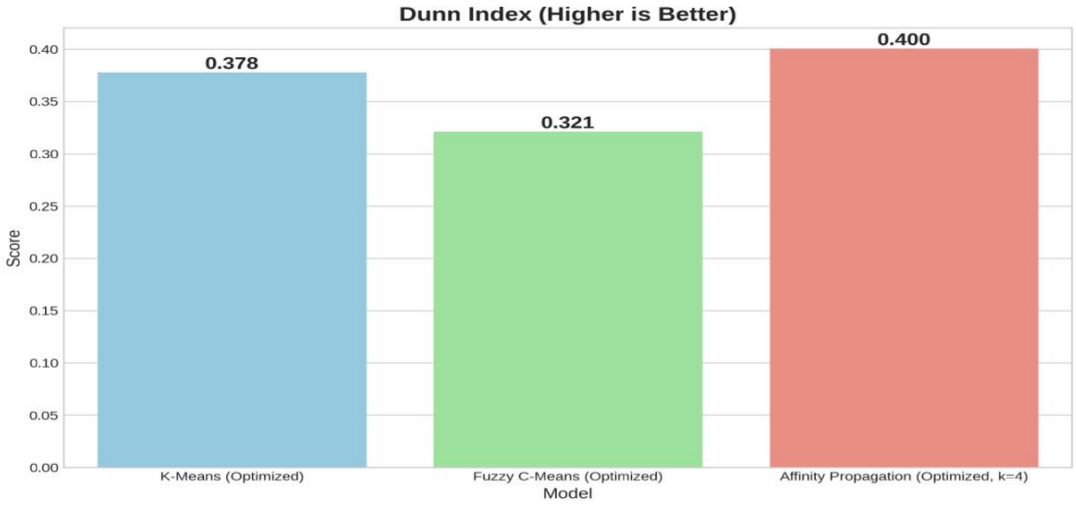


Figure 18. Dunn Index Comparison. (Higher scores mean even the least-separated clusters are still well-defined.)

Consistent with previous metrics, Figure 18 shows that Affinity Propagation achieved the highest Dunn Index score (0.400), indicating superior separation between clusters, even in the least-separated cases. K-Means followed with a score of 0.378, while Fuzzy C-Means recorded the lowest value (0.321), reflecting overlapping cluster boundaries. These findings further support the effectiveness of Affinity Propagation in producing well-defined barangay risk groupings.

To complement these findings visually, the four-dimensional data were reduced to two dimensions using Principal Component Analysis (PCA),

allowing for clear visualization of the final cluster shapes from each model in Figures 19, 20, and 21.

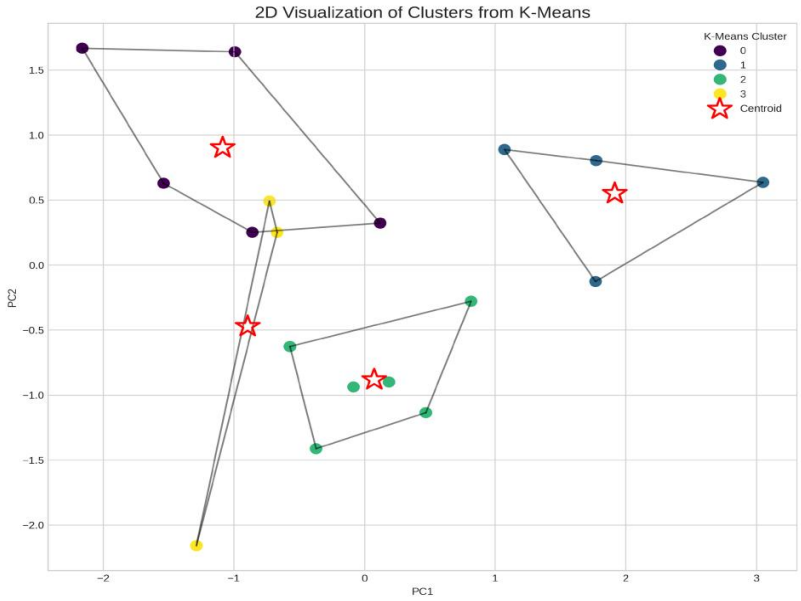


Figure 19. 2D Visualization of Clusters from K-Means.

Figure 19 shows that K-Means effectively captures the main structure of the data, producing compact and well-separated clusters in the PCA space. The clustering pattern indicates stable groupings, although some points appear near cluster boundaries.

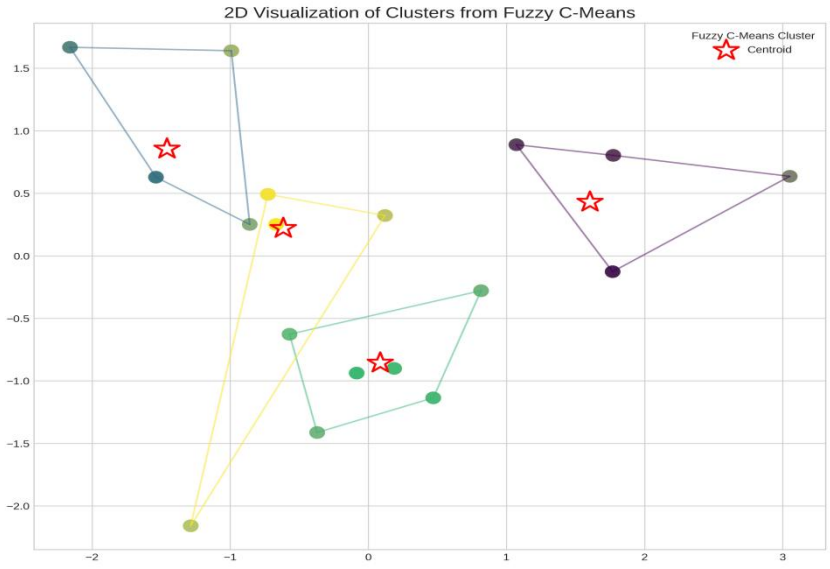


Figure 20. 2D Visualization of Clusters from Fuzzy C-Means.

Figure 20 demonstrates that Fuzzy C-Means produces clustering results very similar to K-Means, confirming that the overall data grouping is consistent. The slight spread of points reflects the algorithm's soft membership property.

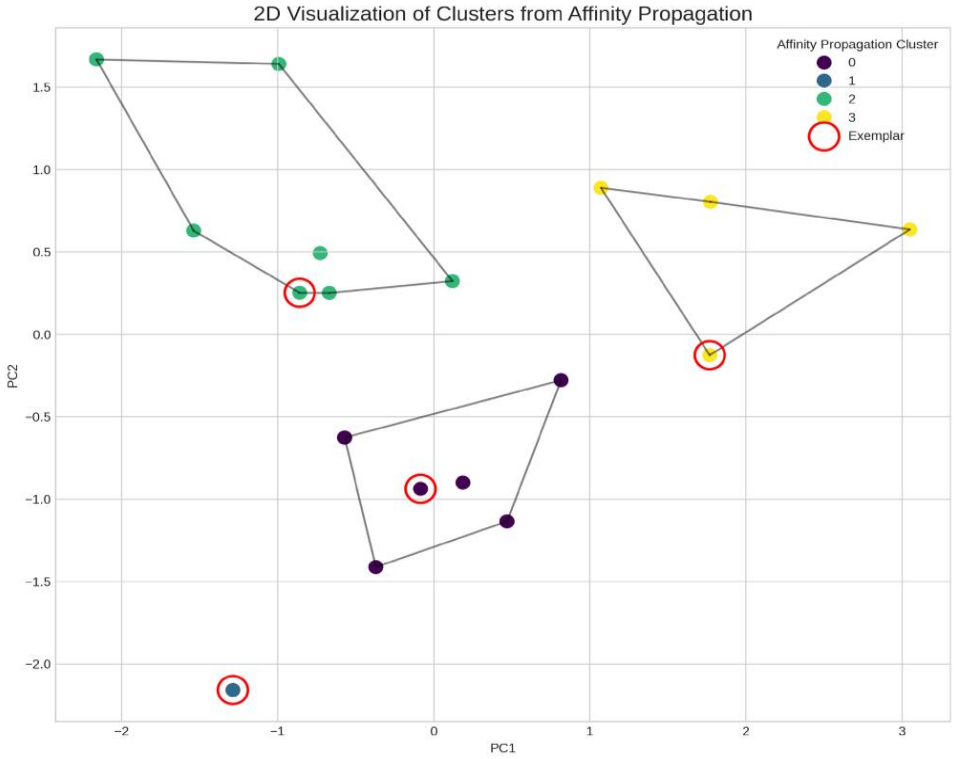


Figure 21. 2D Visualization of Clusters from Affinity Propagation.

Figure 21 highlights a key difference in the Affinity Propagation results. While the major clusters align with those identified by K-Means and Fuzzy C-Means, a small, distinct cluster is isolated. This more apparent separation explains the superior performance of Affinity Propagation on the Davies Bouldin and Dunn Index.

Overall, the researchers chose Affinity Propagation as the final model. It performed best on two out of three key metrics. It produced the most distinct and well-separated clusters, making it the most effective model for identifying different dengue risk profiles across the barangays.

Final Cluster Analysis and Risk Profile Characterization

This section presents the results of the final cluster analysis and risk profile characterization. The analysis follows a logical progression: it starts with the direct, uninterpreted output of the clustering algorithm, continues with the quantitative analysis required to understand its meaning, and culminates in the final, interpreted risk profiles and their visualizations.

Final Cluster Assignment and Visualization

The researchers used the Affinity Propagation algorithm to perform seasonal time-series clustering, allowing the study to group barangays based on their dengue patterns and define their risk profiles. To clearly show these patterns, they plotted the clustering results on a map of Cabuyao City. A qualitative colormap was applied to visually distinguish the clusters, emphasizing that each group represented a distinct, non-ordered group whose epidemiological significance was not yet known.

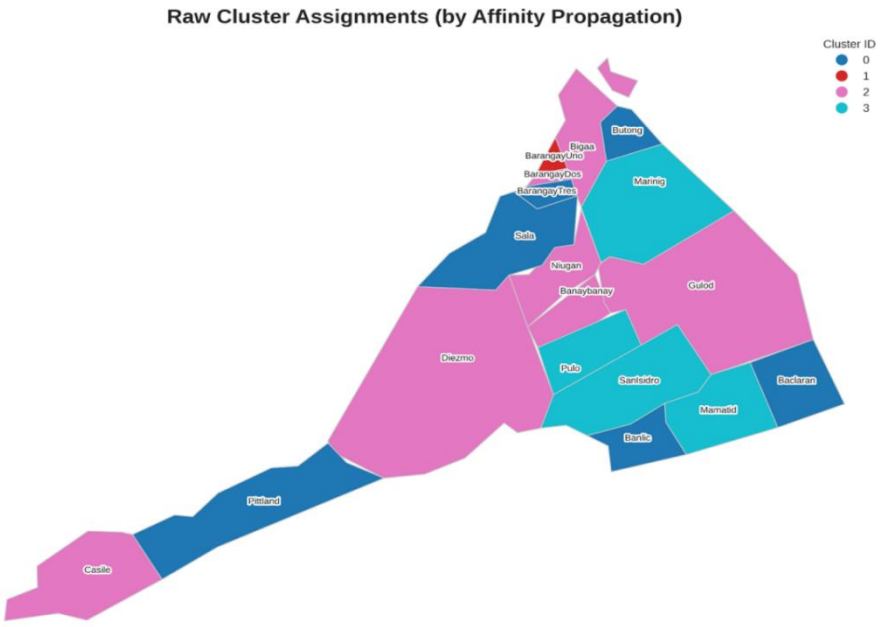


Figure 22. Raw Cluster Assignments (by Affinity Propagation)

Figure 22 illustrates the spatial distribution of the four distinct clusters identified by the algorithm. While the map effectively groups all 18 barangays, the cluster IDs are merely abstract labels that offer no direct information about their public health significance. To interpret these clusters and translate them into meaningful public health profiles, this study conducted further analysis to identify defining characteristics of each group.

Quantitative Cluster Profiling

To interpret the abstract clusters, the researchers generated average profiles based on barangay characteristics within each group. These profiles combined the four temporal dengue features with supplementary demographic and clinical indicators. The temporal features (total cases, incidence rate, rainy season proportion, and average peak month) served as the basis for clustering. At the same time, this study incorporated demographic and clinical indicators during the profiling stage to enhance interpretation. It did not use these additional indicators in the clustering algorithm, but they helped highlight differences in age distribution, sex, severity, and diagnostic confirmation across clusters.

Temporal Features:

- **Feat_Total_Cases** – Average number of dengue cases (burden).
- **Feat_Incidence_Rate** – Average cases per 10,000 people (risk).
- **Feat_Rainy_Proportion** – Share of cases during rainy months (seasonality).
- **Feat_Avg_Peak_Month** – Average month when cases peak (timing).

Demographic & Clinical Features:

- **Youth_Cases_%** – Proportion of cases under 20 years old
- **Female_Pct** – Percentage of female cases.
- **Severity_Index_%** – Share of severe cases (DHF/DSS).
- **Confirmation_Rate_%** – Percentage of lab-confirmed cases.

Table 10. Summary of Dengue Risk Profiles by Cluster.

Final_Cluster	Feat_Total_Cases	Feat_Incidence_Rate	Feat_Rainy_Proportion	Feat_Avg_Peak_Month	Youth_Cases_%	Female_Pct	Severity_Index_%	Confirmation_Rate_%
0	35.67	29.80	0.64	7.44	69.63	41.12	8.06	0.95
1	9.00	24.39	0.44	10.11	77.78	66.67	0.00	0.00
2	37.43	20.93	0.85	8.50	72.90	47.33	7.75	0.00
3	156.25	41.73	0.76	7.71	70.88	44.96	11.67	0.32

Table 10 presents a quantitative summary of the four distinct clusters generated by the Affinity Propagation algorithm, highlighting the key statistical differences underlying the final cluster assignments. The analysis reveals notable variations among the clusters. Cluster 3 shows substantially higher average total cases (156.25), incidence rate (41.73), and severity index (11.67%). In contrast, Cluster 0 exhibits a high incidence rate (29.80) coupled with an early peak month (7.44, mid-July). Cluster 2 exhibits the strongest seasonal dependency, with the highest rainy-season proportion (0.85), while Cluster 1 shows the lowest overall case counts (9.00) and a late seasonal peak (10.11).

These quantitative distinctions demonstrate that the model successfully identified unique temporal and epidemiological patterns across the barangays.

These results support selecting affinity propagation as the final clustering model.

Characterization of Final Dengue Risk Profiles

To translate the geographic distribution and quantitative data into actionable public health intelligence, the researchers synthesized the four identified clusters into distinct risk profiles. Each profile defines its unique epidemiological characteristics and strategic implications, providing a clear framework for targeted interventions.

Cluster 3 emerged as the most critical grouping, distinguished by its substantially higher average total cases, incidence rate, and severity index. These values indicate barangays where transmission is both widespread and clinically severe. Epidemiological studies have documented a relationship between urbanization and more severe dengue outcomes, suggesting that densely populated areas amplify both risk and severity (Anders et al., 2015). For this reason, the researchers defined Cluster 3 as the Very High Risk – Severe Outbreak Zone, encompassing barangays such as Mamatid and San Isidro. This profile demands a dual strategy: sustained, year-round vector control to reduce transmission and enhanced clinical surveillance and readiness in local health centers to manage severe outcomes.

Cluster 0 shows a relatively high incidence rate and an early peak in mid-July. Although its overall burden was lower than that of Cluster 3, its early timing makes this cluster epidemiologically significant, as early peaks often act as precursors to wider outbreaks in surrounding areas (Paireau et al., 2016). The researchers therefore defined Cluster 0 as the High Risk – Early

Season Hotspots, including barangays such as Baclaran and Sala. Strategically, this cluster serves as an early warning signal for the city's broader dengue season, requiring proactive interventions such as early detection, community mobilization, and "search and destroy" activities before the rainy season begins.

Cluster 2 exhibited the strongest seasonal dependency, with the majority of cases occurring during the rainy months. This predictable seasonality reflects the climate-driven nature of dengue transmission in the Philippines, where outbreaks are known to follow increases in rainfall and humidity (Sumi et al., 2016). The researchers defined Cluster 2 as the Moderate Risk – Rainy Season Spike Zone, including barangays such as Banaybanay and Gulod. They timed interventions in these areas for maximum efficiency, intensifying public health campaigns, larviciding, and clean-up drives just before and during the June–November period. The predictability of this pattern allows them to allocate resources more efficiently by concentrating efforts during this specific window.

Cluster 1, by contrast, had the lowest average case count, no severe outcomes, and a late, irregular peak in early October. This cluster represents areas with sporadic, low-level transmission. While the burden is minimal, background transmission zones remain important for surveillance. The researchers defined Cluster 1 as the Low Risk – Background Transmission Zone, represented by Barangay Uno. Although intensive interventions are not required, they maintain routine surveillance and basic awareness campaigns to monitor changes in transmission patterns and ensure that dengue remains under control.

Table 11: Final Dengue Risk Profiles and Strategic Definitions.

Cluster Risk Profiles	Avg. Total Cases	Avg. Incidence Rate (per 10,000)	Rainy Season Cases	Avg. Peak Month	Youth Cases	Female Cases	Severe Cases	Confirmed Cases	Member Barangays
Very High Risk - The Severe Outbreak Zone	156.25	41.73	76.00%	7.71 (Late Jul)	70.90%	45.00%	11.70%	0.30%	MAMATID, MARINIG, PULO, SAN ISIDRO
High Risk - The Early Season Hotspots	35.67	29.8	64.00%	7.44 (Mid Jul)	69.60%	41.10%	8.10%	1.00%	BACLARAN, BANLIC, BARANGAY TRES (POB.), BUTONG, PITTLAND, SALA
Moderate Risk - The Rainy Season Spike Zone	37.43	20.93	85.00%	8.50 (Mid Aug)	72.90%	47.30%	7.80%	0.00%	BANAYBANAY, BARANGAY DOS (POB.), BIGAA, CASILE, DIEZMO, GULOD, NIUGAN
Low Risk - The Background Transmission Zone	9	24.39	44.00%	10.11 (Early Oct)	77.80%	66.70%	0.00%	0.00%	BARANGAY UNO (POB.)

In Table 11, the dengue risk profiles show which barangays have the highest to lowest risk based on past cases and key indicators. The Very High Risk group (Severe Outbreak Zone) includes Mamatid, Marinig, Pulo, and San Isidro, which had the most cases and peaked in late July. The High Risk group (Early Season Hotspots) covers barangays like Baclaran, Banlic, and

Sala, which see cases earlier in the rainy season. The Moderate Risk group (Rainy Season Spike Zone) includes barangays with later seasonal peaks, while the Low Risk group (Background Transmission Zone) has fewer cases and lower overall risk.

This risk grouping helps local health officials focus resources on high-risk areas while monitoring lower-risk barangays, supporting more effective dengue control across Cabuyao City.

Visualizing Final Cluster Profiles

In addition, a radar chart was provided for visual presentation to help illustrate the multidimensional features of each dengue risk cluster. The variables were normalized to the range [0, 1] (min-max normalization) to allow for a fair comparison.

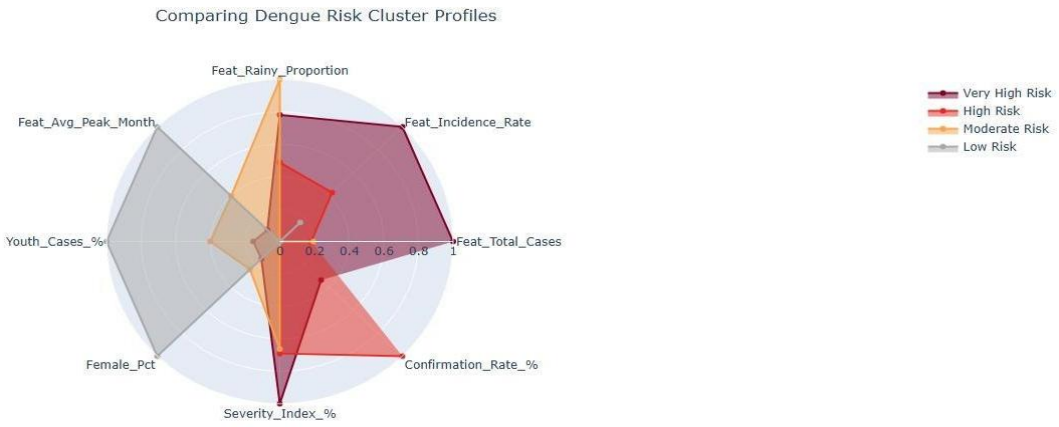


Figure 23. Radar Chart Comparing Dengue Risk Clusters Across Key Features.

Figure 23 illustrates that Cluster 3 exhibited notably higher burden, risk, and severity. Its peak in incidence occurs earlier and is more pronounced among younger age groups than in Cluster 0. Cluster 2 demonstrates an

extreme seasonal pattern, with high scores in peak months, whereas Cluster 1 shows low overall scores and is considered a lower-priority group.

To visually communicate these findings, the researchers created a thematic risk map for Cabuyao City. Each barangay was color-shaded according to its interpreted dengue risk level: Very High Risk (dark red), High Risk (red-orange), Moderate Risk (orange), and Low Risk (gray).

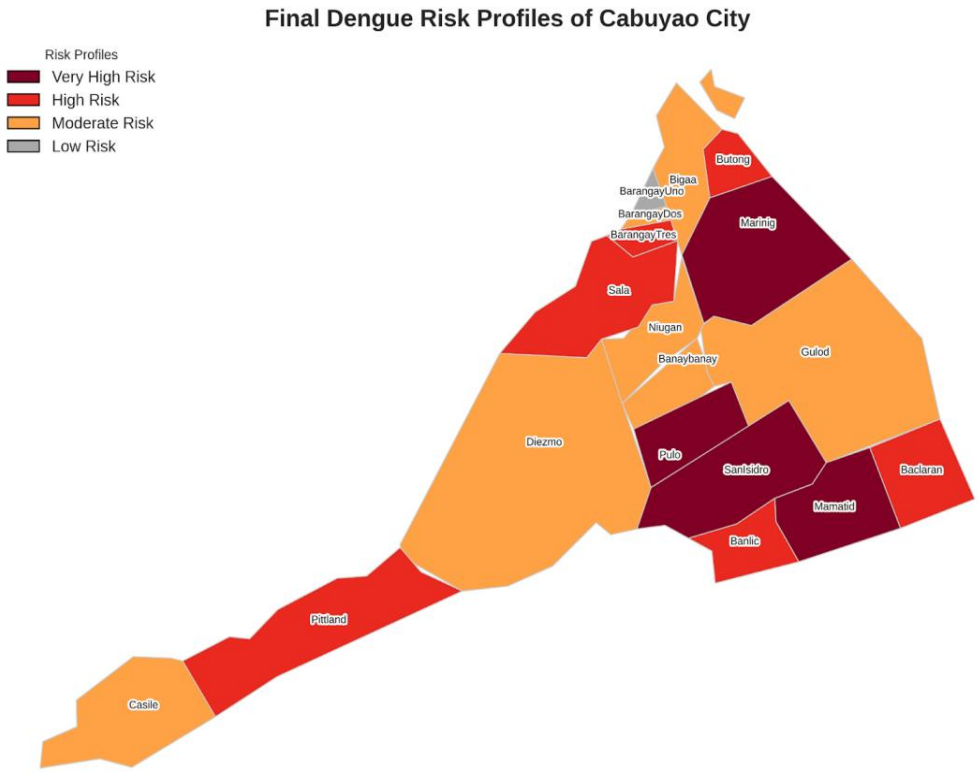


Figure 24. Final Dengue Risk Profiles across Cabuyao City

Figure 24 shows that high-risk barangays, such as Marinig, San Isidro, Mamatid, and Pulo, are densely populated areas with sustained dengue transmission. In contrast, Moderate and Low Risk barangays, such as Casile, Gulod, and Diezmo, are less urbanized and show seasonal patterns. This map helps guide targeted interventions and efficient use of public health resources in Cabuyao City.

CHAPTER V

SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study aimed to identify and characterize high-risk barangays for dengue in Cabuyao City, Laguna, by applying machine learning clustering techniques to dengue incidence data from 2021 to 2022. This study employed a feature-based time-series clustering approach, engineering four key epidemiological features (Burden, Risk, Seasonality, Timing) for each of the 18 barangays. Three centroid-based algorithms—K-Means, Affinity Propagation, and Fuzzy C-Means—were systematically evaluated through a multi-stage process, including baseline analysis, hyperparameter optimization, and a final comparative assessment using the Silhouette Score, Davies-Bouldin Index (DBI), and Dunn Index.

The analysis identified Affinity Propagation as the optimal algorithm, owing to its superior ability to produce compact, well-separated clusters. Using this model, four distinct epidemiological profiles were identified for Cabuyao City: a 'Very High Risk - Severe Outbreak Zone', a 'High Risk - Early Season Hotspot Zone', a 'Moderate Risk - Rainy Season Spike Zone', and a 'Low Risk - Background Transmission Zone'.

To provide context, this study compared these data-driven clusters with historical dengue case records from 2014 to 2024. This review confirmed that the high-risk barangays identified by the model consistency corresponded to

areas with elevated case numbers in previous years, demonstrating the potential of this clustering methodology as a valuable tool for targeted dengue prevention and control.

Conclusions

The primary conclusion of this study is that dengue transmission in Cabuyao City is not uniform but follows four distinct, statistically validated risk profiles. "This study systematically classifies the city's 18 barangays into severe outbreak zones, early-season hotspot zones, rainy-season spike zones, and background transmission zones. These profiles exhibit significant differences in case burden, per-capita risk, seasonality, and clinical severity, confirming that the dengue problem is highly heterogeneous across the city.

This study also clearly established that a feature-based clustering framework is a highly practical approach for characterizing complex epidemiological patterns at the local community level. By carefully engineering key temporal and risk-based features from daily case data, the study effectively demonstrated that standard unsupervised machine learning algorithms can generate meaningful, actionable, and interpretable public health insights, validating this approach for local-level disease surveillance.

Furthermore, among the three algorithms evaluated, Affinity Propagation was identified as the most suitable algorithm for this specific dengue data set. It demonstrated the best overall clustering performance by achieving the lowest Davies-Bouldin Index (a lower score indicating better cluster quality) and the highest Dunn Index (a higher score indicating better cluster quality). The analysis demonstrates the algorithm's superior ability to

produce compact, well-separated clusters, making it crucial for reliably distinguishing between different risk patterns.

The analysis confirms that the cluster profiles align closely with long-term historical patterns. When compared with dengue case records from 2014 to 2024, the barangays classified by the model as "Very High Risk" and "High Risk" were the same areas that repeatedly experienced elevated case counts and outbreak activity in previous years. This alignment between model outcomes and historical data indicates that the clustering results are stable, dependable, and reflective of persistent transmission patterns.

Ultimately, the study's findings clearly demonstrate that clustering analysis can serve as a highly effective decision support tool for dengue prevention and control. The research generated new, actionable, and data-driven insights into distinct epidemiological risks across Cabuyao City, enabling local health authorities to focus their efforts more efficiently, prioritize interventions, and allocate resources with greater precision. By supporting more targeted, evidence-based, and proactive public health strategies, this research contributes directly to the United Nations Sustainable Development Goals, particularly SDG 3 (Good Health and Well-being) and SDG 11 (Sustainable Cities and Communities).

Recommendations

Based on the findings and conclusions of this study, the researchers propose the following recommendations to enhance dengue prevention efforts, strengthen surveillance systems, and support the implementation of evidence-based policies in Cabuyao City and other comparable urban settings.

1. **Implement Profile-Specific Targeted Interventions**

The study strongly recommends that the City Health Office develop and implement tailored intervention strategies for each identified risk profile. This approach shifts away from uniform interventions and supports a more efficient, data-driven allocation of public health resources:

- **For the 'Very High Risk - Severe Outbreak Zone':** Implement sustained, year-round vector control measures and strengthen clinical readiness and surveillance in local health facilities to effectively manage the high case burden and the elevated proportion of severe dengue outcomes.
- **For the 'High Risk - Early Season Hotspots':** Designate these barangays as sentinel surveillance sites for early warning. Deploy resources for proactive "search and destroy" operations and community mobilization ahead of the mid-July peak, enabling early intervention and reducing the likelihood of city-wide transmission.
- **For the 'Moderate Risk - The Rainy Season Spike Zone':** Concentrate public health campaigns, fogging, and clean-up drives to the period immediately preceding and during the rainy season (June-November) for maximum resource efficiency.
- **For the 'Low Risk - Background Transmission Zone':** Maintain routine city-wide surveillance in this area to monitor for any changes in transmission patterns and to serve as a baseline for comparison.

2. Development of a Dynamic Surveillance Dashboard

The static analysis generated in this study may serve as fundation for a

dynamic, real-time monitoring dashboard. By integrating the analytical framework with live or near-real-time case data from the CESU, the dashboard could provide continuously updated insights into emerging hotspots and automatically flag barangays whose current trends are shifting toward a higher risk profile.

3. Integration of Environmental and Climate Factors

Future research should integrate additional environmental and climate variables, including detailed rainfall and temperature data, sanitation indicators, and land-use characteristics derived from remote-sensing imagery (e.g., NDVI for vegetation, NDBI for built-up areas). Incorporating these factors as features in the clustering model could enhance its predictive capability and reveal the specific environmental and climatic drivers behind each dengue risk profile.

4. Replicability and Scalability:

The proposed clustering framework is replicable and adaptable for other dengue-endemic LGUs in the Philippines. By applying the same methodology to local data, municipalities can generate data-driven dengue risk profiles, supporting improved surveillance and decision-making.

5. Future Research and System Development

Future work may explore advanced models, such as LSTMs or transformer-based approaches, to forecast dengue cases. This study also suggests investigating automated hyperparameter optimization techniques to improve model performance and robustness. Additionally, real-time data integration can enhance the overall analysis.

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APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

"CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING"

NATHALIE SHANE DEJUMO, MA. CRISTEL P. ROARING, JABEZ ESTANISLAO have been granted permission to proceed with their proposal presentation.

As their Research Adviser, I confirm that the proponents have met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the proposal presentation.

ROWAN N. ELOMINA
Thesis Project Adviser
November 6, 2025

APPENDIX B

Certificate of Acceptance

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

APPROVAL SHEET

The thesis entitled "**CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING**" prepared and submitted by Nathalie Shane Dejumo, Ma. Cristel P. Roaring, Jabez Estanislao in partial fulfillment requirements for the degree Bachelor of Science in Computer Science specialized in Intelligent Systems, is hereby recommended for approval and acceptance.

ROWAN N. ELOMINA
Thesis Project Adviser

APPENDIX C

Correspondence

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

March 11, 2025

DRA. ELENA C. DIAMANTE
City Health Officer II, City Health Office 1
Cabuyao City, Laguna

THRU: JOHN CARLO GARBIN, RN
Cabuyao CESU

Dear Dra. Diamante:

Good day and greetings!

We are third-year Bachelor of Science in Computer Science students specializing in Intelligent Systems at Laguna State Polytechnic University – Los Baños. Our research seeks to conduct a data-driven and cluster-based analysis of dengue cases in Cabuyao City, Laguna, with the objective to identifying high-risk areas and understanding patterns and contributing factors.

In line with this, we respectfully request access to data on dengue cases within your jurisdiction from 2014 to 2024, including the number of reported cases, affected locations, trends, and implemented public health measures. This information is vital for analyzing patterns and risk factors to enhance our study's findings.

We assure you that all information provided will be used solely for academic purposes and will be treated with strict confidentiality. If there are any procedures or requirements needed to process this request, kindly let us know at your earliest convenience.

We truly appreciate your time and assistance in supporting our research. Should you require further details, please feel free to contact us at 09705684451 or email us at dejumonathalieshane@gmail.com.

Thank you, and we look forward to your positive response.

Very respectfully yours,

Nathalie Shane Dejumo
Jabez Estanislao
Ma. Cristel Roaring

Noted by:

Rowan N. Elomina, PhD
Thesis Adviser

Dr. Loyd S. Echalar
Associate Dean, College of Computer Studies

RECEIVED
JOHN CARLO B. GARBIN, RN
Lic. No. 0093812
3/11/2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

March 11, 2025

Dr. CECIL ARNOLD G. AGUILLO
City Health Officer II, City Health Office 2
Cabuyao City, Laguna

THRU: EUNICE ERICA CRANSTOUN, RN
Cabuyao CESU

Dear Dr. Aguillo:

Good day and greetings!

We are third-year Bachelor of Science in Computer Science students specializing in Intelligent Systems at Laguna State Polytechnic University – Los Baños. Our research seeks to conduct a data-driven and cluster-based analysis of dengue cases in Cabuyao City, Laguna, with the objective to identifying high-risk areas and understanding patterns and contributing factors.

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We assure you that all information provided will be used solely for academic purposes and will be treated with strict confidentiality. If there are any procedures or requirements needed to process this request, kindly let us know at your earliest convenience.

We truly appreciate your time and assistance in supporting our research. Should you require further details, please feel free to contact us at 09705684451 or email us at dejumonathalieshane@gmail.com.

Thank you, and we look forward to your positive response.

Very respectfully yours,

Nathalie Shane Dejumo
Jabez Estanislao
Ma. Cristel Roaring

Noted by:

Rowan N. Elomina, PhD
Thesis Adviser

Dr. Loyd S. Echalar
Associate Dean, College of Computer Studies

RECEIVED BY:
Eunice Erica B. Cranstoun, RN, MPH
Lic. No. 0735668
13-11-2025

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

November 3, 2025

DRA. ELENA C. DIAMANTE
City Health Officer II, City Health Office I
Cabuyao City, Laguna

THRU: JOHN CARLO GARBIN, RN
Cabuyao CESU

Dear Dra. Diamante:

Good day and greetings!

We, the undersigned fourth-year Computer Science students from Laguna State Polytechnic University - Los Baños, are completing our thesis entitled, **"CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING."**

Our study, which utilized the 2021-2022 dengue data provided by your office, has successfully identified four distinct barangay risk profiles based on their seasonal and epidemiological patterns. To ensure our findings are practically relevant, we require validation from a professional with on-the-ground expertise.

In line with this, we respectfully request your assistance in identifying a qualified Public Health Expert or Epidemiologist from your team to act as a validator for our research findings. The validation simply involves a review of our final summary report and answering a structured evaluation tool.

Your office's participation is invaluable for confirming the local validity and public health relevance of our data-driven results.

We sincerely hope for your favorable approval and assistance.

Thank you.

Very respectfully yours,

Nathalie Shane Dejumo
Researcher

Ma. Cristel Roaring
Researcher

Jabez Estanislao
Researcher

Noted by:

Rowan N. Elomina, PhD
Thesis Adviser

Dr. Loyda S. Echalar
Associate Dean, College of Computer Studies

RECEIVED BY:
JOHN CARLO B. GARBIN, RN
Lic. No. 0695812

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

November 3, 2025

Dr. CECIL ARNOLD G. AGUILLO
City Health Officer II, City Health Office 2
Cabuyao City, Laguna

THRU: EUNICE ERICA CRANSTOUN, RN
Cabuyao CESU

Dear Dr. Aguillo:

Good day and greetings!

We, the undersigned fourth-year Computer Science students from Laguna State Polytechnic University - Los Baños, are completing our thesis entitled, **"CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING."**

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Nathalie Shane Dejumo
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Thesis Adviser

Dr. Loyd S. Echalar
Associate Dean, College of Computer Studies

Received by:

Eunice Erica B. Cranstoun RN, MMPHA
Lic. No. 0735668

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

BAGONG PILIPINAS

July 1, 2025

Dr. CECIL ARNOLD G. AGUILLO
City Health Officer II, City Health Office 2
Cabuyao City, Laguna

THRU: EUNICE ERICA CRANSTOUN, RN
Cabuyao CESU

Dear Dr. Aguillo:

Good day and greetings!

We are third-year Bachelor of Science in Computer Science students specializing in Intelligent Systems at Laguna State Polytechnic University – Los Baños. Our research seeks to conduct a seasonal time series analysis of dengue cases in Cabuyao City, Laguna. The objective is to identify and characterize long-term risk profiles for each barangay by applying machine learning clustering techniques.

In line with this, we respectfully request access to an anonymized, line-listed dataset of dengue cases within your jurisdiction from January 1, 2020, to the most recent available date in 2024. For each case, the essential information we require is the date of reporting (or onset) and the barangay of residence. This daily or, if not available, weekly data granularity is vital for our seasonal clustering analysis, which forms the core of our study.

We assure you that all information provided will be used solely for academic purposes, will be treated with strict confidentiality. If there are any procedures or requirements needed to process this request, kindly let us know at your earliest convenience.

We truly appreciate your time and assistance in supporting our research. Should you require further details, please feel free to contact us at 09705684451 or email us at dejumonathalieshane@gmail.com.

Thank you, and we look forward to your positive response. Very respectfully yours,

Nathalie Shane Dejumo
Jabez Estanislao
Ma. Cristel Roaring

Noted by:

Rowan N. Elomina, PhD
Thesis Adviser

Dr. Loyd S. Echalar
Associate Dean, College of Computer Studies

Received by:
Eunice Erica B. Cranstoun RN, MMPHA
Lic. No. 0735668

APPENDIX D

Data Sets (or Primary Data)

Barangay	BACLARAN	BANAY BANAY	BANLIC	BARANGAY DOS (POB.)	BARANGAY TRES (POB.)	BARANGAY UNO (POB.)	BIGAA	BUTONG	CASILE	DIEZMO	GULOD	MAMATID	MARINIG	NIUGAN	PITTLAND	PULO	SALA	SAN ISIDRO	UNSPECIFIED ADDRESS	Total
Month_Name																				
2021-01	2	1	3	0	0	0	0	0	0	0	2	4	13	2	0	8	1	4	0	40
2021-02	3	0	1	0	0	1	1	2	0	1	0	5	3	0	2	6	1	1	0	27
2021-03	0	5	1	0	0	0	1	0	0	0	0	2	3	1	2	2	0	4	1	22
2021-04	0	0	0	0	0	0	1	0	0	0	0	3	1	0	0	1	0	0	0	6
2021-05	0	0	1	0	0	0	0	0	0	0	0	1	0	0	0	0	1	2	0	5
2021-06	1	0	1	0	0	0	0	0	0	0	0	3	0	0	1	0	2	0	0	8
2021-07	0	3	1	0	0	0	1	0	0	0	0	1	1	0	0	0	1	3	0	11
2021-08	0	3	0	0	0	0	0	0	0	0	1	1	2	0	0	0	0	2	0	9
2021-09	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	3	0	8
2021-10	2	2	1	0	0	0	0	1	0	0	2	3	10	1	0	0	1	7	0	30
2021-11	2	9	3	1	0	0	1	2	0	0	1	9	7	1	3	7	1	5	0	52
2021-12	0	2	6	0	0	0	0	2	0	0	2	3	0	0	0	4	2	5	0	26
2022-01	0	1	0	0	0	0	1	2	0	0	0	1	3	1	0	3	2	4	0	18
2022-02	0	3	1	0	0	0	1	4	0	0	0	4	1	1	0	2	3	1	0	21
2022-03	1	2	5	0	0	0	0	2	0	0	1	8	0	0	1	6	0	3	0	29
2022-04	1	2	1	0	0	0	0	0	0	0	0	0	0	0	0	4	2	2	0	12
2022-05	1	1	4	0	4	0	0	1	0	0	0	5	0	0	0	1	1	3	0	21
2022-06	0	3	7	0	1	0	2	0	0	0	1	11	5	1	0	4	3	13	0	51
2022-07	2	7	3	1	2	0	1	3	1	2	3	21	14	4	1	10	3	13	0	91
2022-08	4	9	5	2	2	0	2	9	0	2	7	41	26	11	0	29	8	24	1	182
2022-09	3	11	9	0	2	0	7	4	2	1	5	27	25	12	1	23	3	30	1	166
2022-10	3	8	3	1	1	3	4	7	0	0	13	16	12	22	1	16	3	11	0	124
2022-11	2	13	3	0	1	1	5	4	0	1	3	17	12	10	0	5	8	8	0	93
2022-12	2	5	4	0	1	4	1	5	0	0	6	10	1	12	0	8	0	2	0	61
Grand Total	30	92	64	5	14	9	29	48	3	7	47	196	139	79	12	140	46	150	3	1113

Monthly Dengue Cases per Barangay in Cabuyao City (2021–2022)

Barangay	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Baclaran	8	18	33	8	6	18	11	11	19	7	16
Banaybanay	46	100	96	29	42	78	13	27	65	32	97
Banlic	28	62	34	16	23	74	8	19	45	16	24
Barangay Dos	2	32	6	3	2	4	0	1	4	5	14
Barangay Tres	2	16	9	5	9	9	2	1	14	1	31
Barangay Uno	5	29	8	4	13	2	0	1	8	4	15
Bigaa	5	60	39	15	21	11	4	5	24	19	47
Butong	19	106	40	16	38	46	5	7	41	22	55
Casile	0	1	4	2	1	25	5	0	3	4	4
Diezmo	3	3	6	2	6	5	3	1	6	3	3
Gulod	12	28	48	9	28	30	13	8	39	21	81
Mamatid	97	231	281	64	48	199	21	35	161	29	71
Marinig	33	121	130	40	72	68	21	40	100	35	184
Niugan	20	89	77	39	40	58	11	5	74	32	58
Pittland	5	9	10	7	2	6	3	8	4	5	9
Pulo	52	112	90	34	48	95	19	29	111	45	69
Sala	29	68	26	13	19	29	3	10	36	21	38
San Isidro	36	123	130	28	33	91	14	36	114	14	31
Unspecified Address	8	74	16	12	3	4	--	--	1	--	--
Total	410	1282	1083	346	454	852	156	244	869	315	847

Historical Dengue Cases per Barangay in Cabuyao City (2014–2024)

2020 Cabuyao Population Data

	A	B
1	**Barangay**	**Population_2020**
2	Baclaran	14,606
3	Banaybanay	34,260
4	Banlic	20,646
5	Barangay Dos	1,573
6	Barangay Tres	4,274
7	Barangay Uno	3,690
8	Bigaa	13,665
9	Butong	14,764
10	Casile	3,619
11	Diezmo	6,622
12	Gulod	17,215
13	Mamatid	61,085
14	Marinig	45,343
15	Niugan	38,576
16	Pittland	6,052
17	Pulo	35,113
18	Sala	10,903
19	San Isidro	23,324

Cabuyao GeoJSON File

GID_3	GID_0	COUNTRY	GID_1	NAME_1	NL_NAME_1	GID_2	NAME_2	NL_NAME_2	NAME_3	VARNAME_3	NL_NAME_3	TYPE_3	ENGTYPE_3	CC_3	HASC_3	geometry	coordinates
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Baclaran	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1572,14.2471,121.1677,14.2511,121.1729,14.2399,121.1618,14.2358,121.1572,14.2471
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Banaybanay	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1219,14.2497,121.1202,14.2533,121.1314,14.2624,121.134,14.2557,121.1284,14.2518,121.1219,14.2497
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Banlic	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.143,14.24,121.1432,14.2367,121.1466,14.2311,121.1341,14.2281,121.1336,14.2326,121.1301,14.2344,121.1373,14.2364,121.143,14.24
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	BarangayDos	Poblacion	NA	Barangay	Village	NA	NA	MultiPolygon	121.1276,14.279,121.1229,14.2781,121.12,14.2777,121.1215,14.2797,121.1268,14.2809,121.1276,14.279
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	BarangayTres	Poblacion	NA	Barangay	Village	NA	NA	MultiPolygon	121.12,14.2777,121.1276,14.279,121.1285,14.2761,121.1218,14.2738,121.1179,14.2768,121.12,14.2777
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	BarangayUno	Poblacion	NA	Barangay	Village	NA	NA	MultiPolygon	121.1288,14.2809,121.1215,14.2797,121.1248,14.2862,121.1257,14.2842,121.1288,14.2809
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Bigaa	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1288,14.2809,121.1248,14.2862,121.1265,14.2892,121.1253,14.2936,121.1283,14.2982,121.1351,14.2917,121.1323,14.2889,121.1333,14.2821,121.1291,14.2741,121.1268,14.2809,121.1343,14.2944,121.1318,14.2983,121.1335,14.3001,121.134,14.2977,121.1385,14.2961,121.137,14.2932,121.1343,14.2944
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Butong	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1333,14.2821,121.1323,14.2889,121.1351,14.2917,121.1375,14.2911,121.1426,14.2851,121.1333,14.2821
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Casile	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.0386,14.2117,121.0471,14.2178,121.0527,14.2176,121.0545,14.2172,121.0594,14.2094,121.0469,14.2023,121.0421,14.2035,121.0332,14.2022,121.0336,14.2059,121.0387,14.208,121.0386,14.2117
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Diezmo	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1202,14.2533,121.1245,14.2415,121.1224,14.2357,121.1185,14.2349,121.1163,14.2366,121.1098,14.2305,121.1031,14.2277,121.0963,14.227,121.0875,14.2317,121.087,14.2335,121.1019,14.2603,121.1149,14.2597,121.1171,14.2623,121.1202,14.2533
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Gulod	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1677,14.2511,121.1507,14.245,121.1451,14.2537,121.1391,14.2501,121.1365,14.2563,121.134,14.2557,121.1329,14.2576,121.1319,14.264,121.1339,14.2655,121.1395,14.2642,121.1545,14.2735,121.165,14.2624,121.1677,14.2511
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Mamatid	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1572,14.2471,121.1618,14.2358,121.1466,14.2311,121.1432,14.2367,121.143,14.24,121.1486,14.242,121.1507,14.245,121.1572,14.2471
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Marinig	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1291,14.2741,121.1333,14.2821,121.1426,14.2851,121.1545,14.2735,121.1395,14.2642,121.1339,14.2655,121.1324,14.2645,121.1291,14.2741
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Niugan	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1314,14.2624,121.1264,14.2591,121.1202,14.2533,121.1171,14.2623,121.1204,14.2624,121.1247,14.2671,121.1279,14.2676,121.1291,14.2737,121.1324,14.2645,121.1314,14.2624
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Pittland	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.0594,14.2094,121.0545,14.2172,121.0615,14.2206,121.0645,14.2203,121.0687,14.2244,121.0776,14.2288,121.082,14.2291,121.087,14.2331,121.0901,14.2297,121.0963,14.227,121.0686,14.215,121.0594,14.2094
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Pulo	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1219,14.2497,121.1365,14.2563,121.1391,14.2501,121.1245,14.2415,121.1219,14.2497
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	Sala	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.1179,14.2768,121.1218,14.2738,121.1285,14.2761,121.1279,14.2676,121.1247,14.2671,121.1226,14.264,121.1171,14.2623,121.1149,14.2597,121.1019,14.2603,121.1072,14.2661,121.1132,14.2697,121.1156,14.276,121.1179,14.2768
0	PHL	Philippines	PHL.40_1	Cabuyao	NA	PHL.40.4_1	Cabuyao	NA	SanIsidro	NA	NA	Barangay	Village	NA	NA	MultiPolygon	121.143,14.24,121.1373,14.2364,121.1301,14.2344,121.1266,14.2362,121.1224,14.2357,121.1245,14.2415,121.1451,14.2537,121.1507,14.245,121.1486,14.242,121.143,14.24

APPENDIX E

Relevant Source Code

Importing Required Libraries

```
# --- Core Libraries for Data Manipulation ---
import pandas as pd
import numpy as np

# --- Library for Geospatial Data ---
import geopandas as gpd

# --- Libraries for Data Visualization ---
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.patheffects as pe

# --- Libraries for Machine Learning ---
from sklearn.cluster import KMeans, AffinityPropagation # Add other algorithms as needed
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import davies_bouldin_score

plt.style.use('seaborn-v0_8-whitegrid')
plt.rcParams['figure.figsize'] = (10, 7.5)
plt.rcParams['figure.dpi'] = 300

!pip install scikit-fuzzy
print("Libraries imported successfully.")
```

```
# --- Install necessary libraries ---
!pip install -q scikit-fuzzy kneed

# --- Import models, metrics, and helper functions ---
from sklearn.cluster import KMeans, AffinityPropagation
from skfuzzy.cluster import cmeans
from sklearn.metrics import silhouette_score, davies_bouldin_score, pairwise_distances
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
import plotly.graph_objects as go
from sklearn.preprocessing import MinMaxScaler
import matplotlib.cm as cm
import matplotlib.colors as mcolors
```

Data Cleaning and Preparation

```
print("Unique Barangay names BEFORE cleaning:")
print(df_daily_raw['Barangay'].dropna().unique())

df_daily_raw['Barangay'] = df_daily_raw['Barangay'].astype(str)
df_daily_raw['Barangay'] = df_daily_raw['Barangay'].str.replace(r'\s+', ' ', regex=True)

df_daily_raw['Barangay'].replace({'nan': 'UNSPECIFIED ADDRESS'}, inplace=True)
df_daily_raw['Barangay'].fillna('UNSPECIFIED ADDRESS', inplace=True)

df_daily_raw['Barangay'].replace({'3': 'BARANGAY TRES (POB.)'}, inplace=True)

df_daily_raw['Barangay'] = df_daily_raw['Barangay'].str.upper().str.strip()


print("\nUnique Barangay names AFTER cleaning:")
print(df_daily_raw['Barangay'].unique())

df_daily_raw['DOnset'] = pd.to_datetime(df_daily_raw['DOnset'], errors='coerce')
df_daily_raw['DAdmit'] = pd.to_datetime(df_daily_raw['DAdmit'], errors='coerce')

# RULE-BASED DATE SELECTION
df_daily_raw['ClusterDate'] = df_daily_raw['DOnset']
df_daily_raw['ClusterDate'] = df_daily_raw['ClusterDate'].fillna(df_daily_raw['DAdmit'])

# For cross-year cases, use DAdmit. We use the EXISTING 'Year' column from the raw data.
df_daily_raw['ClusterDate'] = np.where(df_daily_raw['ClusterDate'].dt.year < df_daily_raw['Year'],
                                       df_daily_raw['DAdmit'],
                                       df_daily_raw['ClusterDate'])

print("\nCross-year cases and missing onsets have been handled using the official 'Year' column.")


final_columns = {
    'ClusterDate': 'Date',
    'Barangay': 'Barangay',
    'AgeYears': 'Age',
    'Sex': 'Sex',
    'Outcome': 'Outcome',
    'Type': 'Dengue_Type',
    'CaseClassification': 'Case_Classification'
}
df_clean = df_daily_raw[final_columns.keys()].copy()
df_clean.rename(columns=final_columns, inplace=True)

# Drop any rows where Date or Barangay are still missing
df_clean.dropna(subset=['Date', 'Barangay'], inplace=True)

# Add helper columns for easier analysis
df_clean['Year'] = df_clean['Date'].dt.year
df_clean['Month'] = df_clean['Date'].dt.month
df_clean['Month_Name'] = df_clean['Date'].dt.strftime('%Y-%m')

print("\nFinal clean DataFrame created.")
```

Spatial Analysis

```
total_cases = df_clean[df_clean['Barangay'] != 'UNSPECIFIED ADDRESS'].groupby('Barangay').size().reset_index(name='Total_Cas


total_cases['Barangay'] = total_cases['Barangay'].str.replace(' (POB.)', '', regex=False).str.strip()
print("Cleaned 'Barangay' names in total_cases DataFrame:")
print(total_cases.head())



df_population['Barangay'] = df_population['Barangay'].str.upper().str.strip() # Just make sure it's uppercase

if df_population['Population_2020'].dtype == 'object':
    df_population['Population_2020'] = df_population['Population_2020'].str.replace(',', '', regex=False)
df_population['Population_2020'] = pd.to_numeric(df_population['Population_2020'], errors='coerce')




df_spatial = pd.merge(df_population, total_cases, on='Barangay', how='left')
df_spatial['Total_Cases_21_22'] = df_spatial['Total_Cases_21_22'].fillna(0).astype(int)


print()

df_spatial['Incidence_Rate_per_10k'] = (df_spatial['Total_Cases_21_22'] / (df_spatial['Population_2020'] + 1e-6)) * 10000
df_spatial['Incidence_Rate_per_10k'] = df_spatial['Incidence_Rate_per_10k'].round(2)
print("Spatial data table created successfully:")
print(df_spatial.head())



df_spatial['join_key'] = df_spatial['Barangay'].str.replace(r'[\s]', '', regex=True).str.upper() # Removes all spaces
gdf_map['join_key'] = gdf_map['NAME_3'].str.replace(r'[\s]', '', regex=True).str.upper()


gdf_merged = gdf_map.merge(df_spatial, on='join_key', how='left')


numeric_cols = ['Population_2020', 'Total_Cases_21_22', 'Incidence_Rate_per_10k']
for col in numeric_cols:
    if col in gdf_merged.columns:
        gdf_merged[col] = gdf_merged[col].fillna(0)


print("\nFinal GeoDataFrame prepared for mapping.")
gdf_merged[['NAME_3', 'join_key', 'Total_Cases_21_22', 'Incidence_Rate_per_10k']].head()
```

Feature Engineering for Clustering

```
barangays_to_analyze = sorted(df_clean[df_clean['Barangay'] != 'UNSPECIFIED ADDRESS']['Barangay'].unique())
df_features = pd.DataFrame({'Barangay': barangays_to_analyze})

# --- Feature 1: Overall Magnitude (Total Cases) ---
feature_total_cases = df_clean.groupby('Barangay').size().reset_index(name='Feat_Total_Cases')
df_features = pd.merge(df_features, feature_total_cases, on='Barangay', how='left')

df_features['Barangay_Clean'] = df_features['Barangay'].str.replace(r' \(.+\)', '', regex=True).str.strip()
df_population['Barangay_Clean'] = df_population['Barangay'].str.upper().str.replace(r' \(.+\)', '', regex=True).str.strip()


# --- Feature 2: Incidence Rate (Overall Risk) ---
df_features = pd.merge(df_features, df_population[['Barangay_Clean', 'Population_2020']], on='Barangay_Clean', how='left')

# Calculate and round the incidence rate
df_features['Feat_Incidence_Rate'] = ((df_features['Feat_Total_Cases'] / (df_features['Population_2020'] + 1e-6)) * 10000).round(2)


# --- Feature 3: Seasonality (Rainy Season Dominance) ---
rainy_months = [6, 7, 8, 9, 10, 11]
df_clean['Season'] = np.where(df_clean['Month'].isin(rainy_months), 'Rainy', 'Dry')
seasonal_cases = df_clean.groupby(['Barangay', 'Season']).size().unstack(fill_value=0)
seasonal_cases['Feat_Rainy_Proportion'] = (seasonal_cases['Rainy'] / (seasonal_cases['Rainy'] + seasonal_cases['Dry'] + 1e-6)).round(2)
df_features = pd.merge(df_features, seasonal_cases[['Feat_Rainy_Proportion']], left_on='Barangay', right_index=True, how='left')


# --- Feature 4: Peak Month (Average Month of Cases) ---
weighted_month_sum = df_clean.groupby('Barangay')['Month'].sum()
total_cases_per_barangay = df_clean.groupby('Barangay').size()
feature_avg_peak_month = ((weighted_month_sum / total_cases_per_barangay)).round(2).reset_index(name='Feat_Avg_Peak_Month')
df_features = pd.merge(df_features, feature_avg_peak_month, on='Barangay', how='left')


df_features = df_features.drop(['Population_2020', 'Barangay_Clean'], axis=1)
df_features = df_features.fillna(0)
```

Baseline Performance Analysis

```
# --- Custom Dunn Index function (remains the same) ---
def dunn_index(X, labels):
    unique_labels = np.unique(labels)
    if len(unique_labels) < 2: return 0
    min_inter_cluster_dist = np.inf
    for i in range(len(unique_labels)):
        for j in range(i + 1, len(unique_labels)):
            dist = pairwise_distances(X[labels == unique_labels[i]], X[labels == unique_labels[j]]).min()
            if dist < min_inter_cluster_dist: min_inter_cluster_dist = dist
    max_intra_cluster_dist = 0
    for i in range(len(unique_labels)):
        dist = pairwise_distances(X[labels == unique_labels[i]]).max()
        if dist > max_intra_cluster_dist: max_intra_cluster_dist = dist
    return min_inter_cluster_dist / (max_intra_cluster_dist + 1e-6)

# --- Prepare for tuning ---
possible_k = range(2, 11)
kmeans_scores = []
fuzzy_scores = []

# --- K-Means Evaluation Loop ---
print("Evaluating K-Means Baseline...")
for k in possible_k:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init='auto')
    labels = kmeans.fit_predict(features_scaled)
    kmeans_scores.append({
        'k': k,
        'Silhouette': silhouette_score(features_scaled, labels),
        'Davies-Bouldin': davies_bouldin_score(features_scaled, labels),
        'Dunn Index': dunn_index(features_scaled, labels)
    })
kmeans_scores_df = pd.DataFrame(kmeans_scores).set_index('k')

# --- Fuzzy C-Means Evaluation Loop ---
print("Evaluating Fuzzy C-Means Baseline...")
fp_data = features_scaled.T
for k in possible_k:
    cntr, u, u0, d, jm, p, fpc = cmeans(fp_data, c=k, m=2, error=0.005, maxiter=1000, init=None)
    labels = np.argmax(u, axis=0)
    fuzzy_scores.append({
        'k': k,
        'Silhouette': silhouette_score(features_scaled, labels),
        'Davies-Bouldin': davies_bouldin_score(features_scaled, labels),
        'Dunn Index': dunn_index(features_scaled, labels)
    })
fuzzy_scores_df = pd.DataFrame(fuzzy_scores).set_index('k')
```

Hyper Optimization

```
OPTIMAL_K = 4
optimization_results = []

# --- K-Means Tuning (init parameter) ---
print("Tuning K-Means...")
for init_method in ['k-means++', 'random']:
    kmeans = KMeans(n_clusters=OPTIMAL_K, init=init_method, random_state=42, n_init='auto')
    labels = kmeans.fit_predict(features_scaled)
    optimization_results.append({
        'Model': f'K-Means (init={init_method})',
        'Silhouette': silhouette_score(features_scaled, labels),
        'Davies-Bouldin': davies_bouldin_score(features_scaled, labels),
        'Dunn Index': dunn_index(features_scaled, labels)
    })

# --- Fuzzy C-Means Tuning (m parameter) ---
print("Tuning Fuzzy C-Means...")
for m_val in [1.5, 2.0, 2.5]:
    cntr, u, u0, d, jm, p, fpc = cmeans(fp_data, c=OPTIMAL_K, m=m_val, error=0.005, maxiter=1000, init=None)
    labels = np.argmax(u, axis=0)
    optimization_results.append({
        'Model': f'Fuzzy C-Means (m={m_val})',
        'Silhouette': silhouette_score(features_scaled, labels),
        'Davies-Bouldin': davies_bouldin_score(features_scaled, labels),
        'Dunn Index': dunn_index(features_scaled, labels)
    })

# --- Affinity Propagation Tuning (damping parameter) ---
print("Tuning Affinity Propagation...")
for damp_val in [0.7, 0.8, 0.9]:
    affinity_prop = AffinityPropagation(damping=damp_val, random_state=42)
    labels = affinity_prop.fit_predict(features_scaled)
    num_clusters = len(np.unique(labels))
    # Only score if it finds a reasonable number of clusters
    if num_clusters > 1:
        optimization_results.append({
            'Model': f'Affinity Prop (damp={damp_val}, k={num_clusters})',
            'Silhouette': silhouette_score(features_scaled, labels),
            'Davies-Bouldin': davies_bouldin_score(features_scaled, labels),
            'Dunn Index': dunn_index(features_scaled, labels)
        })
```

Train the Final Optimized models

```
kmeans_final = KMeans(n_clusters=OPTIMAL_K, init='k-means++', random_state=42, n_init='auto')
kmeans_labels = kmeans_final.fit_predict(features_scaled)

cntr, u, u0, d, jm, p, fpc = cmeans(fp_data, c=OPTIMAL_K, m=2.0, error=0.005, maxiter=1000, init=None)
fuzzy_cmeans_labels = np.argmax(u, axis=0)

affinity_prop_final = AffinityPropagation(damping=0.8, random_state=42)
affinity_prop_labels = affinity_prop_final.fit_predict(features_scaled)
num_ap_clusters = len(affinity_prop_final.cluster_centers_indices_)
```

Visualizing Final Cluster profiles

```
scaler_viz = MinMaxScaler()
profile_to_plot = final_profile.copy()
features_for_radar = list(profile_to_plot.columns)
profile_to_plot[features_for_radar] = scaler_viz.fit_transform(profile_to_plot[features_for_radar])

palette = cm.get_cmap('YlOrRd', 8)
cluster_personas = {
    3: {'name': 'Very High Risk', 'color': mcolors.to_hex(palette(7))}, # Darkest Red
    0: {'name': 'High Risk', 'color': mcolors.to_hex(palette(5))}, # Orange
    2: {'name': 'Moderate Risk', 'color': mcolors.to_hex(palette(3))}, # Light Orange
    1: {'name': 'Low Risk', 'color': 'darkgray'}
}
```

APPENDIX F

Evaluation Tool

Evaluation Form – City Health Office I (CHO-I)

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CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING

Nathalie Shane Dejumo, Ma. Cristel P. Roaring, Jabez Estanislao
Researchers

The primary goal of our research is to develop a set of data-driven dengue risk profiles for the 18 barangays of Cabuyao City. Using a feature-based clustering approach on daily dengue case data from 2021-2022, this study aims to identify and characterize groups of barangays that share similar seasonal and epidemiological patterns. The ultimate goal is to provide local health officials with actionable insights to enhance targeted surveillance and prevention strategies.

This questionnaire, referred to in our study as the Evaluation Tool, is designed to help us assess the overall validity and usefulness of our findings. Your honest assessment and professional expertise in public health and epidemiology are essential for validating and contextualizing our model's results.

Our system functions by transforming complex time-series data into a set of key features (Burden, Risk, Seasonality, Timing) and then uses machine learning to find natural groupings. The output is a set of four distinct "risk profiles" that describe the unique dengue patterns found in the city.

Your role as the **Public Health Expert Validator** is to examine the final risk profiles we have generated and determine whether they are clear, meaningful, and useful for real-world public health planning and intervention in Cabuyao City.

Validation Procedure

The core purpose of this validation is to assess whether the data-driven risk profiles are epidemiologically sound and provide actionable intelligence. The validation process will involve the following steps:

1. **Data Provision:**
   The research team will provide you with the following materials for your assessment:
   - **A. The Final Thematic Map:** A map of Cabuyao City showing the geographic distribution of the four final risk profiles.
   - **B. The Final Cluster Profile Summary Table:** A table showing the average characteristics of the barangays within each of the four clusters (the features in this table are defined on the next page).
   - **C. The Interpretation of Final Clusters:** A written analysis describing the "persona" and public health significance of each cluster.

2. **Expert Validation:**
   As the validator, your expertise is essential in determining whether these profiles accurately reflect the dengue situation in Cabuyao and can serve as a valid tool for public health decision-making.

You are expected to provide an objective, professional assessment based on three primary criteria:

- **Clarity and Interpretability:** Measures how clear, understandable, and well-defined the risk profiles are. Do the quantitative descriptions and names make sense?
- **Actionability and Public Health Relevance:** Evaluates whether the findings can be used to inform real-world decisions. Do the profiles provide a useful basis for prioritizing resources or designing targeted interventions?
- **Local Validity (Face Validity):** Assesses the extent to which the discovered profiles and their geographic distribution align with your on-the-ground knowledge and experience of the dengue situation in Cabuyao City.

Evaluation Scale

Rating	Description
5 (Excellent)	The findings are highly relevant, clear, and provide a strong basis for actionable public health strategies. They align well with the expected epidemiological situation.
4 (Very Good)	The findings are relevant and mostly clear; any issues are minor and do not detract from the overall usefulness for public health planning.
3 (Good)	The findings are adequately relevant; some characteristics are clear, but others may be difficult to interpret or apply directly without further refinement.
2 (Fair)	The findings are partially relevant, but contain several unclear or questionable elements that limit their usefulness for public health action.
1 (Poor)	The findings are irrelevant, unclear, or do not appear to align with the real-world public health context.

Explanation of Variables in the Cluster Profile Summary Table

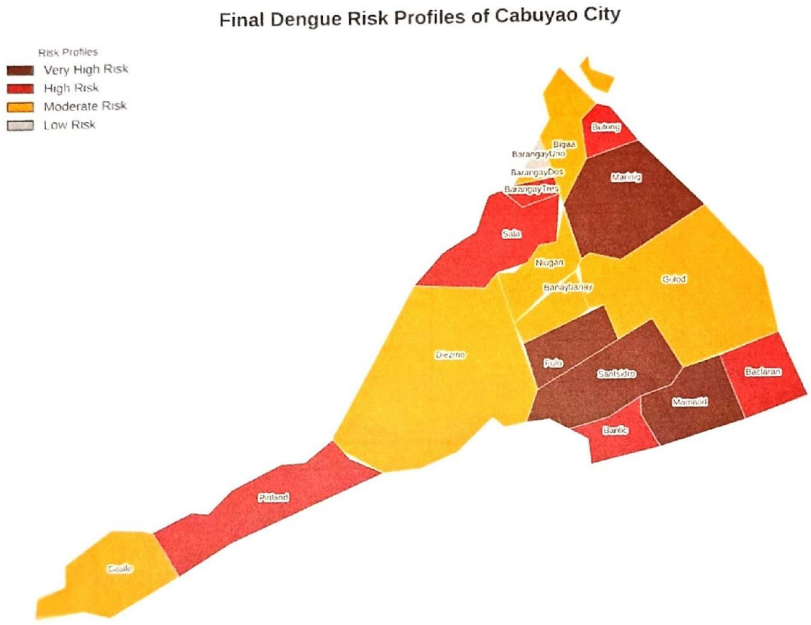
To aid in your evaluation, the columns in the summary table are defined as follows:

Feature Name	Description	Interpretation
Avg. Total Cases	The average number of total dengue cases per barangay in that cluster over the two-year period (2021-2022).	A measure of **Burden.** A higher value means more people were sick.
Avg. Incidence Rate (per 10k)	The average number of cases per 10,000 population.	A measure of **Risk**. A higher value means an individual's chance of getting sick was greater.
Rainy Season Cases	The percentage of a cluster's cases that occurred during the rainy season (June-November).	A measure of **Seasonality.** A value near 85% is highly seasonal; a value near 44% is not.
Avg. Peak Month	The average month of case onset, weighted by case counts.	A measure of **Timing.** A lower number (e.g., 7.4) means an earlier peak (July); a higher number (e.g., 10.1) means a later peak (October).
Youth Cases	The percentage of cases in the cluster that were under 20 years old.	A measure of **Age Distribution.**
Female Cases	The percentage of cases in the cluster that were female.	A measure of **Sex Distribution.**
Severe Cases	The percentage of a cluster's diagnosed cases that were classified as severe (DHF/DSS).	A measure of **Clinical Severity**. A higher value indicates more dangerous outcomes.
Confirmed Cases	The percentage of a cluster's diagnosed cases that were laboratory-confirmed.	A measure of **Diagnostic Certainty.**
Member Barangays	The list of barangays that were grouped into this cluster.	The geographic components of the profile.

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A.The Final Thematic Map: A map of Cabuyao City showing the geographic distribution of the four final risk profiles.



B. The Final Cluster Profile Summary Table: A table showing the average characteristics of the barangays within each of the four clusters.

Cluster Risk Profile	Avg. Total Cases	Avg. Incidence Rate (per 10k)	Rainy Season Cases	Avg. Peak Month	Youth Cases	Female Cases	Severe Cases	Confirmed Cases	Member Barangays
Very High Risk - The Severe Outbreak Zone	156.25	41.73	76.0%	7.71 (Late Aug)	70.9%	45.0%	11.7%	0.3%	MAMATID, MARINIG, PULO, SAN ISIDRO
High Risk - The Early Season Hotspots	35.67	29.80	64.0%	7.44 (Mid Jul)	69.6%	41.1%	8.1%	1.0%	BACLARAN, BANLIC, BARANGAY TRES (POB.), BUTONG, PITTLAND, SALA
Moderate Risk - The Rainy Season Spike Zone	37.43	20.93	85.0%	8.50 (Mid Aug)	72.9%	47.3%	7.8%	0.0%	BANAYBANAY, BARANGAY DOS (POB.), BIGAA, CASILE, DIEZMO, GULOD, NIUGAN

Low Risk - The Background Transmission Zone	9.00	24.39	44.0%	10.11 (Early Oct)	77.8%	66.7%	0.0%	0.0%	BARANGAY UNO (POB.)

C. The Interpretation of Final Clusters: A written analysis describing the "persona" and public health significance of each cluster.

- **Very High Risk - The Severe Outbreak Zone**
  - **Quantitative Profile:** This cluster shows the highest average case count (156.25 cases), the highest risk per person (41.73 per 10,000), and the highest rate of severe cases (11.67%). Its pattern is strongly seasonal, with 76% of cases occurring in the rainy season, and its outbreaks peak early, around late July (average peak month ~7.71).
  - **Interpretation:** The "Very High Risk" designation comes from the combination of high case numbers and high per-person risk. It is named "The Severe Outbreak Zone" because its highest-in-the-city rate of severe cases makes its outbreaks particularly dangerous. This is clearly the city's main dengue hotspot, representing the top priority for public health action. These barangays require intensive, year-round mosquito control, careful monitoring of patients for severe symptoms, and focused community awareness programs.

- **High Risk - The Early Season Hotspots**
  - **Quantitative Profile:** This cluster has the third-highest average case count (35.67 cases) but the second-highest risk per person (29.80 per 10,000). A key difference is a lower rate of severe cases (8.06%). Its seasonal connection is less strong (64% of cases in the rainy season), and it has the earliest seasonal peak of all groups (average peak month ~7.44).
  - **Interpretation:** The "High Risk" designation is due to the high per-person risk, even if the total number of cases is less than other groups. It is named "The Early Season Hotspots" because its most unique and actionable trait is its tendency to start the city's dengue season. While patients here are less likely to have a severe form of the disease, the high case numbers make these established urban barangays a clear second

priority. Prevention efforts and hospital readiness should begin first in these areas.

- **Moderate Risk - The Rainy Season Spike Zone**
  - **Quantitative Profile:** This group has the second-highest moderate case count (37.43 cases) but a lower risk per person (20.93 per 10,000). Their defining trait is an extreme connection to the rainy season, with 85% of their cases occurring then. Their outbreaks peak later than the other high-risk groups (average peak month ~8.50, or mid-August).
  - **Interpretation:** The "Moderate Risk" level reflects its lower per-person risk, despite a higher total number of cases than the "High Risk" group. It is named "The Rainy Season Spike Zone" because its dengue problem is explosive and predictable when the rains come. This makes these barangays ideal targets for precisely-timed preventative actions, such as clean-up drives right before the rainy season begins.

- **Low Risk - The Background Transmission Zone**
  - **Quantitative Profile:** This cluster has the lowest average case count (9.00 cases). Its pattern is not seasonal, with only 44% of cases in the rainy season. It has 0.00% recorded severe cases. The average peak month is very late (~10.11, or early October).
  - **Interpretation:** The "Low Risk" designation comes from the very low case counts and lack of severe outcomes. It is named the "Background Transmission Zone" because the virus does not seem to spread easily or follow a predictable pattern here. These barangays are the lowest priority for targeted action, and routine city-wide health monitoring is likely sufficient.

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Validation Table

Instruction: Following your review of the maps and profile descriptions, this table is provided for you to score the study's findings.

Part A: Overall Assessment
Please rate the overall study results based on the following statements.

Statement	Rating (1-5)	Comments (Optional)
1. The study's overall methodology (feature engineering and clustering) is sound and appropriate for the research goals.	5	
2. The four features used (Total Cases, Incidence Rate, Rainy Season Proportion, Average Peak Month) are a relevant and sufficient set of metrics to define dengue risk in Cabuyao.	5	
3. The final set of four risk profiles provides a clear and useful new way to understand dengue risk in the city.	4	
4. The thematic map is an effective tool for visualizing the geographic distribution of risk.	5	

Part B: Assessment of Individual Risk Profiles
For each of the four risk profiles identified, please provide your assessment based on the three core criteria.

Profile: Very High Risk - The Severe Outbreak Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	

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Profile: High Risk - The Early Season Hotspots

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	

Profile: Moderate Risk - The Rainy Season Spike Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	

Profile: Low Risk - The Background Transmission Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	4	
Local Validity (Does this profile make sense for the barangays included?)	5	

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Validation Certification

November 04, 2025

This letter serves to formally certify that I, JOHN CARLO B. GARBIN, in my capacity as **Public Health Expert Validator**, have completed the validation process for the research results presented by Nathalie Shane Dejumo, Ma. Cristel P. Roaring, and Jabez Estanislao, for the project titled: **"CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING."**

The validation process, conducted on November 4, 2025, focused specifically on the **clarity, public health relevance**, and **local validity** of the dengue risk profiles generated by the feature-based clustering model. This involved reviewing the final thematic map, the quantitative cluster profile summary table, and the written interpretation for each of the four identified risk profiles.

Based on this comprehensive review, I hereby certify that the results as presented in the final report/documentation are consistent with the data provided and the methodology used.

Furthermore, I confirm that the model's outputs were validated to accurately reflect the findings of the study and that the resulting risk profiles are generally consistent with the epidemiological patterns expected for Cabuyao City.

The research team's presentation of these results is an appropriate interpretation of the validated outputs and provides a useful, data-driven framework for public health planning.

JOHN CARLO B. GARBIN, RN
Signature over Printed Name

License/Position: LIC. NO. 0695212 / NURSE / DISEASE SURVEILLANCE OFFICER / CHO-I
Affiliation: CABUYAO CITY HEALTH OFFICE 1
Date of Validation: NOVEMBER 4, 2025

Evaluation Form – City Health Office 2 (CHO-2)

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CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING

Nathalie Shane Dejumo, Ma. Cristel P. Roaring, Jabez Estanislao
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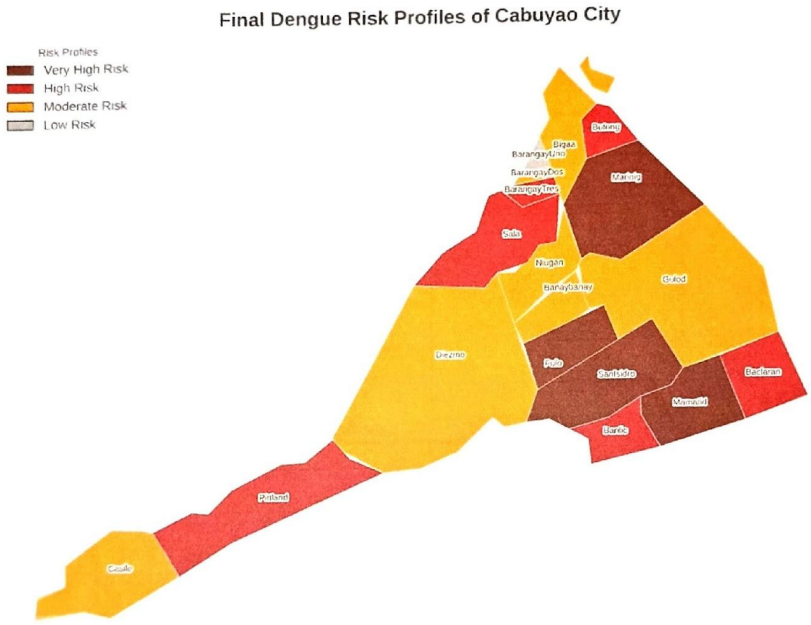
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Moderate Risk - The Rainy Season Spike Zone	37.43	20.93	85.0%	8.50 (Mid Aug)	72.9%	47.3%	7.8%	0.0%	BANAYBANAY, BARANGAY DOS (POB.), BIGAA, CASILE, DIEZMO, GULOD, NIUGAN

Low Risk - The Background Transmission Zone	9.00	24.39	44.0%	10.11 (Early Oct)	77.8%	66.7%	0.0%	0.0%	BARANGAY UNO (POB.)

C. The Interpretation of Final Clusters: A written analysis describing the "persona" and public health significance of each cluster.

- **Very High Risk - The Severe Outbreak Zone**
  - **Quantitative Profile:** This cluster shows the highest average case count (156.25 cases), the highest risk per person (41.73 per 10,000), and the highest rate of severe cases (11.67%). Its pattern is strongly seasonal, with 76% of cases occurring in the rainy season, and its outbreaks peak early, around late July (average peak month ~7.71).
  - **Interpretation:** The "Very High Risk" designation comes from the combination of high case numbers and high per-person risk. It is named "The Severe Outbreak Zone" because its highest-in-the-city rate of severe cases makes its outbreaks particularly dangerous. This is clearly the city's main dengue hotspot, representing the top priority for public health action. These barangays require intensive, year-round mosquito control, careful monitoring of patients for severe symptoms, and focused community awareness programs.

- **High Risk - The Early Season Hotspots**
  - **Quantitative Profile:** This cluster has the third-highest average case count (35.67 cases) but the second-highest risk per person (29.80 per 10,000). A key difference is a lower rate of severe cases (8.06%). Its seasonal connection is less strong (64% of cases in the rainy season), and it has the earliest seasonal peak of all groups (average peak month ~7.44).
  - **Interpretation:** The "High Risk" designation is due to the high per-person risk, even if the total number of cases is less than other groups. It is named "The Early Season Hotspots" because its most unique and actionable trait is its tendency to start the city's dengue season. While patients here are less likely to have a severe form of the disease, the high case numbers make these established urban barangays a clear second

priority. Prevention efforts and hospital readiness should begin first in these areas.

- **Moderate Risk - The Rainy Season Spike Zone**
  - **Quantitative Profile:** This group has the second-highest moderate case count (37.43 cases) but a lower risk per person (20.93 per 10,000). Their defining trait is an extreme connection to the rainy season, with 85% of their cases occurring then. Their outbreaks peak later than the other high-risk groups (average peak month ~8.50, or mid-August).
  - **Interpretation:** The "Moderate Risk" level reflects its lower per-person risk, despite a higher total number of cases than the "High Risk" group. It is named "The Rainy Season Spike Zone" because its dengue problem is explosive and predictable when the rains come. This makes these barangays ideal targets for precisely-timed preventative actions, such as clean-up drives right before the rainy season begins.

- **Low Risk - The Background Transmission Zone**
  - **Quantitative Profile:** This cluster has the lowest average case count (9.00 cases). Its pattern is not seasonal, with only 44% of cases in the rainy season. It has 0.00% recorded severe cases. The average peak month is very late (~10.11, or early October).
  - **Interpretation:** The "Low Risk" designation comes from the very low case counts and lack of severe outcomes. It is named the "Background Transmission Zone" because the virus does not seem to spread easily or follow a predictable pattern here. These barangays are the lowest priority for targeted action, and routine city-wide health monitoring is likely sufficient.

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Validation Table

Instruction: Following your review of the maps and profile descriptions, this table is provided for you to score the study's findings.

Part A: Overall Assessment
Please rate the overall study results based on the following statements.

Statement	Rating (1-5)	Comments (Optional)
1. The study's overall methodology (feature engineering and clustering) is sound and appropriate for the research goals.	5	Methodology is well structured & specifically grounded. Allowing correct grouping of barangay based on dengue risk.
2. The four features used (Total Cases, Incidence Rate, Rainy Season Proportion, Average Peak Month) are a relevant and sufficient set of metrics to define dengue risk in Cabuyao.	5	It effectively captured the key epidemiological dimension of dengue transmission - burden, risk, timing and seasonal.
3. The final set of four risk profiles provides a clear and useful new way to understand dengue risk in the city.	5	Visualization summary could further improve clarity, but overall good.
4. The thematic map is an effective tool for visualizing the geographic distribution of risk.	5	Good. It enhances understanding by clear showing clustering and high risk zone, aiding targeted intervention.

Part B: Assessment of Individual Risk Profiles
For each of the four risk profiles identified, please provide your assessment based on the three core criteria.

Profile: Very High Risk - The Severe Outbreak Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	Profile is clear & easily distinguishes severe outbreak zone.
Actionability and Public Health Relevance	5	Highly relevant, helps prioritize resource deployment & emergency vector control.
Local Validity (Does this profile make sense for the barangays included?)	5	Accurately reflects outbreak prone barangay

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Profile: High Risk - The Early Season Hotspots

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	

Profile: Moderate Risk - The Rainy Season Spike Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	

Profile: Low Risk - The Background Transmission Zone

Criteria	Rating (1-5)	Comments (Optional)
Clarity and Interpretability	5	
Actionability and Public Health Relevance	5	
Local Validity (Does this profile make sense for the barangays included?)	5	Reflects Barangay with consistently low dengue activity

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Validation Certification

November 04, 2025

This letter serves to formally certify that I, EUNICE ERICA B. CRANSTOUN, in my capacity as **Public Health Expert Validator**, have completed the validation process for the research results presented by Nathalie Shane Dejumo, Ma. Cristel P. Roaring, and Jabez Estanislao, for the project titled: **"CLUSTERING ANALYSIS OF TWO-YEAR SEASONAL DENGUE PATTERNS IN CABUYAO CITY, LAGUNA FOR BARANGAY RISK PROFILING."**

The validation process, conducted on November 4, 2025, focused specifically on the **clarity**, **public health relevance**, and **local validity** of the dengue risk profiles generated by the feature-based clustering model. This involved reviewing the final thematic map, the quantitative cluster profile summary table, and the written interpretation for each of the four identified risk profiles.

Based on this comprehensive review, I hereby certify that the results as presented in the final report/documentation are consistent with the data provided and the methodology used.

Furthermore, I confirm that the model's outputs were validated to accurately reflect the findings of the study and that the resulting risk profiles are generally consistent with the epidemiological patterns expected for Cabuyao City.

The research team's presentation of these results is an appropriate interpretation of the validated outputs and provides a useful, data-driven framework for public health planning.

EUNICE ERICA B. CRANSTOUN, RN, MMPHA
Signature over Printed Name

License/Position: 0735668 NURSE / DISEASE SURVEILLANCE OFFICER - II
Affiliation: CITY HEALTH OFFICE II CABUYAO
Date of Validation: 11/04/25

APPENDIX G

Publishable Paper

CLUSTERING ANALYSIS OF SEASONAL DENGUE PATTERNS IN CABUYAO CITY LAGUNA FOR BARANGAY RISK PROFILING

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ABSTRACT

This study analyzed two years of dengue case data in Cabuyao City, Laguna, with the aim of developing clear risk profiles for each barangay. Dengue cases from 2021 to 2022 were examined using three clustering algorithms: K-Means, Fuzzy C-Means, and Affinity Propagation. The analysis focused on key features: total number of cases, incidence rate, seasonal proportion, and peak month. To evaluate the performance of the models, validation measures including Silhouette Score, Davies-Bouldin Index, and Dunn Index were used. The results show that affinity propagation provides the most reliability than other algorithms tested. The findings showed four distinct dengue risk profiles: (1) very high-risk barangays with high cases and severity, (2) high-risk barangays with early seasonal peaks, (3) moderate-risk barangays with strong rainy season patterns, and (4) low-risk barangays with fewer cases and irregular transmission. This study can help improve dengue prevention by guiding local health officials in focusing their actions and resources on the most at-risk barangays.

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INTRODUCTION

Dengue fever is a major health problem in the Philippines, especially during the rainy season. According to WHO [1], an estimated 390 million dengue infections occur worldwide each year, with most cases in Southeast Asia due to rapid urbanization, climate change, and poor control measures [2]. In the Philippines, the Department of Health reported over 200,000 cases in 2022 [3].

In Cabuyao City, Laguna, dengue cases rise seasonally, usually every three years, making it hard for local health agencies to predict outbreaks [4]. In January and February 2025, 192 cases were recorded. To address this, Laguna implemented the 5S strategy: search and destroy mosquito breeding sites, take self-protection measures, seek early consultation, support fogging in hot spots, and sustain hydration [4].

This study used clustering techniques on dengue data from 2021–2022 to identify high-risk areas and seasonal patterns. Findings can help officials improve surveillance and respond more effectively to dengue [5].

1.1 Objectives

The main objective of this study is to develop and evaluate a machine learning framework for creating data-driven dengue risk profiles of the barangays in Cabuyao City using seasonal time-series data. This research specifically aims to:

1. Analyze the spatio-temporal patterns of dengue transmission in Cabuyao City by examining the historical (2014-2024) and seasonal (2021-2022) trends of dengue incidence at the city and barangay levels[6].

2. Develop a feature-based clustering model by engineering key epidemiological metrics (Burden, Risk, Seasonality, Timing) from the daily time-series data[7].

3. Evaluate and select an optimal clustering algorithm from a set of centroid-based methods (K-Means, Fuzzy C-Means, and Affinity Propagation) by conducting a rigorous comparative analysis using multiple internal validation indices[8].

4. Define and map the final barangay dengue risk profiles based on the results of the selected optimal algorithm, and characterize these profiles using their unique temporal, demographic, and clinical severity characteristics to provide actionable insights for public health interventions[5].

Significance of the Study

This study employed clustering methods to develop barangay-level dengue risk profiles in Cabuyao City using data from 2021–2022. The results can inform targeted prevention strategies, enhance local health planning, and support the achievement of SDG goals 3 (good health and well-being) and 11 (sustainable cities and communities) [9].

Local Health Agencies: Identification of high-risk barangays enables focused interventions, including fogging, elimination of mosquito breeding sites, and public awareness campaigns [1].

Residents in Cabuyao: The study promotes community engagement by encouraging precautionary measures such as proper waste management, cleaning stagnant water, and safe water storage [10].

Future Researchers: This research contributes to spatial epidemiology by demonstrating the use of clustering techniques on long-term health data, providing insights for disease risk mapping and public health surveillance [5].

Scope and Limitations of the Study

This study applied and evaluated three clustering algorithms, K-Means, Affinity Propagation, and Fuzzy C-Means to identify dengue risk profiles in Cabuyao City [7]. A feature-based time-series approach was used on 2021–2022 dengue data, incorporating four engineered features: Total Cases (Burden), Incidence Rate (Risk), Rainy Season

Proportion (Seasonality), and Average Peak Month (Timing).

The study was limited to 2021–2022, excluding partial 2023 data to maintain integrity, and relied on the accuracy of official case reports [3]. Other methods, such as shape-based clustering, and external factors like environmental or socio-economic variables, were not included. Model selection was based on three internal validation metrics: Silhouette Score, Dunn Index, and Davies–Bouldin Index [22].

2. THEORETICAL FRAMEWORK OF THE STUDY

Dengue fever, primarily transmitted by *Aedes aegypti* and *Aedes albopictus*, is a major global and Philippine public health concern, with an estimated 390 million infections annually and nearly half of the world's population at risk [11]. In Cabuyao City, Laguna, seasonal outbreaks are influenced by urbanization, poor sanitation, stagnant water, and climatic factors [12,13]. Meteorological variables such as rainfall, temperature, and humidity affect dengue incidence with a lag corresponding to mosquito life cycles, supporting the use of time series analysis for outbreak prediction [14].

Spatio-Temporal Analysis and Clustering

GIS-based spatio-temporal analysis helps identify dengue hotspots and outbreak dynamics, enabling targeted interventions and early warning systems [15]. Clustering methods—including K-Means, Fuzzy C-Means, and Affinity Propagation—classify barangays based on spatial and temporal patterns, capturing seasonality and outbreak timing [3]. This study used a feature-based approach with four epidemiologically relevant features: Total Cases (Burden), Incidence Rate (Risk), Rainy Season Proportion (Seasonality), and Average Peak Month (Timing).

Dengue Prevention and Risk Profiling

Effective dengue control relies on integrated vector management, community participation, and preventive strategies, supported by data-driven risk profiling [3,19]. Time series and clustering analyses allow identification of high-risk areas and periods, guiding precise interventions such as focused vector control, health education campaigns, and community clean-ups [20].

Conceptual Framework

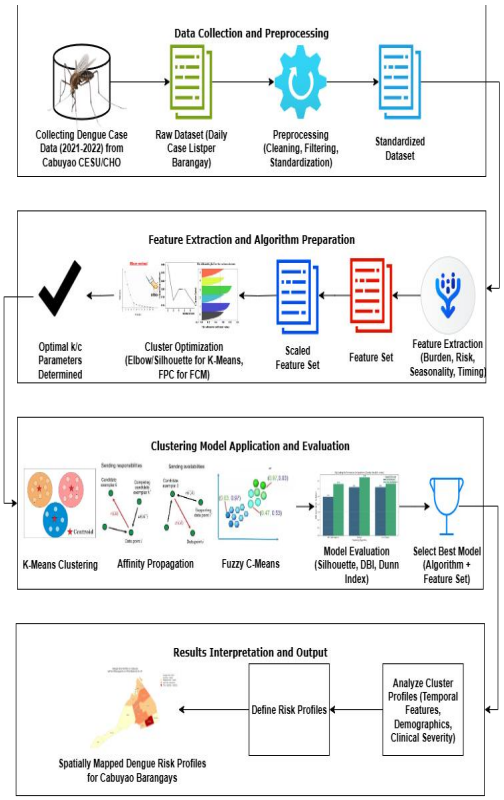


Figure 1. Conceptual Framework illustrating the feature-based clustering process of dengue cases in Cabuyao City from 2021 to 2022.

As shown in Figure 1, The workflow begins with preprocessing 2021–2022 dengue case data from CESU, including cleaning and standardization [21]. Feature-based clustering is applied, and internal validation metrics—Silhouette Score, Davies–Bouldin Index, and Dunn Index—select the optimal algorithm [22]. Cluster assignments are analyzed with demographic and clinical data to generate descriptive risk profiles, visualized as thematic spatial maps for local public health planning and targeted interventions [25].

3. METHODOLOGY

Study Area and Data

The study was conducted in Cabuyao City, Laguna, Philippines, consisting of 18 barangays and characterized by urbanized residential and industrial zones conducive to dengue transmission. The primary dataset comprised a fully de-identified daily dengue surveillance line list from January 2021 to December 2022, obtained from the Cabuyao City Epidemiology and Surveillance Unit, totaling 1,113 confirmed cases. Annual dengue case totals from 2014–2024 were used solely for contextual analysis. Barangay population data were sourced from the Philippine Statistics Authority, and spatial boundary data were obtained from publicly available GIS databases.

Data Preprocessing and Feature Engineering

All records were anonymized prior to analysis. Barangay names were standardized, and missing dates were resolved using available onset or admission data. A feature-based representation was then constructed for each barangay using four epidemiological indicators: total cases (burden), incidence rate per 10,000 population (risk), proportion of cases during the rainy season (June–November) (seasonality), and average peak month of transmission (timing). Features were standardized using Z-score normalization.

Clustering and Model Evaluation

Three unsupervised clustering algorithms—K-Means, Fuzzy C-Means, and Affinity Propagation—were evaluated. K-Means and Fuzzy C-Means were tested across multiple cluster sizes, with four clusters identified as the optimal solution based on internal validation. Affinity Propagation was included to detect natural groupings without predefining the number of clusters. Model performance was assessed using the Silhouette Score, Davies–Bouldin Index, and Dunn Index, and the best-performing model was selected to generate barangay-level dengue risk profiles.

Ethical Considerations

All dengue records were fully de-identified, and only variables relevant to the study objectives were retained. The analysis was conducted to support public health planning without stigmatizing communities, using publicly available data sources and open-source tools to ensure transparency and reproducibility.

4. RESULTS AND DISCUSSION

Historical Trends of Dengue in Cabuyao City

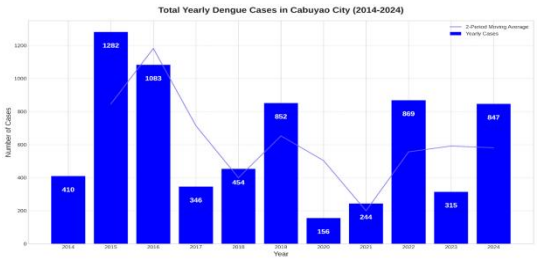


Figure 3: Historical Trends (2014–2020)

To provide context for dengue trends in Cabuyao, we looked at historical data from 2014 to 2020. Figure 3 shows a consistent increase and decrease in dengue cases. There is an increase in cases every 2 to 3 years. Notably, 2020 recorded the lowest number of cases, likely because people were less exposed to dengue habitats due to pandemic restrictions.

Descriptive Epidemiology of Dengue Cases (2021–2022)

To establish a baseline for analysis, this section presents a statistical summary of the 1,113 dengue cases recorded during the 2021 to 2022 study period in Cabuyao City.

Characteristic	Metric	Value
Age Distribution	**0-5 yrs**	184 (16.5%)
	5 > - 19 yrs	607 (54.5%)
	> 19 yrs	322 (28.9%)
Sex Distribution	**Male**	611 (54.9%)
	Female	502 (45.1%)
Outcome	**Survived**	1109

	Died	4
	Case Fatality Rate	0.36%
Clinical Severity	**Dengue Fever (DF)**	989 (90.1%)
	Dengue Hemorrhagic Fever (DHF)	103 (9.4%)
	Dengue Shock Syndrome (DSS)	6 (0.5%)
	Total Severe Cases (DHF+DSS)	109 (9.9%)
Case Classification	**Confirmed**	4 (0.4%)
	Probable	698 (63.6%)
	Suspected	396 (36.1%)

Table 3. Comprehensive Descriptive Profile of Dengue Cases

As shown in Table 3 Dengue cases were concentrated in the 5–19 age group (54.5%), with a slight male predominance (54.9%). The overall case fatality rate was 0.36% (4 deaths), and 9.9% of cases with severity data were classified as severe dengue (DHF/DSS).

Severity **Age_Group**	**Non-Severe (DF)**	**Severe(DHF/DSS)**
0-5 yrs	170 (93.4%)	12 (6.6%)
5>-19 yrs	538 (89.7%)	62 (10.3%)
>10 yrs	282 (89.0%)	35 (11.0%)

Table 4. Age Distribution by Clinical Severity

In Table 4, although children aged 5–19 years had the highest incidence of dengue, the likelihood of developing severe dengue increased with age. Adults (>19 years) showed the highest proportion of severe cases (11.0%), compared with 10.3% in those aged 5–19 years and 6.6% in children under 5 years. This suggests that older patients, while fewer in number, may require closer monitoring for severe outcomes.

Temporal Trends and Barangay-Level Spatial Distribution

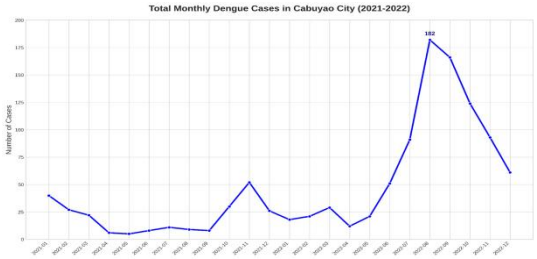


Figure 4. Total Monthly Dengue Cases in Cabuyao City (2021–2022)

Figure 4 shows the total monthly dengue cases in Cabuyao City. Cases peaked in August 2022, consistent with the seasonal pattern where dengue increases during the rainy season and decreases during the dry season.

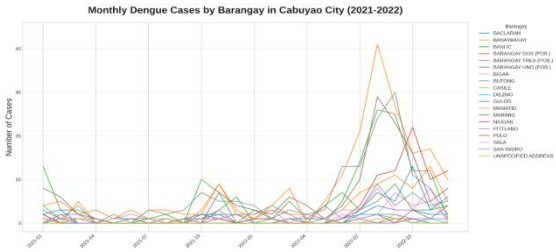


Figure 5. Monthly Dengue Cases in Cabuyao (2021–2022)

Figure 5 illustrates the monthly dengue cases for each barangay, highlighting variation across the city. This helps identify which barangays experience higher seasonal peaks.

Thematic Maps of Dengue Burden and Incidence Rate

To better visualize the spatial distribution of dengue cases in Cabuyao City, thematic maps were created. These maps highlight both the total number of reported cases and the relative risk based on population size.

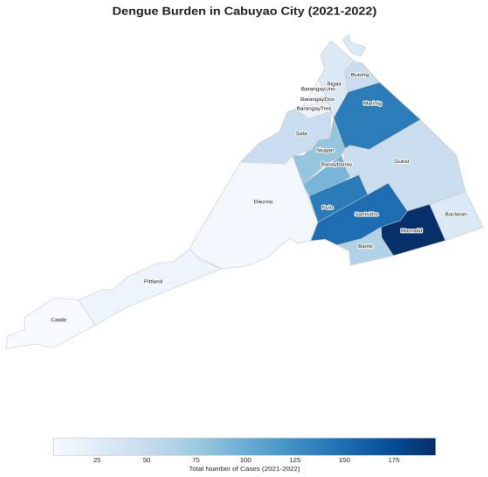


Figure 6. Dengue Case Burden per Barangay (2021–2022)

Figure 6 shows the total dengue burden per barangay. The highest case counts were recorded in Mamatid, San Isidro, Pulo, and Marinig, indicating concentrated transmission in these more urbanized areas. In contrast, barangays such as Casile and Diezmo had considerably fewer cases.

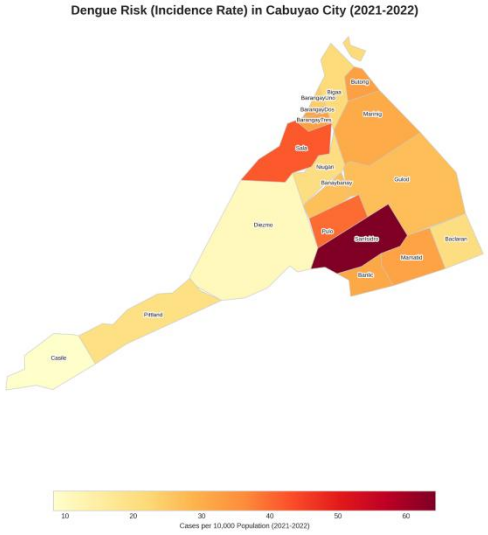


Figure 7. Dengue Incidence Rate per Barangay (2021–2022)

Figure 7 presents the dengue incidence rate per 10,000 population, offering a more risk-adjusted perspective. San Isidro and Pulo show the highest incidence rates, reflecting greater relative risk even after accounting for population size. Meanwhile, Casile, Pittland, and Diezmo remain in the low-risk category.

Evaluation and Selection of Clustering Algorithm

Before grouping barangays by similar dengue patterns, the raw data were converted to four main numerical features: **Number of cases, Incidence Rate, Proportion during Rainy Season,** and **Average Peak Month**. This information is displayed in Table 5. We cleaned the data, normalized it to per barangay, and scaled it where possible. This step is important to prevent biases in clustering algorithms.

Barangay	**Feat_Total_Cases**	**Feat_Incidence_Rate**	**Feat_Rainy_Proportion**	**Feat_Avg_Peak_Month**
BACLARAN	30	20.54	0.67	7.53
BANAYBANAY	92	26.85	0.76	8.36
BANLIC	64	31.00	0.58	7.59
BARANGAY DOS (POB.)	5	31.79	1.00	8.80
BARANGAY TRES (POB.)	14	32.76	0.64	7.64
BARANGAY UNO (POB.)	9	24.39	0.44	10.11
BIGAA	29	21.22	0.79	8.10
BUTONG	48	32.51	0.62	8.00
CASILE	3	8.29	1.00	8.33
DIEZMO	7	10.57	0.86	7.43
GULOD	47	27.30	0.77	9.17
MAMATID	196	32.09	0.77	7.88
MARINIG	139	30.66	0.82	7.64
NIUGAN	79	20.48	0.78	9.28

PITTLAND	12	19.83	0.58	6.50
PULO	140	39.87	0.68	7.53
SALA	46	42.19	0.72	7.41
SAN ISIDRO	150	64.31	0.79	7.79

Table 5. Final Feature Table for Clustering (Sorted by Barangay).

Internal Validation of K-Means and Fuzzy C-Means (k = 2 to 10)

To evaluate how well clustering algorithms highlight meaningful patterns in the data, we applied K-Means and Fuzzy C-Means using various cluster counts (k = 2 to 10). These methods also require specifying the number of clusters in advance. The results from the internal validation steps help determine the best number of clusters.

Three metrics were used for evaluation:

- Silhouette Score (higher is better): This measures how well each item in a cluster fits with its group and how distinct it is from other clusters.
- Davies-Bouldin Index (DBI) (lower is better): This measures how compact the clusters are and how well they are separated.
- Dunn Index (higher is better): This value clusters that are compact and well separated.

K-Means Baseline Evaluation Score

K	Silhouette	Davies-Bouldin	Dunn Index
2	0.202	1.338	0.136
3	0.282	1.145	0.247
4	0.282	1.019	0.378
5	0.297	0.796	0.378
6	0.238	0.779	0.378
7	0.288	0.574	0.457
8	0.225	0.603	0.438
9	0.213	0.637	0.445
10	0.144	0.593	0.445

Fuzzy C-Means Baseline Evaluation Scores

k	Silhouette	Davies-Bouldin	Dunn Index
2	0.228	1.455	0.229
3	0.282	1.145	0.247
4	0.303	0.981	0.321
5	0.150	1.007	0.378
6	0.151	0.978	0.362
7	0.088	0.891	0.362
8	0.163	0.878	0.362
9	0.057	0.906	0.254
10	0.045	0.777	0.254

Table 6. Internal Validation Scores for K-Means and Fuzzy C-Means (k = 2 to 10)

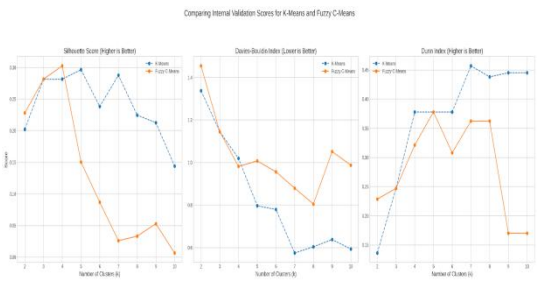


Figure 8. Internal Validation Scores for K-Means and Fuzzy C-Means

The optimal structure for clustering K-Means and Fuzzy C-Means is at k=4 as evident from the validation metrics. At k=4, Fuzzy C-Means achieved the highest Silhouette Score and the lowest DBI. This suggests that the clustering had good cohesion

and separation. K-Means, on the other hand, performed somewhat better in the Dunn Index. However, both methods have performed quite

Hyperparameter Optimization

Each algorithm underwent hyperparameter tuning to ensure optimal performance:

- **K-Means**: Compared k-means++ vs. random initialization
- **Fuzzy C-Means**: Fuzziness coefficient (m) tested at 1.5, 2.0, and 2.5
- **Affinity Propagation**: Tested damping factors of 0.7 and 0.8

Model	Silhouette	Davies-Bouldin	Dunn Index
K-Means (init=k-means++)	0.282	1.019	0.378
K-Means (init=random)	0.301	0.822	0.400
Fuzzy C-Means (m=1.5)	0.303	0.981	0.321
Fuzzy C-Means (m=2.0)	0.303	0.981	0.321
Fuzzy C-Means (m=2.5)	0.272	1.018	0.253
Affinity Prop (damp=0.7, k=4)	0.301	0.822	0.400
Affinity Prop (damp=0.8, k=4)	0.301	0.822	0.400

Table 7. Internal Validation Scores from Hyperparameter Optimization (k = 4)

Table 7 shows both Fuzzy C-Means undertaken at m = 1.5 and m = 2.0, a Silhouette Score value of 0.303, which is the highest among all the techniques studied. K-Means and Affinity Propagation thus achieved fairly at k=4 therefore had performance degradation at high fuzziness level

the best values of the Davies-Bouldin Index of 0.822 along with a maximum Dunn Index of 0.400, suggesting that they are the best for compactness and separation of clusters. FCM had the lowest scores with m = 2.5 on all three metrics and

Final Model Selection

The three models were compared based on the final internal validation results. The performance is summarized in Table 8 and Figure 9.

Model	Silhouette	Davies-Bouldin	Dunn Index
K-Means (Optimized)	0.282	1.019	0.378
Fuzzy C-Means (Optimized)	0.303	0.981	0.321
Affinity Propagation (Optimized, k=4)	0.301	0.822	0.400

Table 8. Final Model Performance Comparison

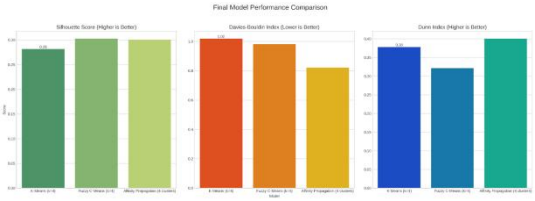


Figure 9. Final Model Performance Comparison Across Metrics

Based on Table 8, and Figure 9, Affinity Propagation scored the best in the Davies-Bouldin Index (0.822) and the Dunn Index (0.400), where clusters should be compact and well-separated. FCM gave the highest Silhouette Score (0.303), indicating that clusters are more strongly interconnected. K-Means results were consistent across all measures but did not outperform other model testing measures.

Overall, Affinity Propagation became the final model in this study, as its performance outweighed other models regarding separation and compactness. It was the most successful in distinguishing among different dengue risk profiles found within the barangay.

Final Cluster Assignment and Visualization

To understand the spatial patterns of the results, the cluster assignments from the best-performing model, Affinity Propagation plotted on a map of Cabuyao City. A qualitative colormap was used to emphasize that the clusters represent distinct, non-ordered groups.

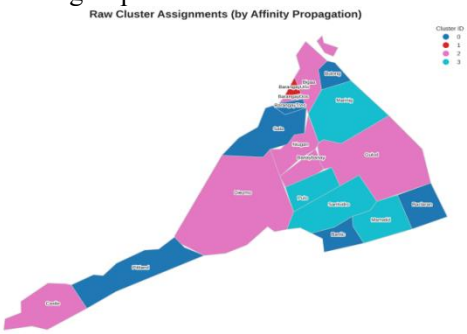


Figure 10. Raw Cluster Assignments (by Affinity Propagation)

Figure 10 shows the spatial distribution of dengue risk clusters in Cabuyao City. Cluster 3, which includes Mamatid and San Isidro, stands out as the main hotspot with the highest burden and severity. Cluster 2, including Diezmo and Gulod, saw noticeable peaks during the rainy season while maintaining moderate risk levels overall. Cluster 0 showed earlier-season outbreaks with high incidence rates, implying that an early intervention should be imposed. Transmission in Cluster 1, represented only by Barangay Uno, remained low and irregular, keeping it at a low risk but still requiring ongoing monitoring. These patterns allow for targeted dengue control strategies for each cluster.

Quantitative Cluster Profiling

To understand what defines each cluster, average profiles were created based on the barangays grouped together. These profiles combine time-related dengue features with added sociodemographic and clinical indicators to show how each cluster differs in disease burden, timing, and severity.

Temporal Features

- **Feat_Total_Cases** – Average number of dengue cases (*burden*).
- **Feat_Incidence_Rate** – Average cases per 10,000 people (*risk*).
- **Feat_Rainy_Proportion** – Share of cases during rainy months (*seasonality*).
- **Feat_Avg_Peak_Month** – Average month when cases peak (*timing*).

Demographic & Clinical Features

- **Youth_Cases_%** – Proportion of cases under 20 years old (*age pattern*).
- **Female_Pct** – Percentage of female cases.
- **Severity_Index_%** – Share of severe cases (DHF/DSS).
- **Confirmation_Rate_%** – Percentage of lab-confirmed cases.

Final_Cluster	Feat_Total_Cases	Feat_Incidence_Rate	Feat_Rainy_Proportion	Feat_Avg_Peak_Month	Youth_Cases_%	Female_Pct	Severity_Index_%	Confirmation_Rate_%
0	35.67	29.80	0.64	7.44	69.63	41.12	8.06	0.95
1	9.00	24.39	0.44	10.11	77.78	66.67	0.00	0.00
2	37.43	20.93	0.85	8.50	72.90	47.33	7.75	0.00
3	156.25	41.73	0.76	7.71	70.88	44.96	11.67	0.32

Table 9. Summary of Dengue Risk Profiles by Cluster

Table 9 presents the Analysis of clusters using time, spread, and severity comparison. Cluster 3 is the highest risk as it is the cluster with the highest number of cases (156.25), the highest incidence rate (41.73), and the best share of severe cases (11.67%). Cluster 0 is also characterized by its incidence rate (29.80) and peaks early in July, necessitating early interventions. Cluster 2 has a strong rainy-season pattern with 85% of cases being in August, while Cluster 1 registers the least number of cases (9.00) and peaks in October, with no severe cases. These insights provide a strong, data-driven foundation for designing dengue prevention and control strategies that fit the unique profile of each cluster.

Visualizing Final Cluster Profiles

In addition, we provided a radar chart for visual presentation to help illustrate the multi-dimensional features of each dengue risk cluster. The variables were normalized by scaling them all from 0 to 1 (min-max

normalization) with a weight indicating the degree of presence of that feature, in terms of presence (either incidence or severity), or seasonality ratio.

Risk classification by color:

- Very High Risk - Cluster 3 (Dark red)
- High Risk - Cluster 0 (Orange)
- Moderate Risk - Cluster 2 (Light orange)
- Low Risk - Cluster 1 (Gray)



Figure 11. Radar Chart Comparing Dengue Risk Clusters Across Key Features

Figure 11 shows that Cluster 3 has increased in particular in terms of burden, risk, and severity. The peak is earlier in terms of incidence and higher among young people than in Cluster 0. In particular, for Cluster 2, there is a very strong seasonal pattern with high scores in peak months, and Cluster 1 scores low overall and is considered a lower priority.

Thematic Map of Final Dengue Risk Profiles

Finally, we developed the final map for Cabuyao City to illustrate dengue risk levels in all barangays. Barangays are color-shaded by risk category: Very High Risk (dark red), High Risk (red-orange), Moderate Risk (orange), and Low Risk (gray). These categories have been defined directly based on our clustering results and, hence, differences in case numbers, incidence rates, timing of outbreak, and severity.

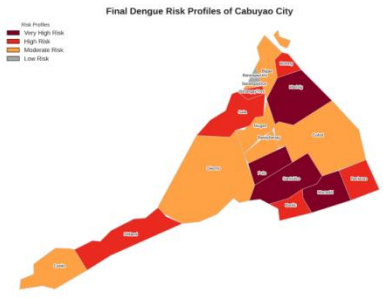


Figure 12. Final dengue risk profiles across Cabuyao City

In Figure 12, high-risk barangays such as Marinig, San Isidro, Mamatid, and Pulo are mostly found in densely populated or urbanized areas, where dengue transmission is continuous and severe cases remain high. Meanwhile, barangays under Moderate and Low Risk, like Casile, Gulod, and Diezmo, are generally less urbanized and show clearer seasonal dengue patterns. This thematic map provides a valuable tool for guiding targeted dengue interventions and optimizing the allocation of public health resources in Cabuyao City.

5. SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

Summary

This study focuses on identifying high-risk barangays in Cabuyao City, Laguna that are vulnerable to dengue outbreaks by applying machine learning clustering techniques on dengue incidence data from 2021 to 2022. There were three algorithms applied (K-Means, Affinity Propagation, and Fuzzy C-Means) that grouped barangays with the same seasonal pattern of dengue cases. In order to truly assess which model was the best, the Silhouette Score, Davies-Bouldin Index (DBI), and Dunn Index were used as three validated internal metric. Within the three algorithms Affinity Propagation was the most consistently able to produce visually compact and separated clusters, which subsequently is the best method to use for dengue risk profiling in this current study.

To provide a broader context, dengue case records from 2014 to 2024 were reviewed to see if the current clustering results matched historical trends. The findings showed that barangays marked as high-risk in the models were often the same areas that reported high dengue cases in previous years. This consistency shows the potential of clustering analysis as a tool for targeted dengue prevention and control.

By identifying recurring high-risk areas, the study provides useful insights for tracking outbreaks, allocating resources, and implementing timely interventions. It also supports the United Nations Sustainable Development Goals. especially Goal 3: Good Health and Well-being and Goal 11: Sustainable Cities and Communities. It promotes healthier communities and improves urban planning to prevent disease.

Conclusions

1. **Implementation of clustering algorithms to dengue incidence patterns**.

This research demonstrated that K-Means, Affinity Propagation and Fuzzy C-Means could be successfully applied to cluster the 18 barangays of Cabuyao City based on their seasonal and temporal patterns of dengue incidence.

2. **Identifying the best clustering algorithm.**

Among the three algorithms, Affinity Propagation was found to have the best clustering performance as it had a lower Davies-Bouldin Index (DBI), higher Silhouette Score, and a moderate Dunn Index resulting in clusters that were compact and well-separated.

3. **Verification of the clustering results against historical dengue report.**

A comparison with past dengue incidence report from the years 2014 to 2024, indicated that the same high-risk barangays would be the recurring areas, confirming the output of the model as dependable and stable.

4. **Importance of clustering analysis for dengue prevention and control.**

The findings illustrate that clustering analysis can be utilized as a decision-support tool for dengue prevention and control, to be able to provide new data about risk, and allow local health authorities to focus their efforts, prioritize actions and share resources and plan for policies to work towards the United Nations Sustainable Development Goals (SDG 3 and SDG 11) .

Recommendations

Targeted intervention - High-risk barangays identified through clustering should be the target of enhanced vector control; it should encompass fogging, larvicides, and community clean-ups; especially before/during peak dengue season.

Regular monitoring of data - Establishing a regular system for dengue surveillance would allow for periodic clustering for further pattern analysis and subsequent identification of revisions in risk and relevant interventions.

Integration of Environmental factors - Future research should incorporate additional variables, particularly rainfall, temperatures, sanitation, and areas of population density, so that clustering provides a more robust basis for informed prediction.

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APPENDIX H

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Vocabulary

Metrics compared to other Grammarly users

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APPENDIX I

TURNITIN REPORT

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APPENDIX J

Curriculum Vitae

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Education

Laguna State Polytechnic University - Los Baños Campus **Los Baños, Laguna**
Bachelor of Science, Computer Science; Specialization in Intelligent Systems 2022-2026

Academic Honors: Dean's Lister

Relevant Coursework: Data Structures and Algorithms, Natural Language Processing, Advanced Machine Learning

Grace Christian Community Schools Inc. *Calauan, Laguna*
Senior High School - Humanities and Social Sciences (HUMSS) Strand *Graduated 2020*
Junior High School *Graduated 2018*

Balayhangin Elementary School *Calauan, Laguna*
Primary School *Graduated 2014*

Projects

Clustering Analysis of Dengue Patterns in Cabuyao City *(Ongoing)*

- Developed a data-driven model to analyze two-year seasonal dengue patterns using machine learning techniques.
- Applied clustering algorithms to profile barangays based on risk levels, providing actionable insights for public health initiatives.
- Utilized Python and its data analysis libraries to process, model, and interpret epidemiological data.

Smart Attendance using RFID *(June 2025 – July 2025)*

- Collaborated in a team to develop an automated attendance checker using RFID technology for a summer class project.
- Integrated a machine learning model to analyze attendance data and predict future student class punctuality.

Experience

HITT | Cryptocurrency, NFT Gaming *Remote*
Community Manager / Scholarship Manager / Scholar *August 2021 – July 2022*

- Functioned as both a player-scholar and manager, actively earning through gameplay while overseeing a community of 20+ players.
- Tracked and recorded player earnings and managed profit-sharing distributions using a web-based tool.
- Onboarded new players by demonstrating gameplay mechanics and leveraging established guides from major guilds (e.g., YGG); delegated further training to experienced players to foster community leadership.
- Organized and executed community engagement events, including mini-tournaments, and managed the distribution of prizes to winners.
- Adapted community strategies in response to fluctuating market conditions of digital assets (SLP/AXS token).

Skills & Interests

Technical: Python (with machine learning/data analysis libraries), Data Analysis, Machine Learning, Google Colab, Visual Studio Code, Microsoft Excel, Google Sheets.
Language: English (Proficient), Filipino (Native).

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Education

Laguna State Polytechnic University - Los Baños Campus **Los Baños, Laguna**

Bachelor of Science, Computer Science; Specialization in Intelligent Systems 2022-2026

Matara High School *Libon, Albay*

Senior High School - Technical-Vocational-Livelihood (TVL) Strand *Graduated 2022*

Junior High School *Graduated 2020*

Bangiawon Elementary School *Oas, Albay*

Primary School *Graduated 2015*

Projects

Clustering Analysis of Dengue Patterns in Cabuyao (*Ongoing)*

- Developed a data-driven model to analyze two-year seasonal dengue patterns using machine learning techniques.
- Applied clustering algorithms to profile barangays based on risk levels, providing actionable insights for public health initiatives.
- Utilized Python and its data analysis libraries to process, model, and interpret epidemiological data.

SafePaws *(June – July 2025)*

- Developed a real-time pet tracking system using ESP32, GPS, and Wi-Fi modules, featuring a web dashboard for location monitoring and activity alerts.
- Integrated a K-Nearest Neighbors (KNN) model to interpret and predict pet movement behaviors.

Experience

Family Business *Remote*

Operations & Sales Assistant *July 2025 – Present*

- Managed and recorded inventory data to ensure accurate stock tracking.
- Handled point-of-sale transactions, processed payments, and assisted customers with purchases.
- Delivered customer service support, addressing inquiries and ensuring a positive buying experience.

Skills & Interests

Technical: Python (basic data handling/cleaning), MS Office & Google Workspace, Google Colab, Microsoft Excel,

Language: English (Basic), Filipino (Native).

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Education

Laguna State Polytechnic University - Los Baños Campus **Los Baños, Laguna**
Bachelor of Science, Computer Science; Specialization in Intelligent Systems 2022-2026

Relevant Coursework: Data Structures and Algorithms, Natural Language Processing,

Saint Ignatius Academy *Cabuyao, Laguna*
Senior High School – Information and Communication Technology (ICT) Strand *Graduated 2022*

Pulo National High School *Cabuyao, Laguna*
Junior High School *Graduated 2020*

San Isidro Elementary School *Cabuyao, Laguna*
Primary School *Graduated 2016*

Projects

Clustering Analysis of Dengue Patterns in Cabuyao City *(Ongoing)*

- Developed a data-driven model to analyze two-year seasonal dengue patterns using machine learning techniques.
- Applied clustering algorithms to profile barangays based on risk levels, providing actionable insights for public health initiatives.
- Utilized Python and its data analysis libraries to process, model, and interpret epidemiological data.

Smart Attendance using RFID *(June 2025 – July 2025)*

- Collaborated in a team to develop an automated attendance checker using RFID technology for a summer class project.
- Integrated a machine learning model to analyze attendance data and predict future student class punctuality.

Experience

Business *Remote*
Freelance Book Reseller / Rent Monitoring system *July 2025 – Present*

- Present Managed an online sales platform, handling customer data and order tracking.
- Utilized social media tools to market and promote products.
- Strengthened problem-solving, digital literacy, and record-keeping skills.
- Designed and developed a system to track tenant information, rent payments, and due dates.
- Implemented features such as overdue alerts, payment status tracking, and mode of payment options.
- Enhanced skills in data processing, automation, and system design.

Skills & Interests

Technical: Data Analysis, Machine Learning, Google Colab, Microsoft Excel, Google Sheets
Language: English (Proficient), Filipino (Native).